

AN ARCHIVAL ALMA SEARCH FOR UNRESOLVED SPECTRAL-CARRIER TECHNOSIGNATURES TOWARD NEARBY STARS

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ABSTRACT

We have searched the public ALMA archive for artificial narrowband emission from nearby stars. “Narrowband” describes the assumed transmitter, not ALMA’s resolving power: these are carriers unresolved within 15.3–1953 kHz channels. The limits are commensurable with Hz-resolution SETI as total powers, but the channel widths differ by about seven decades. The primary experiment seeks a drifting carrier in 403 fine-channel windows toward 60 systems over 47.7 GHz of unique sky frequency, a median 1.84 GHz per system. A secondary search of 1252 coarse windows adds 70.4 GHz of spectral-excess coverage that cannot discriminate drift.

Sensitivity is calibrated by injection through the search and trigger, with the control-rank gate applied as a measured multiplier. The median system’s best window reaches 1.5×10^{15} W EIRP, but the distribution matters more: of 60 systems, 3 reach 10^{14} W, 22 reach 10^{15} W and 55 reach 10^{16} W. These are upper bounds on reach; the window-to-window transfer of the calibration is uncertain by $\times 0.48$ – $\times 1.35$.

We identify no independently confirmed narrowband carrier. 13 windows were flagged by a screen comparing the star with 512 controls; the screen prioritises and does not test significance. 9 are circumstellar or foreground CO, including the β Pictoris disc recovered blind. The other 4 are not independently confirmed: none is a point source at the stellar position in the visibilities, where the same fit recovers β Pictoris CO at 6.1–9.5 σ , and none recurs.

Two limitations bound this. Only 42 of 82 systems have epochs over a day apart, so the result constrains persistent carriers and says little about intermittent ones; and the targets were observed for other purposes, chiefly disc and planet-formation programmes, so no inference to the 40 pc stellar population is attempted. No independent re-implementation exists, so the result rests on internal validation alone.

Subject headings: extraterrestrial intelligence, radio lines: stars, submillimetre: stars, surveys, astrobiology

1. INTRODUCTION

A technological civilisation, signalling deliberately or going about its affairs, may imprint the electromagnetic spectrum detectably, a *technosignature* (Sheikh 2020; Vidal et al. 2026). Six decades of radio searches cover many thousands of stars yet sample only 6×10^{-18} of the haystack of transmitter frequency, bandwidth, power and duty cycle (Wright et al. 2018). Two gaps motivate this work. The millimetre and submillimetre regime is very sparsely searched, and everything above that axis’s ~ 115 GHz upper limit more sparsely still; and few searches reprocess *existing* interferometric archives, although those archives already contain a spectrum of every target they point at.

This paper is an archival ALMA search of nearby stellar systems over a largely unexplored frequency regime, with an explicit empirical characterisation of its sensitivity and selection function. It is not, and does not attempt to be, a survey of the local stellar population: the sample is whatever ALMA happened to observe, and §6.6 sets out how far that is from representative. The signal

class searched is an *unresolved spectral-carrier technosignature*: emission confined to a small number of spectral channels, spatially unresolved at the stellar position on the relevant baselines, and persistent enough to recur between epochs. Each qualifier is a real restriction, since a carrier broader than the searched channel width, one that fills the synthesised beam, or one that appears in a single epoch and never again would not be recovered here whatever its power. The fine channels used are 15.3–1953 kHz wide, orders of magnitude broader than a centimetre-wave narrowband backend, so “narrowband” throughout describes the assumed transmitter and never the resolving power of the data (§4); no hertz-scale spectral resolution is claimed. We mine the archive, correcting for proper motion and Doppler drift, for such carriers and for anomalous millimetre continuum. Channelisation sets which of two search classes a window belongs to: 76 per cent of the searched windows carry no drift discrimination at all, and only 403 support the fine-channel search that “narrowband SETI” normally means.

Three contributions follow. The first is the systematic exploitation of archival ALMA stellar fields, 113 target/band datasets mined from public data with no new observing time. The second is spatial-control screen calibration for interferometric SETI, in which the same statistic is evaluated at the star and at 512 control positions so that the false-alarm rate is measured and not as-

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sumed. The third is a quantitative accounting of the frequency, power and drift domain actually searched, which gives the non-detection explicit boundaries. One worked case runs through the paper, a crossing towards β Pictoris, which is the largest star-versus-control contrast here and is attributed to the disc’s known circumstellar CO across two transitions and 8 execution blocks (§5.2). Figure 3 is the roadmap: it gives the selection from catalogue to searched sample on the left, and the decision chain from window to disposition on the right.

Two properties of the design bear on how every number below should be read. The statistic and the molecular-line mask were defined on these data, so the rank probabilities are exploratory rather than pre-specified; §4.1 and §C.1.1 set out what is claimed instead, with repository timestamps and an out-of-sample test. And the statistic compares a single position with single positions, which rules out the obvious region-maximum alternative on grounds of exchangeability; both forms are scored for every window and released, and Appendix K.2 audits the comparison.

One convention holds throughout: **wherever this paper says “sensitivity” without naming another quantity, it means P_{90}^{sel}** . Four power scales appear below and only the last is a sensitivity: the trigger level P_{trig} , the power P_{eff} a transmitter must radiate to reach it, the trigger’s own 90 per cent recovery P_{90} , and P_{90}^{sel} , the power the *complete* automated selection recovers nine times in ten. P_{90}^{sel} is the only one of the four quoted in the abstract, the discussion and the conclusions; Appendix E defines them and every other term used here in a particular sense.

The targets were not chosen for a technosignature search but are the stars that archival disc and planet-formation programmes happened to observe, so the sample is enriched in F and G stars and depleted in M dwarfs relative to the 40 pc census, with only 21 of 5 908 catalogued M dwarfs within 40 pc in it at all. Every statistical statement below therefore concerns *the ALMA archive-selected stellar sample within 40 pc* and not the solar neighbourhood; in particular the survey cannot constrain the local habitable-zone planet population, which M dwarfs dominate (§6.6).

2. BACKGROUND

Radio searches, from Cocconi & Morrison (1959) and Drake (1960) through the programmes reviewed by Tarter (2001) and Project Phoenix (Backus & Project Phoenix Team 2002), divide into wide-field commensal and targeted surveys (Zhang et al. 2020; Czech et al. 2021; Morrison et al. 2023; Li et al. 2026; Johnson et al. 2023), with wide-field low-frequency coverage from the MWA (Tremblay et al. 2024) and a single-dish C- and K-band programme at 6 and 18 GHz with the Sardinia Radio Telescope (Manunza et al. 2025), with machine-learning interference rejection now central (Ma et al. 2023; Choza et al. 2024; Jacobson-Bell et al. 2025; Poznanski et al. 2025; Garrett et al. 2026), a capability our classical threshold search lacks. Essentially all of that work lies below ~ 30 GHz, above which only Mason et al. (2024) has searched, in ALMA Band 3. There are physical reasons to go higher: interstellar scattering and dispersion fall steeply with frequency, a given aperture delivers a tighter beam per watt of EIRP, and directional

communication, radar and beamed power are already engineered as coherent mm-wave emission. Atmospheric transmission, receiver noise and phase stability raise the cost without forbidding it.

Mason et al. (2024) searched archival Band 3 windows toward 28 “stellar bycatch” stars, observed for other science and usable for free, placing $\text{EIRP}_{\text{min}} > 7 \times 10^{17}$ W for the nearest and identifying the challenges of high-frequency work: drift rates scale with frequency, smearing erodes sensitivity without a drift search, and the dense forest of molecular lines raises the confusion background. Every window reported here was re-reduced and re-searched from the raw archive data with the pipeline of §4, so the numbers to cite for any star in common are this paper’s.

Figure 1 places this survey among published searches. The vertical axis is physically commensurable: for a carrier narrower than every programme’s channel the minimum detectable EIRP is a total radiated power, not a spectral power density, so the same quantity is plotted for each. What differs is not the unit but the convention — each programme adopts its own detection threshold and its own completeness level — and the difference runs one way here. The value plotted for this survey is P_{90}^{sel} , a 90 per cent recovery through the complete automated selection, whereas most published limits are nominal thresholds at which recovery is not stated; comparing them like for like therefore flatters the others rather than this work. The figure also shows that ALMA occupies a different combination of frequency and spectral resolution from cm-wave SETI. Two numerical comparisons are worth stating, and they differ. Against centimetre-wave work, essentially all of which lies below ~ 30 GHz, a search spanning 89.6–873.1 GHz raises the frequency by about two orders of magnitude. Against the only prior millimetre search, Mason et al. (2024) in Band 3, the increase is about a factor of nine, and that is the honest comparator for novelty of regime. Any EIRP advantage toward the nearest systems is a d^2 effect and not an instrumental one, decomposed in §6.2. This survey samples far fewer stars, far fewer epochs and a channelisation seven orders of magnitude coarser than a dedicated narrowband search; it is complementary to those surveys rather than a deeper version of them.

A non-detection must become a quantitative exclusion statement: Wright et al. (2018) formalised the “Cosmic Haystack” search volume, Margot et al. (2023) its conversion into an upper bound on the transmitter-hosting fraction, and Sheikh et al. (2025) asked where Earth’s technosignatures would be detectable. That bound is valid only if the sample represents the underlying population, which interest-selected samples do not: most surveys select by proximity, spectral type, planets or Earth Transit Zone membership, and the ALMA-specific form, disc selection, skews the sample young and dynamically active. We instead take every star within 40 pc lying in the usable field of at least one public ALMA observation (§3). That sample is *work-list complete but epoch-incomplete*, and this paper uses no other completeness formulation: every star on the frozen work list that proves genuinely covered is searched, but not every execution block those stars hold is in this release, and no completeness is claimed for the local population itself. The 168 sample entries are ~ 1 per cent of the catalogued

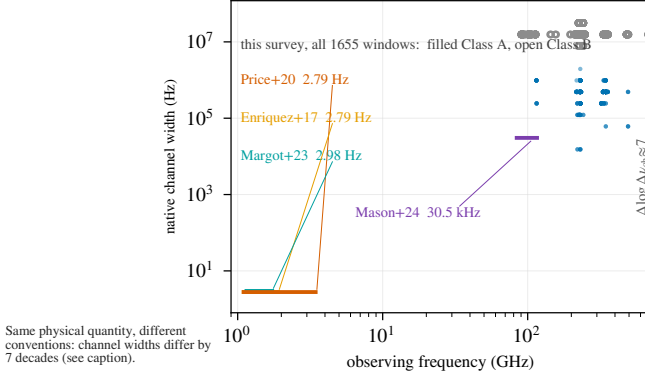


FIG. 1.— Where this survey sits in spectral resolution. Native channel width against observing frequency for every searched window (filled: Class A, fine-channel; open: Class B, coarse-channel), with the comparison programmes as horizontal segments: Enriquez et al. (2017), Margot et al. (2023), Price et al. (2020) and Mason et al. (2024). The distinctive point of this survey is its frequency regime, not its depth: the channel widths plotted span about seven orders of magnitude — which is what the $\Delta \log \Delta \nu_{\text{ch}} \approx 7$ annotation marks — so the channelisation, not the plotted power, is what differs most between programmes. The corresponding EIRP-versus-distance comparison, which requires that warning to be read correctly, is Fig. 11 in Appendix E.

population within 40 pc (the 90 actually searched ~ 0.5 per cent), and the residual coverage selection is characterised as a completeness function (Table 2).

3. SAMPLE SELECTION

The sample is *work-list complete but epoch-incomplete* within 40 pc (Table 12), a much weaker condition than volume completeness. The execution-block accounting closes as follows, and every number in this paper refers to one of these populations.

1. 656 public execution blocks lie in the parent member observing-unit sets of the sample stars. This is the population in scope, and it splits in two.
2. 479 of them have been downloaded, calibrated and searched; 177 have not. Every one of the 177 is repeat coverage: each sits in an observing-unit set from which another block was already taken, and the deduplication rule below keeps one block per star and tuning. 20 of them the archive offers as raw visibilities with no pipeline calibration at all. That is the first identity, $479 + 177 = 656$, which splits finer as $401 + 78 + 177 = 656$ into science-sample blocks, the reserved hold-out of §C.1.1, and blocks never searched.
3. 484 blocks were processed in total, 4.1 TB of raw data through one frozen pipeline. This is a *different* population from the 479: it is those plus 5 whose member-OUT progenitor link the archive does not expose (3 of them in the science sample), so they cannot be placed in the ledger above. That is the second identity, $479 + 5 = 484$. The two counts must not be added to each other.
4. Of the 484, 404 form the *frozen science sample* reported here, 77 are the reserved calibration hold-out of §C.1.1, and 3 left no surviving window after the quality cut.

No claim is made to have searched the whole ALMA archive, only every star on the work list; Table 25 gives the provenance of every dataset used anywhere in this paper.

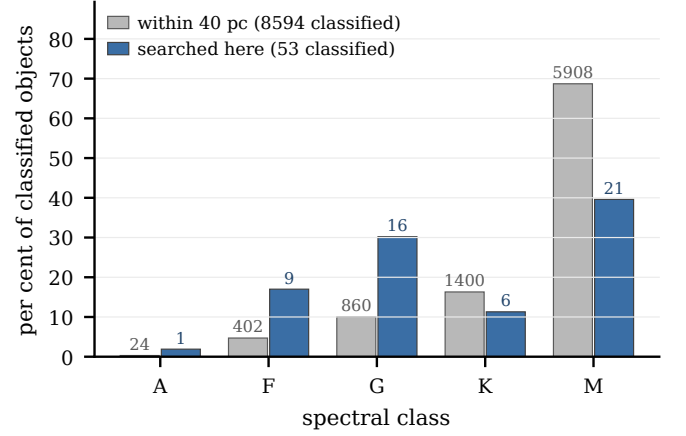


FIG. 2.— The archival selection, which no limit in this paper escapes. Grey: the spectral-class composition of the Gaia DR3 objects within 40 pc that carry a temperature. Blue: the same for the searched sample. Counts are printed above each bar. The sample is enriched in F and G stars by factors of four and three and depleted in M dwarfs by a factor two, so 21 of the 5908 classified M dwarfs within 40 pc carry a searched window. Because M dwarfs host most of the habitable-zone planets in the solar neighbourhood, this is the single most important limitation on what the null result means (§6.6).

The procedure admits every star within 40 pc of the Sun (Gaia DR3 parallaxes, distances direct inversions $d = 1/\pi$; every entry $\pi \geq 25$ mas with a parallax measured to better than 20 per cent, which is the condition the archive query applied) whose proper-motion-corrected position at the observation epoch falls in the quoted field of view of at least one public ALMA execution block. The star need not sit at the pointing centre, and §A.3 shows the quoted field of view admitted 53 stars no observation in fact contains, so “covered” is a property of the selection and not a guarantee. Disc, planet and spectral type enter as descriptive metadata only, and units are defined in §A.2. This gives 168 unique target stars, processed nearest-first; because trigger thresholds scale with d^2 at fixed noise that order is sensitivity-ranked, so the prevalence statements of §6 are conditional on it.

Composition follows from the archival selection and is tabulated against the reference census in Table 14. The sample is F- and G-enriched and deficient in M dwarfs, and 285 of the 1655 windows come from 3 systems; §6.6 gives the figures and their cost. The spectral-class proxy is `teff_gspphot`, available for 53 of 90 sample stars and missing preferentially for the coolest objects, so the M fraction is a lower limit and the earlier-type ratios upper limits. The debris-disc hosts contribute dusty continua, kept from masquerading as anomalies by the continuum lane’s screen (§4). One catalogue entry is worth flagging, since the released catalogue and the text would otherwise appear to disagree: “ γ Lupi” resolves by parallax to HD 139664 at 17.40 pc, not to the naked-eye star, and is relabelled accordingly, with no measured value changed. Appendix K lists the other cross-match and exclusion edge cases.

“Usable public ALMA coverage” is a five-criterion decision tree, frozen absent an erratum.

No star or window was excluded by an unlisted criterion, and none for short integration.

The rest of the paper is organised around that figure.

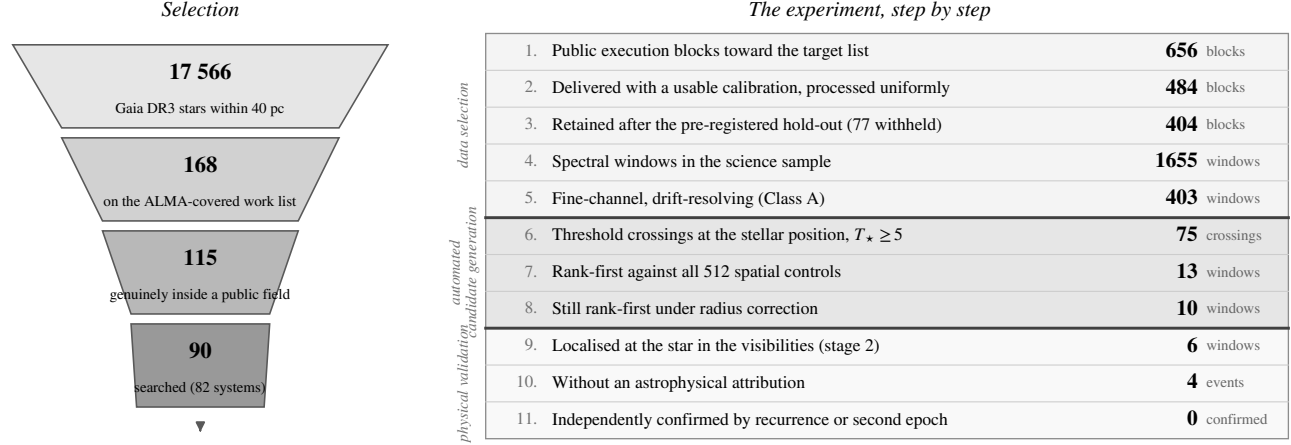


FIG. 3.— Left: the selection funnel. The three denominators the text keeps apart are the catalogued population, the 168-entry ALMA-covered work list, the ~ 115 stars a public field genuinely contains, and the 90 searched. Right: the experiment in eleven steps with the number surviving each, matching §4.3 line for line — from the public execution blocks in scope, through the pre-specified hold-out and the window inventory, to the candidate tests. The 512-control comparison at step 7 is a candidate-generation screen and not a test of the null; step 8 is the same screen with the radial gradient removed. Every count is read from the generated macros, so the figure, the list and the released catalogue cannot disagree.

TABLE 1

THE ARCHIVAL SELECTION BY SPECTRAL CLASS, INCLUDING THE COVERAGE THAT MATTERS. $N_{40\text{ pc}}$ COUNTS GAIA DR3 OBJECTS WITHIN 40 PC CARRYING A TEMPERATURE; N_{searched} THOSE WITH AT LEAST ONE SEARCHED WINDOW; AND THE LAST COLUMN THOSE WITH AT LEAST ONE *Class A* WINDOW, WHICH IS THE ONLY COVERAGE THAT CAN RESOLVE A DRIFTING CARRIER. 13 M DWARFS HAVE CLASS A COVERAGE, OF 5 908 CLASSIFIED WITHIN 40 PC.

Class	$N_{40\text{ pc}}$	N_{searched}	f_{searched}	with Class A
A	24	1	0.0417	1
F	402	9	0.0224	4
G	860	16	0.0186	15
K	1 400	6	0.0043	6
M	5 908	21	0.0036	13

Its right-hand panel is given step by step, with every number attached, in §4.3: the first five steps assemble the data and its inventory and the remaining six test it, the boundary being the threshold crossing, from which point everything is the candidate sequence of §4.1.

4. DATA AND METHODOLOGY

Three abbreviations recur below: QA2 is ALMA’s second-stage quality assurance, the stage that delivers calibrated data; a member observing unit set (MOUS) is the scheduling container holding one or more execution blocks; and TDM and FDM are the correlator’s time- and frequency-division modes, giving coarse and fine channelisation. Symbols, including $T_\star$, T_{trig} and the power scales, are defined in Appendix E.

One shared archive-ingestion stage feeds two independent search lanes, and Table 4 gives the data path of the primary lane in order. For each target the calibrated archival execution blocks covering the star’s Gaia-propagated position are mined within a bounded budget. Their visibilities feed (i) the *primary* lane, a Doppler-drift-corrected spectral-carrier search following Mason et al. (2024) and Enriquez et al. (2017), which converts each non-detection into an equivalent isotropic radiated power limit $\text{EIRP}_{\text{trig}} = 4\pi d^2 S_{\text{min}} \Delta\nu$. The product $S_{\text{min}} \Delta\nu$ is an integrated power only if the channel’s effective noise bandwidth equals its separation, which for a smoothed correlator it does not; the response

factor C_{response} corrects that, calibrated against unresolved tones of known amplitude injected into real visibilities and recovered through the unmodified pipeline. The measured factor is $\times 2.31$ at the median, spanning $\times 2.02$ – $\times 2.69$ across correlator modes, against the $\times 2.29$ predicted analytically for the Hanning response (Appendix G); the two medians agree to one per cent, and the spread is window-to-window variation in the correlator response rather than an error on the estimator. So $P_{\text{eff}} = S_{\text{min}} \Delta\nu C_{\text{response}} C_{\text{smear}}$ recovers the injected power, with S_{min} the primary-beam-corrected trigger-level flux density defined below, $\Delta\nu$ the native channel width and d the distance (worked numerically in Appendix F). EIRPs are quoted to two significant figures, the precision the flux scale supports. Five terms enter the systematic budget, each multiplying S_{min} and hence $\text{EIRP}_{\text{trig}}$ directly.

- (i) The ALMA absolute flux scale: 7 per cent band-weighted, up to 20 per cent at Bands 9 and 10, and epoch-dependent.
- (ii) Residual atmospheric phase after water-vapour correction and phase referencing: a coherent point-source loss $\exp(-\sigma_\phi^2/2)$, 5–20 per cent at 230–345 GHz. It is common to star and annulus within the isoplanatic patch and so no threat to exchangeability, but it is *not* common to the injected tones that calibrate P_{90} , which is why §5.3 carries it as a one-sided bias.
- (iii) The primary-beam model at the stellar offset: negligible on axis, 1.9 per cent at the worst pointing error.
- (iv) Visibility calibration: bounded empirically at 4.5 per cent. An inconsistent delivered amplitude scale would make the ratio of each window’s measured per-integration rms to its radiometric prediction scatter from block to block beyond what thermal noise explains. Over 2271 windows in 559 blocks the per-block median of that ratio is 0.956 with a robust scatter of 4.5 per cent, against 0.8 per cent within a block. Including it raises the combined budget from 8 to 9 per cent, the figure quoted throughout.

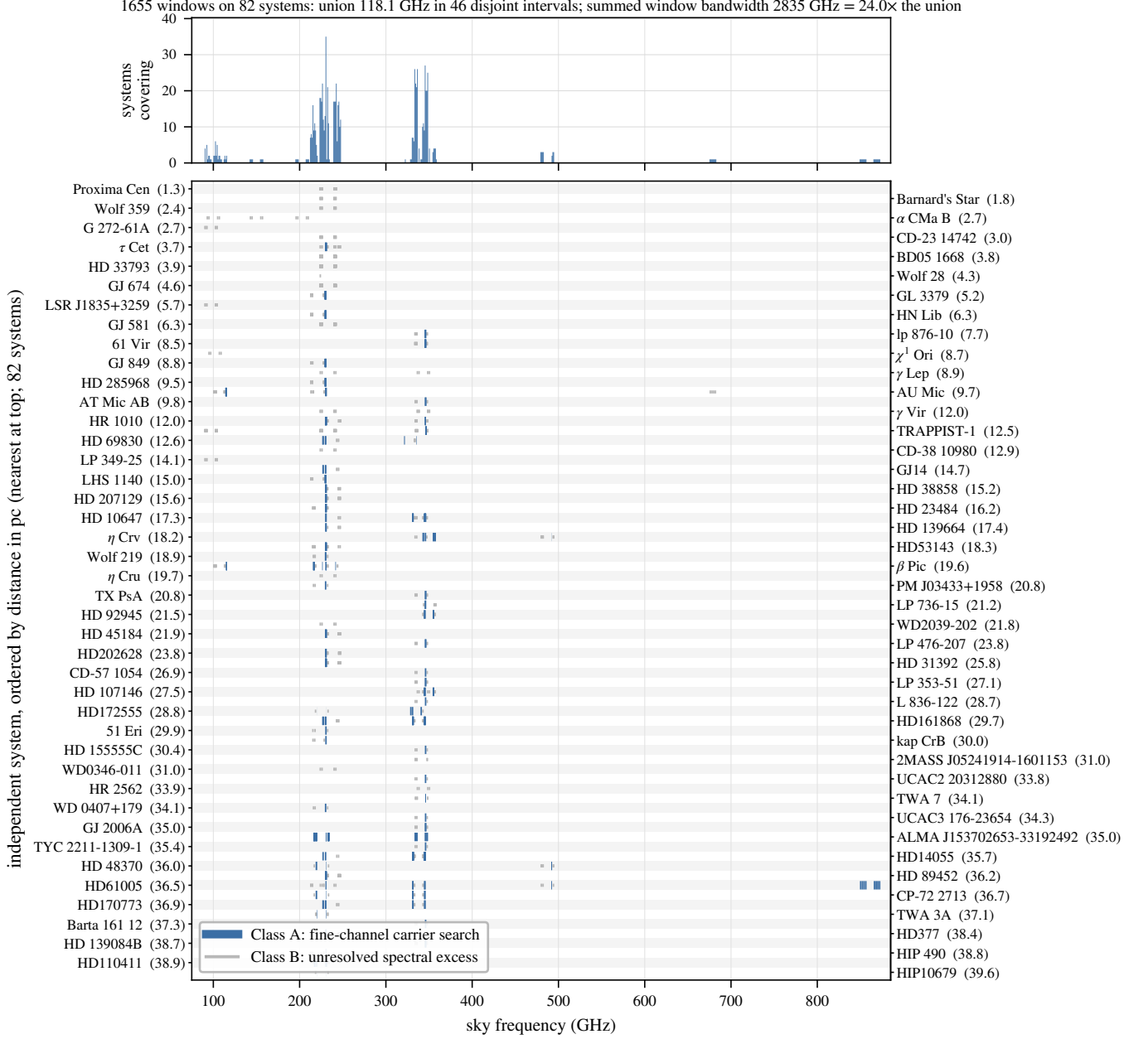


FIG. 4.— Where this survey searched. *Lower panel*: each searched window is a horizontal segment on its host system's row, rows ordered by distance and labelled alternately on the left and right axes; heavy blue marks the fine-channel carrier search (Class A) and light grey the coarse-channel spectral-excess search (Class B). *Upper panel*: the number of systems covering each gigahertz of sky frequency; the threshold distribution is Fig. 1. The coverage is narrow, clustered and repeatedly revisited: 46 islands concentrated near 230 and 345 GHz, with summed window bandwidth 24.0 $\times$ the union. The union measures frequency coverage, the summed bandwidth integrated search effort.

- (v) The instrumental spectral response: median $\times 2.29$, the largest of the five by an order of magnitude, carried explicitly as P_{eff} by the reading rule below.

Gaia distance errors are negligible, 0.2 per cent typically. The same visibilities also feed an ancillary line-excluded continuum lane, which bears on no technosignature limit and is not reported here. Two survey classes are reported, distinguished by the dimensionless drift parameter

$$\eta_{\text{drift}} = |\dot{\nu}_{\text{max}}| \tau_{\text{track}} / \Delta \nu_{\text{ch}}, \quad (1)$$

the channel-width fraction the *maximum* searched drift $|\dot{\nu}_{\text{max}}|$ traverses over the full on-source track τ_{track} at native channel width $\Delta \nu_{\text{ch}}$. The track is the right time-scale

because the de-drift step sums along it; τ_{track} is written throughout, leaving T for the search statistic alone, and η_{smear} (Appendix F) is the same ratio over one integration. A window is *drift-resolving* when $\eta_{\text{drift}} \gtrsim 1$, carrying a carrier across enough channels for its motion to be measured. The drift ceiling inside η_{drift} is a search-grid boundary, the widest trial drift the stack evaluates, and not a prior on transmitter drift rates: carriers drifting faster are missed, and §6.1 gives the resulting selection function (Fig. 6).

Class A is the fine-channel spectral-carrier search: the 403 windows of native width 15.259–1953.125 kHz (median 488.3 kHz; 60 per cent at the 488.3-kHz mode), spanning $\eta_{\text{drift}} = 0.17$ –422 (median 14.7), and the only

TABLE 2

THE SURVEY AS SEARCHED, AND THE PAPER’S HEADLINE NUMBERS IN ONE PLACE. EVERY COUNT QUOTED IN THIS PAPER RECONCILES WITH IT. THE TWO COMPLETENESS EXPERIMENTS MEASURE DIFFERENT QUANTITIES AND ARE KEPT IN SEPARATE TABLES. EVERY FREQUENCY UNION QUOTED HERE IS AN AGGREGATE OVER A CLUSTERED, REPEATEDLY REVISITED COVERAGE, WHOSE ACTUAL DISTRIBUTION IS FIG. 4; THE UNION SHOULD NOT BE READ AS CONTINUOUS COVERAGE OF ITS ENDPOINTS.

Quantity	Value
Stars within 40 pc with qualifying public ALMA coverage	168
genuinely pointed at; never observed (§A.3)	115; 53
Stars searched; independent systems	90; 82
Star/band datasets	113
Spectral windows searched	1655
12 m array; ACA 7 m	601; 1054
extracted; repeats, defective, withheld	1725; 45, 13, 12
Class A fine-channel / Class B coarse-channel	403 / 1252 (76% Class B)
Execution blocks	404
Distance range	1.30–39.63 pc
$\Delta\nu_A$, unique frequency union, Class A (the drifting-carrier experiment)	47.7 GHz
$\Delta\nu_B$, full Class B union (overlaps Class A by 20.1 GHz)	90.5 GHz
$\Delta\nu_{B\text{-only}} = \Delta\nu_{A\cup B} - \Delta\nu_A$, coverage Class B adds	70.4 GHz
$\Delta\nu_{A\cup B}$, total archival coverage, of which only $\Delta\nu_A$ is drift-resolving usable after the molecular-line mask	118.1 GHz (46 intervals, 89.6–873.1 GHz; Fig. 4)
Windows by ALMA band (3/4/5/6/7/8/9/10)	112/4/4/1072/427/24/4/8
Native channel widths	15.3 kHz–31.25 MHz; median 15.625 MHz
Drift-rate ceilings	1.1–10.5 kHz s ^{−1} (12.0–13.3 Hz s ^{−1} GHz ^{−1})
On-source time per window	21–5274 s (median 2812 s)
Nominal $S_{\min}$	0.41 mJy–1.30 Jy (median 9.8 mJy)
90 per cent selection P_{90}^{sel} , Class A, per window	8.7×10^{13}–1.4×10^{17} W (median 2.8×10^{15}) ^a
per system, best window	8.7×10^{13} – 4.4×10^{16} W (median 1.5×10^{15})
90 per cent trigger P_{90} , Class A	6.3×10^{13} – 1.1×10^{17} W (median 2.4×10^{15})
Effective $P_{\text{eff}} = P_{\text{trig}} C_{\text{resp}} C_{\text{smear}}$, per window, Class A	5.1×10^{13} – 9.1×10^{16} W (median 1.9×10^{15})
Class B	3.7×10^{13} – 6.4×10^{17} W (median 1.4×10^{15})
Nominal trigger P_{trig} , Class A (15.3–1953 kHz)	2.2×10^{13} – 6.8×10^{16} W (median 9.4×10^{14})
Class B (7.812–31.25 MHz)	1.6×10^{13} – 2.8×10^{17} W (median 6.0×10^{14})
Completeness, by class and experiment	Table 13 (amplitude); Table 13 (dwell)
Threshold crossings ($T_{\star} \geq T_{\text{trig}}$ at the stellar position)	75 windows (71 Class A, 4 Class B; Table 1)
Star-exceeds-ring windows ($T_{\star} \geq T_{\text{trig}}$)	13 windows (13 Class A, so all of them)
circumstellar or foreground CO	9
unattributed, absent in the repeat block	4
Persistent, independently confirmed unresolved spectral-carrier technosignatures, within the domain searched	0

^a Power and frequency quantities are given to two significant figures, except where a channel or sky frequency must be matched to the catalogue and is given to the precision the grid supports. The internal consistency checks quoted elsewhere to a per cent (§K.1.2) bound *relative* quantities within one dataset; the absolute flux scale rests on ALMA’s delivered calibration.

class with measured end-to-end recovery of a drifting carrier. *Class B* is the coarse-channel spectral-excess search: the 1252 windows, 76 per cent of the sample, of native width 7.812–31.25 MHz (1212 at the 15.625-MHz mode), spanning $\eta_{\text{drift}} = 0.004$ –1.35 (median 0.51), sensitive to persistent near-stationary excess power but carrying no drift discrimination, and never described as a narrow-band search here.

η_{drift} is the physical criterion and the classes are labelled by the instrument. Channelisation is bimodal, with no window between 1953.125 kHz and 7.812 MHz, an empty gap of factor 4.0, so any boundary inside the gap assigns identical membership. The two criteria do not coincide exactly: 38 long-integration Class B windows reach $\eta_{\text{drift}} \geq 1$, only marginally (maximum 1.35), where a one-to-two channel traverse supports no discriminative drift fit after baseline removal, while 13 short Class A windows fall below unity and are conservatively kept in Class A. On the physical criterion alone 428 windows of 1655 reach $\eta_{\text{drift}} \geq 1$ against 403 in Class A, so “drift-resolved” is used of a class only where the two agree. Each window’s η_{drift} , η_{smear} , trial drift count and maximum searched acceleration $a_{\text{max}} = 3.60$ – 4.00 m s^{−2} are released catalogue columns, so any reader may reclassify on the physical criterion alone. Results are given per class first and combined only where statistically appropriate (§5.2).

Both lanes exclude catalogued molecular transitions at tolerances suited to their statistics. The continuum lane masks the common species at native resolution before integrating; the carrier lane applies a ± 50 km s^{−1} stellar-frame velocity exclusion mask (§5.2.4) to any threshold crossing. **That half-width was set after the β Pic-toris crossings had been inspected, and is the one element of the analysis not fixed in advance.**

Two results bear on whether that matters. First, reclassifying all 75 crossings at ± 20 , ± 30 , ± 50 and ± 100 km s^{−1} admits no new unattributed event at any width, and CP−72 2713 remains the only unmasked one throughout; this is a one-sided test and is reported as such (§5.2.4). Second, this paper does not claim a measured demonstration that the half-width is doing no work. Such a demonstration would have to compare each crossing with its nearest catalogued transition in the *stellar* frame, after the chain of Appendix F; the catalogue’s uncorrected sky-frame offset is a different quantity and cannot stand in for it. The comparison is also unavailable for most events: the pipeline stored a crossing frequency for only 3 of the line-attributed and 1 of the unattributed stage-1 windows, so for the remaining 9 the offset is a window-level number rather than a property of the crossing, and one of them reads $-12\,161$ km s^{−1}. Re-running the frozen search to recover every crossing cell, which §5.2.1 costs, would supply both the frequencies and a stellar-frame offset for each; until then the insensitivity of the classification to the half-width rests on the one-sided widening test above and not on a measured separation. The Keplerian bound, $\lesssim 13$ km s^{−1} where these molecules survive beyond 10 au, is the physical scale that motivates masking at all, and is a stellar-frame quantity.

The mask is a classification mask, not a detection mask. It is applied after the search, to decide how an event that has already been found should be interpreted, and it removes nothing from the detection stage: ev-

ery masked window is searched exactly as an unmasked one is, and no threshold moves. What it defines is the frequency space *excluded from the exclusion statement*. The survey covers 118.1 GHz of unique sky frequency in 46 clustered islands (Fig. 4); the effective default quoted throughout is the ≈ 113.3 GHz surviving the mask, and no constraint is placed on carriers inside the excluded velocity regions, not because they could not have been detected but because a detection there could not have been distinguished from the molecular emission the mask marks.

All flux densities and EIRP thresholds are total intensity (Stokes I), summed over the two parallel-hand cross-correlations with QA2 weights. No cross-hand extraction or circularity test was performed, so the search performs no polarimetric discrimination and the quoted sensitivities assume the delivered parallel-hand calibration represented the total-intensity response adequately. We do not claim strict polarisation blindness: incomplete parallactic-angle sampling, unequal flagging or calibration of the two hands, and instrumental leakage can all break the equivalence between a strongly polarised and an unpolarised signal of the same total power. The archive records both parallel hands for 198 of the 200 blocks whose polarisation setup it resolves, but the retained products do not record per-correlation flagging inside QA2, so we cannot say what fraction kept both hands unflagged all the way through calibration; one that lost a hand would be $\sqrt{2}$ less sensitive than its threshold implies, with nothing here to catch it.

The three sensitivity quantities of Table 12 are defined here in full. The trigger power P_{trig} is five times the window’s per-channel rms at its native channel width, for a transmitter at the star’s position matching the searched morphology: the level at which the search fires, and what every reference to the trigger below means. It is an internal pipeline quantity, not a sensitivity to compare with a transmitter and not a significance.

The effective unresolved-carrier threshold P_{eff} is P_{trig} corrected for the instrument’s spectral response. ALMA applies online Hanning smoothing, which leaves only 0.38 to 0.50 of a sub-channel tone’s power in the peak channel depending on where in the channel it falls, median 0.44 (Fig. 5, Appendix F). Nominal thresholds are therefore optimistic by $\times 2.00$ to $\times 2.67$, median $\times 2.29$, and $P_{\text{eff}} = 2.29 P_{\text{trig}}$ is the number to compare with any transmitter power. It follows from the 0.25/0.5/0.25 kernel and is window-specific ($\times 1.33$ – 2.00 for the 146 windows with online channel averaging), and injection through the response kernel reproduces it to 1.1 per cent (median $\times 2.31$; Appendix G). It is a sensitivity correction rather than a completeness, it is not small, and it is the principal physical threshold.

The completeness limit P_x , the power at which measured recovery reaches x per cent, is a different quantity again: $P_{50} = 1.10 P_{\text{trig}}$ and $P_{90} > 2 P_{\text{trig}}$ on the calibration window’s drifting class (Appendix G). Transfer to other bands, channelisations and integration times is measured on 28 further configurations, where P_{50} spans 0.64 – $1.13 \times$ its median; P_{95} remains *not determined*. This survey has a measured reference recovery function whose transferability is bounded by measurement, not a measured Class A completeness function. Every headline

EIRP is quoted beside its native channel width, and the catalogue carries P_{trig} and both response-corrected values per window. Three signal morphologies are detectable, and §6.1 quantifies each against real windows: a continuous frequency-stationary carrier, a continuously drifting one, and an intermittent one.

4.1. Final candidate definition and statistical interpretation

The spatial screen was frozen before the candidate list was examined, in the form that ranks the stellar position against 512 single-position controls under one global noise normalisation. Validation showed that form to be non-exchangeable, with the stellar rank displaced from the uniform expectation. The cause is a defect in the control geometry, and it is confined to one stratum. The extraction code computes the primary beam from a fixed 12 m dish diameter for every window, with no dependence on which array observed the block, so for an ACA 7 m window the control annulus is placed at 58 per cent of the radius it should occupy. At 230 GHz its inner edge falls at 3.8 arcsec rather than 6.6 arcsec, inside a typical ACA synthesised main lobe of 7.3 arcsec. Controls placed there are not independent of the star: they sit on the star’s own synthesised beam and collect its flux.

Three measurements identify this as the mechanism. First, the departure is confined to the affected stratum: the stellar rank of the 601 12 m windows, whose annulus is correctly placed, is consistent with uniformity ($D = 0.027$, $p = 0.77$), while the 1054 ACA windows are not ($D = 0.10$, $p < 0.001$). Second, the excess sits where the contamination is: pooling the stored control vectors by fractional radius, the innermost annulus bin stands $+0.11$ above the outer half in ACA windows and only $+0.01$ in 12 m windows. Third, the direction is right: contaminated controls are inflated, so the star ranks *low*, which is the sense observed. An effect of the primary beam itself would act on both arrays equally; this does not, which rules out a primary-beam noise gradient as the cause.

The remedy is to re-extract the ACA windows on the correct 7 m beam with an inner radius outside the synthesised main lobe. That requires the raw visibilities and a full re-search, and is not done here; §6.3 states what rests on it. A radius-corrected normalisation applied to the stored statistics does not substitute for it: tested prospectively on 778 windows that played no part in building it, the correction does not restore exchangeability (§5.2.2), which is what one expects if the defect is contamination of particular controls rather than a smooth function of radius.

The originally frozen screen therefore remains the primary candidate-generation statistic, the statistic that decides which windows are examined, and “primary” is used in that sense throughout. Which normalisation gives the preferred comparison for the false-alarm *rate* is a separate question, treated with both forms side by side in §5.2; both are reported for every window in the released catalogue. A tail quantity computed against the screen’s own exchangeability-violating null is written $\mathcal{P}_{\text{scr}}$, never p : it is the probability of the observed count under a stated and imperfect null, and p is reserved for genuine significance levels such as the Kolmogorov–Smirnov tests on rank distributions.

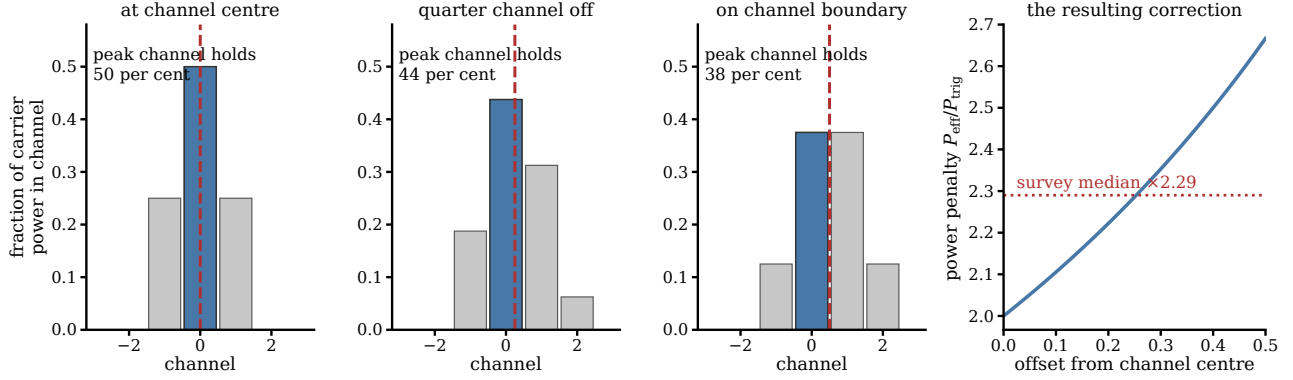


FIG. 5.— Why the nominal channel threshold is not the transmitter threshold. ALMA applies online Hanning smoothing by default, so an intrinsically unresolved carrier is spread over neighbouring channels and the peak channel holds only part of its power: half at a channel centre, 0.38 on a boundary (first three panels; blue is the peak channel, the dashed line the carrier). The search threshold is set in units of the channel noise, so the power a transmitter must radiate to reach it is larger than the nominal figure by the reciprocal of that fraction (right-hand panel). This is the whole of the difference between P_{trig} and P_{eff} in Table 12.

TABLE 3

HOW TO INTERPRET A CANDIDATE. THE FOUR LEVELS AT WHICH A WINDOW IS JUDGED, AND WHAT EACH CAN ESTABLISH. EVERY STATISTIC DEFINED IN THIS PAPER SERVES ONE OF THESE ROLES.

Level	Question	Role
Detection	Does a cell exceed the trigger T_{trig} ?	generates crossings
Screen	Is the star above all 512 controls?	prioritisation only
Localisation	Is the emission on the star in the visibilities?	principal validation
Recurrence	Does it repeat in an independent epoch?	confirmation

Neither form is a significance test. Ranking the star against 512 control positions drawn from the same field is a powerful way to decide which windows deserve examination, but $1/(512+1) = 1.95 \times 10^{-3}$ is the *resolution* of the ensemble, the smallest rank it can express, and not the probability that noise produced the window. Against a survey-wide Bonferroni scale

$$\alpha_{\text{survey}} = \frac{0.05}{1655} = 3.0 \times 10^{-5}, \quad \frac{1.95 \times 10^{-3}}{\alpha_{\text{survey}}} = 65, \quad (2)$$

that floor is 65 times too coarse (Appendix J), so a technosignature claim here requires independent recurrence or independent astrophysical confirmation and never rank alone.

What disposes of an event is the physical evidence, and the decisive test is interferometric rather than statistical: a carrier at the stellar position must appear coherently in the real part of the visibilities after phase rotation to that position, with an imaginary part consistent with zero. That is a physical statement about where the emission is, it does not rest on the exchangeability assumption the spatial screen violates, and every disposition in this paper turns on it, together with coincidence with a catalogued transition in the stellar frame and recurrence in an independent epoch.

The control rank is not a detection significance and not a p -value. The smallest rank 512 controls can express is $1/(512+1) = 1.95 \times 10^{-3}$, which is 65 times coarser than this survey's Bonferroni scale (Eq. 2). No window can therefore be established as a technosignature by rank alone, however extreme it looks. Every disposition in this paper rests on the third and fourth levels of Table 3.

The selection logic is five steps, in order. The first two generate candidates; the third is the primary physical test.

- (i) Threshold crossing. A channel $\times$ drift cell at the stellar position reaches $T_{\star} \geq T_{\text{trig}}$. A fixed trigger power, not a calibrated p -value.
- (ii) Spatial prioritisation. The stellar statistic exceeds all 512 control positions in the same window. A candidate-generation mechanism: it decides which windows are examined and carries no significance of its own, for the reasons above.
- (iii) Visibility-domain localisation. Each event's own channel is followed along its recovered drift rate, the local continuum is removed per integration, and the visibilities are phase-rotated to the star: emission centred there gives a positive real part with an imaginary part consistent with zero (§5.2.1). This is the first test in the chain to ask a physical question about the sky rather than a comparative question about an ensemble, and the first a window can fail on evidence.
- (iv) Astrophysical identification. The event is checked against catalogued molecular transitions in the stellar frame and against instrumental signatures.
- (v) Independent recurrence. It repeats at the same tuning in an *independent epoch*, or acquires independent confirmation.

The visibility test was applied to the 13 windows the screen flagged rather than to all 75 threshold crossings, for an operational reason: the pipeline deletes each measurement set once its search completes and retains only the derived products, so the test could be run only while a block was in flight, which in practice meant on windows the screen had just promoted. 2 execution blocks still hold visibilities and neither contains a threshold crossing, so no crossing window can now be tested from data in hand, and extending the test to the remaining 62 crossing windows means re-downloading and recalibrating 52 blocks, 0.40 TB, about 18 h of processing on this pipeline's measured cost model. The frequencies and drift rates needed to place each fit survive in the retained search products, so that work is re-calibration rather than a re-search; it is not done here, and until it

is, the statement below applies to 13 windows and not to 75. The asymmetry is one-sided: the windows tested are those the image-plane screen selected, so the test as applied vets the screen’s output rather than replacing it. Stage (iii) is worth its cost because it uses no control ensemble at all, runs on recalibrated raw data, can fail a window the image-plane statistic promotes, and carries an internal positive control, the same fit recovering the β Pictoris CO emission at the stellar position in 6 of its 8 windows at $6.1\text{--}9.5\sigma$. Every stage-1 event is carried through all five stages in Table 7, and nothing short of (v) is called a candidate.

All delivered channels of each spectral window were searched, with no edge-channel or bandpass-roll-off trimming before extraction, so residual bandpass structure near a window edge is carried in the calibration-residual term of the systematic budget rather than removed by flagging. The distance of every threshold crossing from its window edges is a released catalogue quantity, reported for the flagged windows in §5.2.

The primary statistic is formed on extracted spectra, never on images or visibilities, from the QA2-calibrated measurement set as delivered, per integration and per native channel, with the archive’s flagging carried through untouched and no regridding (Table 4, step 3). What the extraction returns at a propagated sky position $\mathbf{x}$ is the real part of the weighted vector average of the phase-shifted visibilities, the natural-weighted dirty-map value there; there is no deconvolution, so a quoted S_{peak} in a field with extended emission is a dirty-map amplitude. The primary-beam correction enters the quoted thresholds and fluxes (§F.1) and leaves untouched the extracted amplitudes the statistic ranks. Each per-integration spectrum $I(\mathbf{x}, t, \nu)$ is then baseline-subtracted by a block median along the *frequency* axis, with no temporal median at any stage (§4.2). The residual dynamic spectrum is stacked along each trial linear drift rate $\dot{\nu}$ with inverse-variance weights, and the statistic is the maximum of that stack over the channel $\times$ drift grid,

$$T(\mathbf{x}) = \max_{\dot{\nu}, \nu} \frac{\sum_t I(\mathbf{x}, t, \nu + \dot{\nu}t) / \sigma^2(t, \nu)}{[\sum_t 1 / \sigma^2(t, \nu)]^{1/2}}, \quad (3)$$

a matched filter whose denominator is the noise of the stacked spectrum. $T(\mathbf{x})$ is neither assumed nor shown to be Gaussian or χ distributed under the null, which is why T_{trig} denotes a trigger level and not a tail probability; Appendix J quantifies the departure. It reduces to a single-channel amplitude divided by one noise scale only when σ is the same for every integration and channel; otherwise the weights vary along the track. The scale $\sigma(t, \nu) = 1.4826 \text{ MAD}_{i=1\dots 512}[I_i(t, \nu)]$ is $1.4826\times$ the median absolute deviation taken *across the control positions* at each integration t and channel ν , making it global in position, local in frequency and time, and shared by the star and every control. The across-position median enters only in forming that deviation and is never subtracted from the numerator, so T is an absolute stacked signal-to-noise carrying a cross-position noise scale, and measures no local contrast.

This form is used rather than a region maximum because T is evaluated at one position, identically at the star and at each control, and that is what makes the rank interpretable. If the stellar side took a maximum

over n_{src} positions while the controls remained single positions, the two sides would not be exchangeable: under the null the region maximum exceeds all N_{ctrl} controls with probability $n_{\text{src}}/(n_{\text{src}} + N_{\text{ctrl}})$ rather than $1/(N_{\text{ctrl}} + 1)$, which for $n_{\text{src}} = 135$ and $N_{\text{ctrl}} = 512$ is 21 per cent rather than 0.2, a floor under which most crossings flag whatever the data contain. The argument is combinatorial and holds for any statistic, noise distribution and field; §K.2 gives the comparison and Table 21 audits every window either form flags. A line filling the primary beam drives star and annulus up together instead of cancelling.

The searched drift-rate range is the larger of (a) a generic frequency-scaled default ($12\text{--}13 \text{ Hz s}^{-1} \text{ GHz}^{-1}$), the Mason et al. (2024) fiducial with a threefold safety margin, and (b) for confirmed exoplanet hosts, the maximum line-of-sight orbital acceleration of the innermost known planet, so a transmitter on a short-period planet cannot drift out of the window. Drift rate maps onto line-of-sight acceleration through the first-order Doppler relation

$$a_{\text{los}} = c \frac{\dot{\nu}}{\nu}, \quad (4)$$

so the ceiling is quoted in cross-band-comparable form $\dot{\nu}/\nu$: $12\text{--}13 \text{ Hz s}^{-1} \text{ GHz}^{-1}$ is $(1.2\text{--}1.3) \times 10^{-8} \text{ s}^{-1}$, that is $|a_{\text{los}}| = 3.6\text{--}3.9 \text{ ms}^{-2}$ independent of observing frequency, the acceleration at $\sim 0.04 \text{ au}$ around a solar-type star. Earth-motion contributions lie two orders of magnitude below and are subsumed by the grid. The grid is uniform in $\dot{\nu}$ over n_{drift} steps ($2\text{--}1944$ across windows, recorded per window), so a carrier midway between trial rates leaves a residual drift traversing at most 0.26 of a native channel over the longest fine-class track, an amplitude loss an order below the recovery slope measured between $T_{\star} = 5$ and 10. Two limits of the model matter more. The rates searched are *apparent* topocentric ones, so the survey does not separate a deliberately chirped carrier from an accelerating platform and constrains only linear apparent drift inside the ceiling; and an emitter need not be tied to a known planet at all, a free-flying platform or rotating structure accelerating as it chooses. For circular orbits the ceiling covers a transmitter co-orbiting an Earth analogue by three orders of magnitude and Proxima b comfortably, while TRAPPIST-1 b ($a_{\text{los}} = 4.00 \text{ ms}^{-2}$) sits at it and c (2.14 ms^{-2}) reaches it at modest eccentricity, which raises the periastron value by up to $(1+e)/(1-e)^2$ (Fig. 6). TRAPPIST-1 transits, so its inclination is measured and effectively edge-on, and the edge-on figure is the relevant one: TRAPPIST-1 b exceeds the ceiling over 5–29 per cent of its orbital phase, depending on which end of the $12.0\text{--}13.3 \text{ Hz s}^{-1} \text{ GHz}^{-1}$ ceiling applies, and is searchable through the rest of it.

Across the 1655 windows maximum searched drift rates span $1.1\text{--}10.5 \text{ kHz s}^{-1}$ and trial drift rates per window $2\text{--}1944$ (median 4), the largest being the narrow, finely channelised β Pictoris Band 6 window (§D.1.2); 11 windows are narrower than 100 MHz. The threefold margin matters: at $1\times$ the search would have been blind to two of its threshold-crossing windows, including the AU Mic crossing, while at $5\times$ no disposition changes. Transmitters drifting faster than the $12.0\text{--}13.3 \text{ Hz s}^{-1} \text{ GHz}^{-1}$ ceiling remain unconstrained. Intra-integration smearing is negligible for 446 of the 455 windows that carry a value,

TABLE 4

THE STATISTICAL DATA PATH, IN ORDER. STEPS 1–8 RUN IDENTICALLY AT THE STELLAR POSITION AND ALL 512 CONTROLS; ONLY STEP 9 ONWARD DISTINGUISHES THEM.

1	calibrated measurement set from the ALMA archive
2	stellar position propagated to the observation epoch (Gaia DR3)
3	spectral extraction at that position, native channelisation
4	per-integration baseline removal: block median along frequency ($W = 65$)
5	stack along each trial linear drift rate
6	robust noise map $\sigma(t, \nu)$: MAD across the 512 controls
7	$T(\mathbf{x})$ from Eq. 3, an inverse-variance-weighted stack
8	repeat 3–7 at 512 control positions on a ring
9	threshold: $T_* \geq 5$ (a crossing)
10	rank test: $T_* > \max_i T_i$ (stage-1 flagged window)
11	molecular-line mask, $\pm 50 \text{ km s}^{-1}$ stellar frame
12	vetting: recurrence and astrophysical attribution

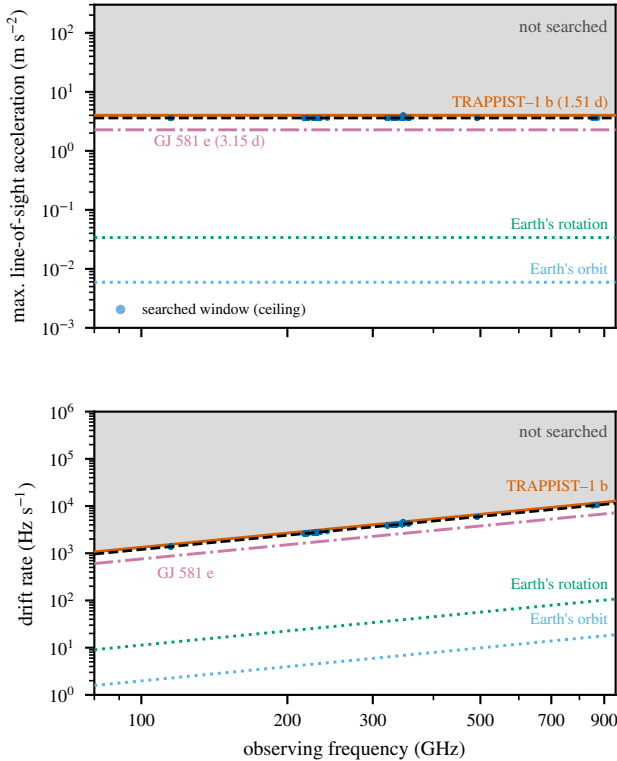


FIG. 6.— What the drift search reaches, as a physical selection function. *Upper*: the drift ceiling of each of the 403 Class A windows (points), converted to line-of-sight acceleration, $|a| = c\dot{\nu}/\nu$, against observing frequency; shaded above is unsearched. The boundary is flat because the grid was built at constant acceleration, 3.60 m s^{-2} . *Lower*: the same boundary in the units the search works in, where it is not flat, running 1.4 kHz s^{-1} at 115 GHz to 10.5 kHz s^{-1} at 872 GHz. Horizontal lines are reference accelerations: Earth’s rotation (0.034 m s^{-2} at the equator) and orbit (0.0059 m s^{-2}), and the two most demanding known planets of any target, TRAPPIST-1 b (4.00 m s^{-2}) and GJ 581 e (2.27 m s^{-2}). TRAPPIST-1 b is the one case the grid does not contain, missing by 0.01 per cent.

at most 1.10 channels and median 0.001; 9 have $\eta_{\text{smear}} < 0.99$, and 5 of those, all 15.3 kHz Band 6 windows and so among the most drift-capable here, carry effective thresholds $1.65\text{--}1.74 \times$ nominal, 2 being stage-1 flagged. Channel widths are effective resolutions from the measurement sets’ spectral window tables (Appendix F).

4.2. Ancillary and validation analyses

Nothing here generates candidates; it is validation, and the detail is in the appendices. Two demonstrations carry the weight. Synthetic tones of known amplitude injected

into real visibilities and recovered through the unmodified pipeline — 1200 trials across six window configurations, three bands and both classes (Appendix G) — establish that the machinery recovers a drifting narrow-band carrier, the searched morphology; and β Pictoris CO is recovered and localised in two transitions and 8 blocks (§D.1.2), establishing that it finds a real feature at a known stellar position. The baseline step is a per-integration median along frequency (Table 4, step 4) and so cannot remove a one- or two-channel feature. Appendix K reports the three quantified checks that bound undetected pipeline faults.

4.3. The experiment in summary

The experiment runs in eleven steps, drawn with the number surviving each one in the right-hand panel of Fig. 3: 656 public execution blocks whose fields contain a catalogued star within 40 pc; 484 of them delivered with a usable calibration and put through one frozen pipeline with no per-target tuning; 404 in the science sample once 77 are withheld by a rule fixed before any was looked at, a hash of the block identifier, and 3 yield no usable window; 1655 spectral windows, de-duplicated by identifier; 403 of them Class A, the primary experiment, the remaining 1252 supporting only a search for unresolved spectral excess; 75 threshold crossings, where $T_* \geq 5$ somewhere on the drift grid; 13 stage-1 windows, where the star also outranks all 512 spatial controls; 10 of those surviving the radius correction, which loses 4 and gains 1 without changing any disposition; 6 localised at the star by the drift-following visibility fit; 4 without an astrophysical attribution, the other 9 being carbon monoxide and 8 of those the β Pictoris debris disc; and 0 independently confirmed candidates. Table 7 carries every stage-1 event through the last four steps, and the survivor count at every step is a column filter on the released catalogue, reproduced by a script in the data release.

5. RESULTS

The 1655 windows are the size of the processing exercise, not of the technosignature search. The primary experiment is the fine-channel search of 403 Class A windows toward 60 systems; the 1252 Class B windows are a complementary search for channel-confined excess power. Results are reported per class, and the combined count appears only where it describes the archive processed rather than the sensitivity achieved.

The choice of detection statistic makes the non-detection more conservative, not less: the symmetric form of Eq. 3 flags a strict subset of what the region-maximum form flags, 13 windows against 66, with none

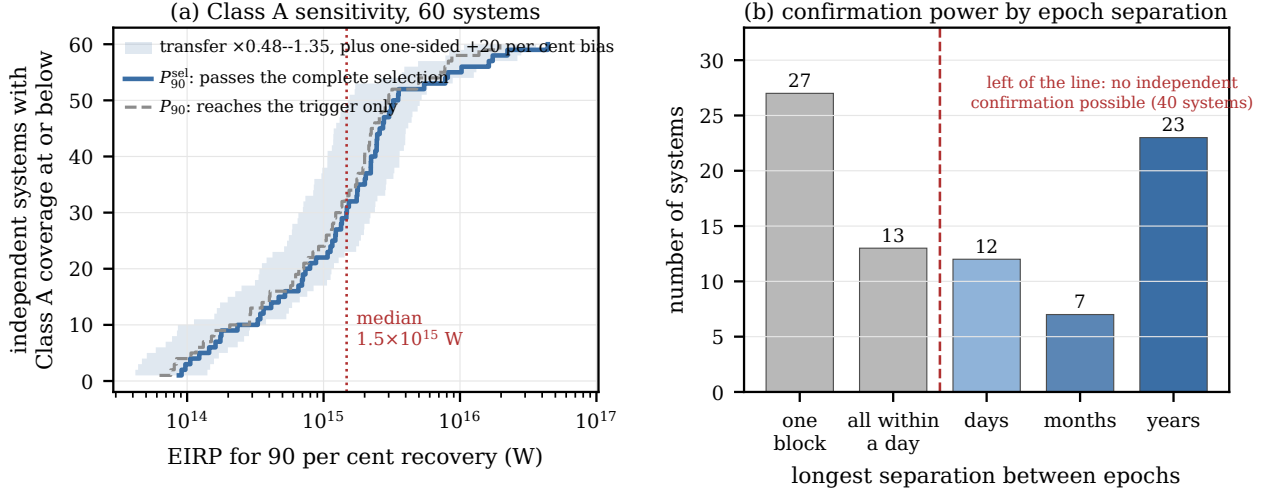


FIG. 7.— (a) Class A sensitivity. Cumulative number of independent systems whose best window reaches a given EIRP, for P_{90}^{sel} , the power passing the complete automated selection (solid), and for P_{90} , which reaches the trigger only (dashed). The shaded band is the systematic budget: the transfer bracket $\times 0.48\text{--}1.35$, which dominates, plus the one-sided $+20$ per cent atmospheric-decorrelation bias, which is why it is asymmetric. (b) The temporal selection function: systems grouped by the longest separation between their searched epochs, an independent epoch being defined as a separation of more than 1 day. **For the 27 systems with a single block and the 13 whose blocks all fall within one day — left of the dashed line — independent confirmation was not possible at all, whatever the sensitivity in panel (a). Recurrence over years can be tested for 23 systems.**

flagged by the symmetric form alone (§K.2, Table 21). Per-target values, including each band’s searched frequency range, are in the released catalogue (Data Availability).

5.1. Barnard’s Star and Wolf 359

Barnard’s Star and Wolf 359 are the second- and fourth-nearest stellar systems, and we find no prior mm or submm technosignature limit for either, so these appear to be the first. Both are non-detections in public Band 6 archival data. The nominal EIRP_{trig} trigger thresholds are $6.2 \times 10^{13}\text{--}9.3 \times 10^{13}$ W for Barnard’s Star over 12 windows and $7.3 \times 10^{13}\text{--}1.1 \times 10^{14}$ W for Wolf 359 over 12, with continuum upper limits of 0.575 and 0.453 mJy.

5.2. Technosignature search

Two ranking conventions run in parallel through this section: the frozen screen, which ranks the stellar position against 512 controls under a single global noise normalisation, and a radius-corrected variant, which standardises the probes within radius bins. Neither is exchangeable, so neither is a significance test, and the frozen screen is primary because it was fixed before the candidate list was examined (§5.2.2). **The choice does not affect the conclusion:** both yield zero confirmed candidates and differ only in the count of unattributed, non-recurring events, 4 under the frozen screen and 2 under the radius-corrected one.

All 113 star/band datasets yielded at least one carrier-lane result, 1655 spectral-window measurements after the catalogue audit removed 45 duplicate rows, Bands 6 and 7 dominating (Table 15). Nominal trigger thresholds span $1.6 \times 10^{13}\text{--}2.8 \times 10^{17}$ W at a median of 7.1×10^{14} W (Figs. 4 and 1), the deepest reached toward the nearest systems — 1.6×10^{13} W for the UV Ceti pair at 2.7 pc — though depth tracks integration time and channelisation as well as distance.

The statistics come strictly before the astrophysics, in

three named stages that do not promote one another. A *crossing* is one channel $\times$ drift cell at the stellar position reaching $T \geq 5$: 75 of the 1655 windows contain one, and 95 per cent of them are Class A — a trials effect rather than a physical one, since a median Class A window holds 2.6×10^5 channel $\times$ drift cells against 468 for a median Class B window. A *stage-1 flagged window* is one whose stellar statistic also exceeds all 512 controls: 13 do, in 6 systems and 7 star-band pairs. A local scramble test, which asks only whether a feature exceeds the noise at its own position, clears the control-ring maximum too in 3 of the 4 windows, so candidacy turns on the star-versus-control comparison. A *candidate* survives every stage of §4.1 and then recurs or is independently confirmed. There are none.

The outliers fall into two groups. 9 of the 13 sit within $\pm 50 \text{ km s}^{-1}$ of a molecular transition in the star’s own frame and carry independent CO evidence: 8 are β Pictoris, in 5 Band 3 and 3 Band 6 windows drawn from 8 execution blocks; one is the published foreground emission at HD 48370. That is 6 astrophysical sources, not 13 events, and the recurrence is the point: a frozen pipeline recovers the same disc in two transitions and 8 independent blocks without being told it is there.

The remaining 4 have no astrophysical attribution. They are rejected on physical grounds, and it is worth putting that before any counting argument. In the drift-following visibility fit none of the four shows a source at the stellar position: the largest $|\text{Re}/\sigma|$ among them is 0.93, while the same procedure applied to the β Pictoris CO windows recovers the disc at $6.1\text{--}9.5\sigma$. None of the four recurs in a repeat observation of the same star at the same tuning. Those two failures, not the false-alarm arithmetic, are why none is called a candidate; the counting argument below is supporting information, and it rests on a screen this paper has shown to be non-exchangeable.

With that said, how many should chance produce? A stage-1 event requires the stellar position both to reach

the trigger and to outrank every control, so the expectation must be conditioned on both: exactly one probe per window is the ensemble maximum, and the probability that the stellar position would have been that maximum, treated as exchangeable with the probes, is $1/(512 + 1)$ in windows whose control maximum reaches $T_{\text{trig}} = 5$ and zero in the rest. The condition is restrictive and informative: only 460 of the 1614 windows with a stored ensemble have a control maximum reaching the trigger at all, and 391 of those are Class A. *This is why every stage-1 event in the survey is Class A*, and the choice of reference class largely dissolves once the trigger condition is imposed. An expectation built on the rank alone predicts a different quantity — the number of windows in which the star would *rank first* — which is why the figures collected in Table 17 (Appendix J) appear to disagree.

Resampling execution blocks with replacement, because windows within a block share weather, calibration and correlator setup, and applying the measured non-exchangeability of the stellar position as a factor $\times 1.4$, gives 1.3 expected over the 1614 windows with stored ensembles, 95 per cent interval 0–4. Against that, 4 unattributed events are observed under the frozen screen, $\mathcal{P}_{\text{scr}} \approx 0.04$, and 2 under the radius-corrected statistic, $\mathcal{P}_{\text{scr}} \approx 0.4$ (probabilities to one significant figure; the tail factor is itself known only to 1.2–1.9). Either way we describe the count as a marginal excess and draw no inference from it. It cannot be dismissed as an artefact of the screen’s position-dependent rank bias: that bias places the star *low* in its control distribution, and a bias in that direction suppresses star-first outcomes rather than manufacturing them. Neither number is evidence about any individual event.

An excess this size needs only a small unmodelled tail in the control ensemble, and §C.1.2 shows that one exists. The same noise model sets the recovery fractions of §G.1, however, so a tail large enough to remove the excess would make the quoted completeness optimistic by roughly the same factor; the two cannot be traded independently, and neither correction has been applied.

The four hosts are not a quiet sample. 3 of the 4 carry activity indicators in SIMBAD (Table 5, queried 2026 September 22) and 1 is catalogued as multiple; CP–72 2713 in particular is a K7Ve star flagged as an emission-line, rotationally variable X-ray source, the profile of a star that flares. Magnetically active stars do produce coherent bursts at centimetre wavelengths, and although extrapolating that to the millimetre is not straightforward, a stellar-activity origin for these events cannot be excluded on the evidence assembled here. It would not change their disposition: an active star is still a compact source *at the stellar position*, and the visibility test finds no such source for any of the four.

All four have a persistence test and none passes it. Each has 1 to 16 further execution blocks of the same star whose tuning covers the event’s frequency, 23 in total (Table 8), and the feature recurs in none: the largest stellar statistic anywhere in them is $T_{\star} = 5.42$, 0 exceed their own window’s control maximum and 0 are stage-1 flagged windows. The baselines, however, are days rather than years: only 4 of the repeats are separated from their event by a resolvable interval, running 0–2 d, and the remaining 19 fall on the same day or within the

TABLE 5
ACTIVITY AND MULTIPLICITY STATUS OF THE 4 HOSTS OF UNATTRIBUTED STAGE-1 EVENTS, FROM SIMBAD OBJECT-TYPE LISTS QUERIED 2026 SEPTEMBER 22; DISTANCES ARE INVERTED PARALLAXES. ACTIVITY INDICATORS ARE X-RAY, EMISSION-LINE, FLARE AND ROTATIONAL-VARIABILITY CLASSIFICATIONS; “NONE CATALOGUED” MEANS SIMBAD CARRIES NO SUCH FLAG, NOT THAT THE STAR IS KNOWN TO BE INACTIVE.

Host	Sp. type	d (pc)	Activity indicators	Multiplicity
61 Vir	G6.5V	8.5	rot. var.	multiple
HD 23484	K2V	16.2	X-ray	none
HD 14055	A1Vnn	35.7	none	none
CP–72 2713	K7Ve	36.7	em.-line, rot. var., X-ray	none

same scheduling block. *These tests therefore exclude a transmitter persistent over hours to days, and bound no duty cycle measured in months or years.* Repeat coverage is matched on the window rather than on the crossing frequency, which the release records only where the pipeline stored a peak. CP–72 2713 is examined in detail (§A.3.1) because its repeat block is the deepest and closest in configuration.

5.2.1. The visibility-domain localisation test

Every stage-1 window has been tested in the visibility domain, twice, and none of the unattributed ones behaves like emission at the star. The image-plane rank is a screen; the physical question is whether the emission sits at the stellar coordinates, and the visibilities answer it directly. Phase-rotating every visibility to the star turns emission centred there into a constant real offset, so its real part measures the source; emission elsewhere in the field is suppressed in the real part and in general leaves a non-zero imaginary part. The two parts do different work and should not be read symmetrically. *The real part carries the detection*; the imaginary part is a discriminant against *displaced* emission only, as HD 48370 illustrates. A small imaginary part is not by itself evidence that emission is present at the phase centre, because for pure noise the imaginary part is centred on zero whether or not a source is there. The test separates *at the star* from *displaced from the star*, not unresolved from resolved. All 13 stage-1 execution blocks were recalibrated from the raw archive data — 4 unattributed and 9 attributed to carbon monoxide — and the test applied at each event’s recorded peak frequency, with eight control positions on the annulus the image-plane test uses (Table 6).

The attributed events are the test’s controls, and they are why the null result on the other four can be believed. 7 of the 9 return emission at the stellar position at 10–59 σ , every one above its own controls: the β Pictoris CO belt is recovered in both transitions and in seven separate execution blocks. Of the 2 that do not, one is HD 48370, whose real part is large (53 σ) but whose imaginary part is 7.9 σ from zero — exactly what emission displaced from the phase centre produces, and independently known to be a foreground cloud resolved several beams away; the other is a β Pictoris Band 3 block whose measured peak frequency does not survive in the frozen release, so the test had to be run at the frequency measured in its sibling blocks and returns nothing.

Against that, the 4 unattributed events, panel by panel in Fig. 8: 0 of them shows emission at the star. The largest real part is 2.5 σ ; 61 Vir does not exceed its own

controls; and HD 14055 returns a 2.5σ real part with a 3.2σ imaginary part, which a source centred on the star would not produce — though a phase-calibration residual or a small astrometric error produces the same signature. At this significance the uniform test does not localise the excess in either direction, and it is not equally sensitive to the two groups: it uses a single channel and no continuum subtraction, identically for every event, which is what makes the thirteen comparable, and a broad line fills that channel completely whereas a drifting narrow-band carrier occupies it for only part of the track — the tracks here sweep 4–23 channels for the unattributed events, losing as much as $\times 2.0$ – $\times 4.7$ in signal-to-noise.

A *drift-following, continuum-subtracted estimator* was therefore run on all 13 events as well, taking the channel the carrier occupies at each integration along its recovered drift rate and subtracting a local continuum from reference channels either side of the track before phase-rotating to the star (Table 6). It is decisive in both directions. β Pictoris is recovered at the stellar position in 6 of its 8 windows at 6.1 – 9.5σ ; HD 48370 returns a large real part, 5.9σ , with an imaginary part 2.7σ from zero, the displaced-source signature its foreground identification predicts, and so does not count as a recovery at the star; and 1 window cannot be measured this way, its drift track sweeping more channels than the continuum reference leaves. The four unattributed events show nothing: the largest $|\text{Re}/\sigma|$ among them is 0.93, and no control position or control frequency anywhere in the thirteen exceeds 2.14σ .

CP–72 2713 needs particular care with the continuum. Its stellar continuum is 10.9 mJy at 5.5σ , comparable to any narrowband excess it could carry, so a fit that leaves the continuum along the drift track in place reports the continuum instead: it returns 5.5σ , while removing it per integration returns -0.86σ at the stellar position with an imaginary part of 0.02σ . The difference is the continuum, not a carrier.

The average is taken over the measurement set’s own visibility rows, which carry one point of each conjugate pair rather than the Hermitian completion of the uv -plane. The imaginary part is therefore an informative quantity: had the conjugates been included it would vanish identically and the test would reduce to its real part alone.

Table 9 gives each star–band pair with the evidence dispositioning it.

5.2.2. Which normalisation is authoritative

The authoritative normalisation is the originally frozen screen. Every number in this paper’s abstract, results and conclusions is computed under it. The radius-corrected alternative was built, tested prospectively and *not* adopted, because it failed the validation set for it (Appendix C). Where the two are compared, the comparison is labelled as such.

Neither form is exchangeable, so neither assigns the unattributed events a calibrated false-alarm probability: the spatial statistic is a *prioritisation diagnostic* deciding which windows are examined further, not a detection statistic deciding which are real, and the disposition of every event rests on the visibility-domain test and on recurrence. The evidence for the displacement, the radial mechanism, the pseudo-star control and the prospective test that returned FAIL is set out in Appendix C.

The per-window rank probability, its survey-level counterpart, the pseudo-star and stratum tests, and the trials budget in cells are calibration rather than result and are collected in Appendix J; two statements from them carry the main line of argument, that the tail rate is measured and that no window in this survey approaches the survey-wide correction.

5.2.3. Dispositions of the remaining outlier windows

The third stage-1 flagged window, towards HD 48370 in Band 6, is the largest excess here in absolute terms and the smallest in relative ones, $T_\star = 27.10$ against a control-ring maximum of 26.83 — star and ring rising together, as emission filling the primary beam does. The velocities agree: Cataldi et al. (2023) report prominent CO(2→1) towards HD 48370 at $v_{\text{bary}} \approx 41\text{ km s}^{-1}$ and flag it as possible foreground cloud contamination, and our own frame chain returns $v_{\text{bary}} = +41.2\text{ km s}^{-1}$ for this crossing. Three checks close the circumstellar alternative: the same authors report the disc itself undetected in CO and in carbon; the sight line runs through the outer Galaxy, where a flat rotation curve reaches the measured velocity only at a kinematic distance of 2.07 kpc; and the block’s $^{13}\text{CO}(2\rightarrow 1)$ window has a control maximum of 20.4 against 4.6 at the star, so the field carries bright ^{13}CO that the star does not, resolved 1.7 synthesised beams away.

5.2.4. Line exclusion: definition and cost

The exclusion regions are defined before any crossing is classified, and those frequencies are *excluded from the experiment* rather than vetoed within it. Each catalogued transition of 11 species is masked to $\pm 50\text{ km s}^{-1}$ in the stellar frame; Appendix I gives the kinematic justification for that half-width and the species criterion, and Appendix K.4.3 the cost species by species. The statistic is computed inside the masked channels and the mask applied afterwards, at the reporting stage: the threshold and rank tests run over every delivered channel, and the mask then removes the masked intervals from the *reported* search domain, because a crossing coincident with a catalogued transition cannot be dispositioned as artificial on these data alone.

That ordering makes a robustness check possible, and removing the mask entirely adds no candidate. Of the 75 threshold crossings, 17 fall inside a tube, of which 9 are stage-1 flagged windows already dispositioned as CO in Table 9 and the remaining 8 do not exceed their own control ensembles. The mask costs 4.1 per cent of the unique-frequency union, 4.7 per cent of gross bandwidth, and nothing else. One caveat of principle survives: those rest frequencies are the ‘magic’ channels Cocconi & Morrison (1959) argued a communicator might choose, so a transmitter deliberately parked on one lies in unsearched reported space and this survey would not claim it either way. Those channels are retained in the release for anyone bringing independent spatial or time-domain discrimination.

5.3. What this experiment excludes, and what it does not

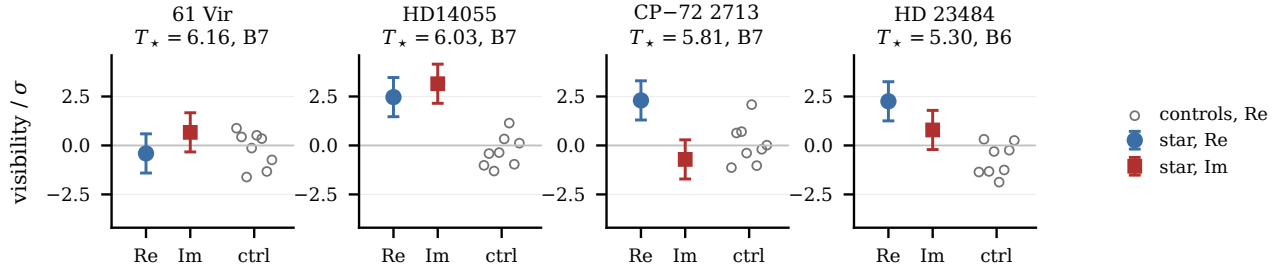
Excluded. For a continuously transmitting, unresolved carrier lying outside the molecular-line mask, within the

TABLE 6

THE DRIFT-FOLLOWING, CONTINUUM-SUBTRACTED FIT AT THE STELLAR POSITION, FOR ALL 13 STAGE-1 WINDOWS (§5.2.1); ‘CTRL’ IS THE LARGEST REAL PART RETURNED BY EIGHT POSITIONS ON THE CONTROL ANNULUS OR FOUR OFF-EVENT FREQUENCIES IN THE SAME WINDOW. THE ONE β PICTORIS WINDOW MARKED UNREACHABLE HAS A DRIFT TRACK WIDER THAN THE CONTINUUM REFERENCE LEAVES.

Star	B	N_{vis}	Re/ σ	Im/ σ	ctrl	verdict
<i>Unattributed</i>						
61 Vir	7	12 375	-0.93	+0.33	+1.18	nothing at the star
HD 14055	7	13 365	+0.56	-0.91	+0.38	nothing at the star
CP-72 2713	7	405 860	-0.86	+0.02	+0.98	nothing at the star
HD 23484	6	15 015	-0.57	-0.35	+1.30	nothing at the star
<i>Attributed to CO (positive controls)</i>						
β Pic	3	285 327	+8.13	+0.80	+1.14	source at the stellar position
β Pic	3	288 435	+9.52	-0.06	+2.14	source at the stellar position
HD 48370	6	9 832	+5.93	+2.74	+0.75	displaced: phase inconsistent
β Pic	3	421 890	+2.42	+0.43	+1.15	nothing at the star
β Pic	3	274 700	+6.12	+0.53	+1.84	source at the stellar position
β Pic	3	180 600	+8.01	-1.20	+1.63	source at the stellar position
β Pic	6	347 613	+8.47	+1.04	+1.49	source at the stellar position
β Pic	6	398 475	+6.34	-0.05	+1.39	source at the stellar position
β Pic	6	405 900	+0.00	+0.56	+1.53	track too wide to reference

unattributed events (shared scale)



events attributed to CO: the test's controls (panels autoscaled)

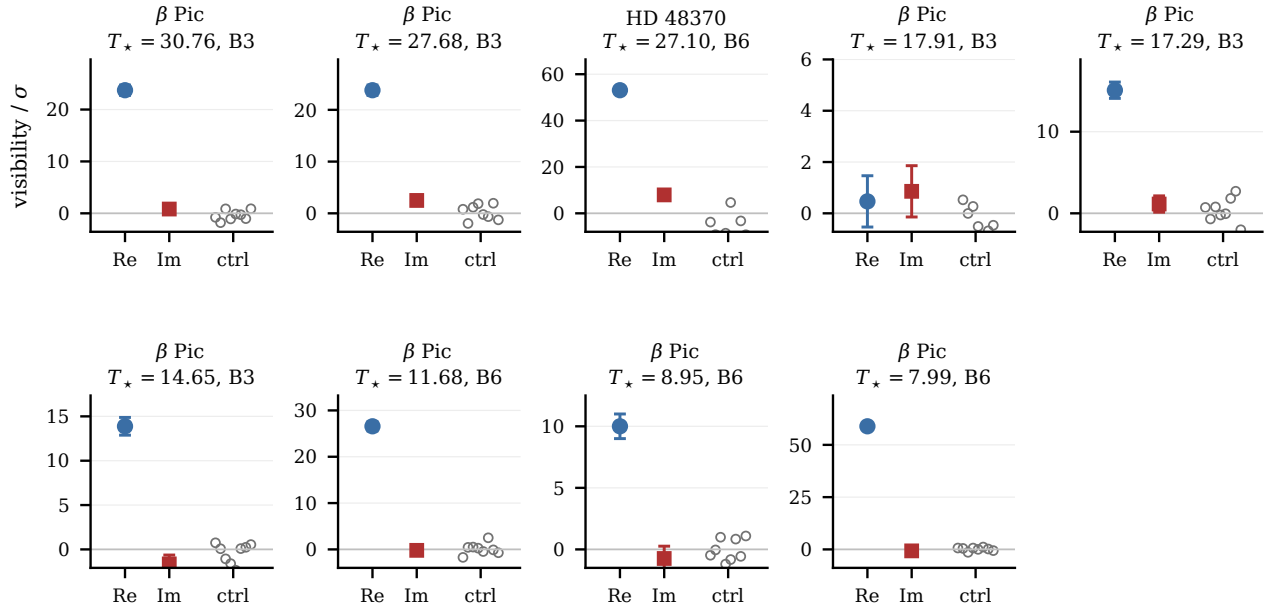


FIG. 8.— The visibility-domain test on all 13 stage-1 windows, one panel each, in units of the measurement uncertainty: the real part of the visibility phase-rotated to the stellar position (blue circle), which emission centred there makes positive and significant; its imaginary part (red square), which such emission leaves consistent with zero; and the real parts returned by eight control positions on the same annulus the image-plane screen uses (open circles). *Top row*: the 4 unattributed events, on one shared scale; none reaches 3σ . *Lower rows*: the 9 events attributed to carbon monoxide, the test's controls, each panel autoscaled because they reach 59σ .

TABLE 7

EVERY STAGE-1 EVENT AND ITS FATE: THE 13 WINDOWS THAT PASS THE DETECTION THRESHOLD AND THE SPATIAL SCREEN, CARRIED THROUGH THE FIVE STAGES OF §4.1. *screen*: THE STAR OUTRANKS ALL 512 CONTROLS, TRUE OF ALL BY CONSTRUCTION. *localised*: THE VISIBILITY FIT PLACES THE EMISSION AT THE STELLAR POSITION (§5.2.1); 6 DO. *attributed*: THE CROSSING LIES WITHIN $\pm 50 \text{ km s}^{-1}$ OF A CATALOGUED TRANSITION IN THE STELLAR FRAME WITH INDEPENDENT SUPPORTING EVIDENCE; 9 DO. *recurs*: THE FEATURE REAPPEARS AT THE SAME TUNING IN ANOTHER BLOCK *and* SATISFIES THERE THE SAME TWO CONDITIONS THAT DEFINE A STAGE-1 EVENT, REACHING THE TRIGGER AND OUTRANKING THAT BLOCK'S OWN CONTROLS; 7 DO AND 1 HAS NO REPEAT COVERAGE. THE ONE β PICTORIS WINDOW THAT FAILS IS THE BAND 6 TUNING IN WHICH THE RESOLVED CO RING FILLS THE CONTROL ANNULUS AND OUTSHINES THE STAR, WHICH IS THE BEHAVIOUR EXTENDED EMISSION PRODUCES AND NOT A FAILURE OF THE RECURRENCE ITSELF. 0 EVENTS PASS EVERY STAGE.

^a ONE β PICTORIS BLOCK HAS A DRIFT TRACK WIDER THAN THE CONTINUUM REFERENCE LEAVES AND CANNOT BE TESTED THIS WAY.

Star	B	T_*	screen	localised	attributed	recurs
β Pic	3	30.76	✓	✓	✓	✓ (4, $T_* = 27.7$)
β Pic	3	27.68	✓	✓	✓	✓ (4, $T_* = 30.8$)
HD 48370	6	27.10	✓	×	✓	no repeat
β Pic	3	17.91	✓	×	✓	✓ (4, $T_* = 30.8$)
β Pic	3	17.29	✓	✓	✓	✓ (4, $T_* = 30.8$)
β Pic	3	14.65	✓	✓	✓	✓ (4, $T_* = 30.8$)
β Pic	6	11.68	✓	✓	✓	✓ (5, $T_* = 8.9$)
β Pic	6	8.95	✓	✓	✓	✓ (5, $T_* = 11.7$)
β Pic	6	7.99	✓	n/a ^a	✓	×
61 Vir	7	6.16	✓	×	×	×
HD 14055	7	6.03	✓	×	×	×
CP-72 2713	7	5.81	✓	×	×	×
HD 23484	6	5.30	✓	×	×	×

TABLE 8

THE 4 STAGE-1 FLAGGED WINDOWS WITH NO ASTROPHYSICAL ATTRIBUTION, ONE ROW EACH. T_* IS THE SEARCH STATISTIC AT THE STELLAR POSITION AND 'CTRL' THE LARGEST OF THAT WINDOW'S 512 CONTROL STATISTICS; $|\dot{\nu}|_{\text{max}}$ IS THE WIDEST DRIFT THE STACK EVALUATED FOR THAT WINDOW; P_{eff} IS THE RESPONSE- AND SMearing-CORRECTED THRESHOLD POWER. 'REPEAT BLOCKS' COUNTS OTHER EXECUTION BLOCKS OF THE SAME STAR WHOSE TUNING COVERS THIS WINDOW, THE LARGEST STATISTIC ANY OF THEM RETURNS, THE RANGE OF TEMPORAL SEPARATIONS, AND THE DEPTH OF THE DEEPEST REPEAT. THE ELAPSED TIMES ARE THE LIMITATION RATHER THAN THE DEPTHS: 4 SEPARATIONS RESOLVE AT ALL AND NONE EXCEEDS 2 d. ALL FOUR WERE ALSO TESTED IN THE VISIBILITY DOMAIN (TABLE 6) AND AGAINST THE INTERFERENCE SIGNATURES OF §M. ^a THE RELEASED CROSSING FREQUENCY IS STORED ONLY WHERE THE PIPELINE RECORDED A PEAK; FOR THESE THE WINDOW IS GIVEN INSTEAD. ^b THE ARCHIVE INDEXES 200 OF THE 404 SEARCHED BLOCKS BY EXECUTION BLOCK AND THE REST ONLY BY MEMBER OBSERVING UNIT, SO A SEPARATION WITHIN ONE UNIT CANNOT BE RESOLVED AND IS REPORTED AS UNDER A DAY. ^c THE DEEPEST REPEAT CROSSES THE TRIGGER, $T_* = 5.42$, BUT DOES NOT OUTRANK ITS OWN WINDOW'S CONTROLS AND SO DOES NOT REPRODUCE A STAGE-1 EVENT.

Star	EB	B	ν (GHz)	$ \dot{\nu} _{\text{max}}$ (Hz s ⁻¹)	T_*	ctrl	P_{eff} (W)	epoch (UT)	N_{rep}	Δt	T_*^{rep} (Δ)	recurs?
61 Vir	Xc079b5_X82f	7	344.87–346.71 ^a	4150	6.16	5.88	9.4×10^{14}	2017-05-17 ^b	3	2 d ^b	4.83 (−1.32)	no
HD14055	Xff3d8_Xd46	7	330.39–332.23 ^a	3976	6.03	5.96	1.9×10^{16}	2022-10-16 ^b	16	0–1 d ^b	5.42 (−0.61)	no ^c
CP-72 2713	Xff0235_X4a6d	7	344.2697	4141	5.81	5.68	1.9×10^{15}	2022-10-02	1	< 1 d ^b	4.70 (−1.11)	no
HD 23484	X10b22d7_X2445	6	229.61–231.45 ^a	2766	5.30	5.25	1.7×10^{15}	2023-08-06 ^b	3	2 d ^b	4.99 (−0.30)	no

TABLE 9

EVERY STAGE-1 FLAGGED WINDOW OF THE COMPLETED SWEEP, ONE ROW PER STAR-BAND PAIR, GENERATED FROM THE RELEASED CATALOGUE. N_{win} IS THE NUMBER OF FLAGGED WINDOWS IN THAT PAIR; THE REMAINING COLUMNS DESCRIBE ITS DEEPEST WINDOW, WHERE T_* IS THE SEARCH STATISTIC AT THE STELLAR POSITION (Eq. 3), 'CTRL' THE LARGEST OF THAT WINDOW'S 512 CONTROL STATISTICS, ν THE CROSSING CHANNEL, AND $\Delta\nu$ THE OFFSET OF THAT CHANNEL FROM THE NEAREST REST FREQUENCY OF THE MASKED SPECIES *in the topocentric frame*, AS THE RELEASED CATALOGUE STORES IT. THE DISPOSITION RULE IS APPLIED IN THE *stellar* FRAME, AFTER THE FRAME CHAIN OF §F.1, WITH THOSE OFFSETS IN TABLE 20: FOR β PICTORIS THEY LIE WITHIN 0.46 km s^{-1} OF SYSTEMIC WHILE THE TOPOCENTRIC COLUMN READS ABOUT −28, THE DIFFERENCE BEING THE BARYCENTRIC AND SYSTEMIC TERMS. A PAIR IS DISPOSITIONED AS CIRCUMSTELLAR OR FOREGROUND EMISSION WHEN THE STELLAR-FRAME OFFSET SATISFIES $|\Delta\nu| \leq 50 \text{ km s}^{-1}$ AND INDEPENDENT EVIDENCE EXISTS, AND AS UNATTRIBUTED OTHERWISE.

Star	B	N_{win}	ν (GHz)	T_*	ctrl	$\Delta\nu$	disposition
β Pic	3	5	0.0000	30.76	7.34	−28	circumstellar CO
HD 48370	6	1	230.5117	27.10	26.83	−34	foreground CO
β Pic	6	3	230.5161	11.68	9.12	−28	circumstellar CO
61 Vir	7	1	0.0000	6.16	5.88	+449	unattributed
HD14055	7	1	0.0000	6.03	5.96	−12161	unattributed
CP-72 2713	7	1	344.2697	5.81	5.68	−1323	unattributed; absent in epoch 2
HD 23484	6	1	0.0000	5.30	5.25	+328	unattributed

searched frequency range and inside the linear-drift grid, and present during one of the archival epochs, the fine-channel search recovers the signal at least nine times in ten above P_{90} : 6.3×10^{13} W of EIRP in the deepest Class A window and 2.4×10^{15} W at the median one, a per-window property and a column of the released catalogue. No such carrier was found toward any of the 82 systems searched.

P_{90} is not the survey’s sensitivity, because a window does not select a signal for vetting merely because the trigger fires: the stellar statistic must also exceed every one of that window’s 512 control positions, and those maxima sit above the trigger, running 4.78 – 26.83σ across Class A with a median of 5.79σ . The survey’s sensitivity is therefore each window’s P_{90} scaled by its own gate, $\times 1.16$ the median P_{90} over the 60 Class A systems: the complete automated selection recovers a Class A carrier nine times in ten above $P_{90}^{\text{sel}} = 8.7 \times 10^{13}$ – 1.4×10^{17} W per window, and per system, on each system’s best window, 8.7×10^{13} – 4.4×10^{16} W with a median of 1.5×10^{15} W. That per-system median is the figure quoted in the abstract, in Table 2 and in the conclusions; P_{90} remains in the catalogue as the matched-filter completeness, for a reader comparing filters rather than surveys. Figure 9 shows the same completeness against duty cycle and Fig. 7 the per-system spread and the epoch coverage that limits confirmation; Appendix E defines the chain from P_{trig} to P_{90}^{sel} .

Read every such value as an upper bound on reach rather than a central estimate. The injection campaign covers 28 configurations rather than all 403, for a computational reason — each injected window is re-searched end to end through the frozen pipeline over the full drift grid — so the window-to-window transfer is reported as a measured bracket rather than removed: across the injected windows P_{90} ranges over $\times 0.48$ to $\times 1.35$ of its median-window value, so a single quoted threshold carries a factor of about two of spread and the catalogue gives each star its own bracket. Widening the campaign is the recommendation of §L that would most directly improve the headline number.

Table 10 is the budget. P_{90} is linear in every factor that builds it, and in d^2 , so each entry is a percentage of the answer. The independent errors combine to ± 9 per cent, dominated by the absolute flux scale: ALMA’s delivered absolute calibration is a few per cent at the low bands and up to 20 per cent at Bands 9 and 10, and the Class A sample is weighted toward Bands 6 and 7. Parallax, the finite size of the injection campaign and residual pointing are all small. Three further terms are carried separately, being of a different kind: the unknown position of a carrier within a channel, which no calibration removes and which leaves -12 to $+17$ per cent; atmospheric decorrelation, a *one-sided bias* of $+5$ to $+20$ per cent, since the injected tones that calibrate P_{90} are added to already-calibrated visibilities and suffer none of it while a real carrier does; and the window-to-window spread, -52 to $+35$ per cent, which is the largest term in the budget by a factor of several. The second significant figure of P_{90} is therefore meaningful for a single injected window and not for the survey. For the median Class A window, Wolf 219 Band 6, with $P_{90} = 2.37 \times 10^{15}$ W, the two-sided terms combine in quadrature to 2.18 – 2.55×10^{15} W, the one-sided terms take the upper end to 3.06×10^{15} W, and

TABLE 10
SYSTEMATIC UNCERTAINTY BUDGET ON P_{90} . THE FIRST BLOCK ARE INDEPENDENT ERRORS ON THE QUOTED NUMBER AND COMBINE IN QUADRATURE; THE SECOND ARE CARRIED SEPARATELY: THE SUB-CHANNEL PHASE IS AN UNKNOWN ABOUT THE TRANSMITTER, THE DECORRELATION TERM IS A *one-sided bias* THAT MAKES P_{90} OPTIMISTIC AND CANNOT MAKE IT PESSIMISTIC, AND THE WINDOW-TO-WINDOW TERM IS A TRANSFER UNCERTAINTY FOR THE WINDOWS THE CAMPAIGN DID NOT INJECT INTO.

Term	Origin	ΔP_{90} (%)
Absolute flux scale	ALMA calibration, band-weighted	± 7
Distance	Gaia parallax, as d^2	± 0.2
Injection statistics	4032 trials, bootstrap	± 2.0
<i>Combined</i>	in quadrature	± 8
Decorrelation	one-sided; injections suffer none	$+5 / +20$
Pointing	one-sided; worst case, $0.6''$ off axis	$+1.9$
Sub-channel phase	irreducible; median-corrected	$-12 / +17$
Window-to-window	transfer uncertainty off-grid	$-52 / +35$

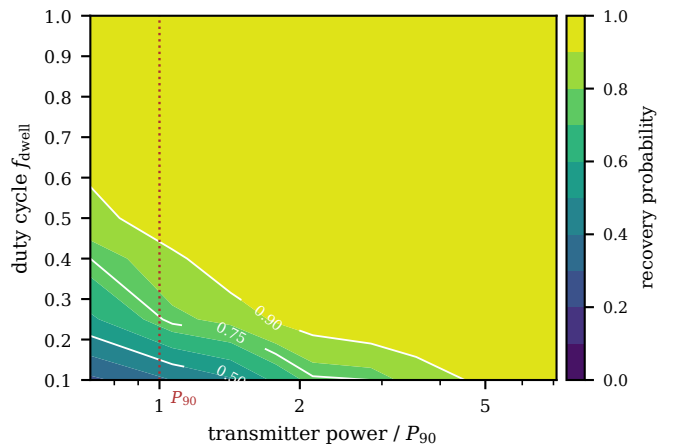


FIG. 9.— The duty-cycle selection function, measured. Recovery probability against transmitter power, in units of P_{90} , and duty cycle f_{dwell} , from 3888 injection trials. The abscissa is in units of P_{90} , which is the nominal trigger power multiplied by $\times 2.29$ for spectral response and smearing and then by $\times 1.2$ from the injection campaign, so a value of 1 here is $\times 2.81$ the trigger power. Recovery falls away below $f_{\text{dwell}} \simeq 0.3$. The dwell injections are zero-drift, so this selection function is measured for a stationary carrier and transferred to the drifting primary experiment; the joint drifting-and-intermittent cell of Table 13 remains unmeasured.

the transfer bracket spans 1.15 – 3.21×10^{15} W about the same central value; the three are not added together.

Not excluded. A transmitter absent during the archival observation, or on for less than about a tenth of the time. Emission inside the ± 50 km s $^{-1}$ molecular-line mask, at any strength. Drift outside the searched grid, or drift curved enough to leave a channel within the track. Signals broad enough to be removed with the continuum baseline. Anything below the window’s own completeness threshold. And any emission from the 17 566 stars within 40 pc that ALMA has not observed, which is almost all of them.

Two blind spots remain. The search is Stokes I only: a fully polarised carrier appears in I at full strength and is not missed, but the discrimination polarisation would offer against unpolarised astrophysical emission is unavailable, and the gain a per-hand search would buy is narrower than it first appears. ALMA’s feeds are linear (XX and YY), so splitting the parallel hands helps only a carrier whose linear polarisation happens to align with one feed, and rotation of the parallactic angle along a

track dilutes even that. A *circularly* polarised carrier — the electron-cyclotron maser case discussed in §D.1.2 — divides equally between XX and YY and gains nothing at all from the parallel hands; detecting its circularity requires the cross-hands, XY and YX, which this analysis does not extract. The 198 of 200 resolvable blocks that deliver both parallel hands (50 per cent of the 404) therefore support a partial test only. And the frequency coverage is the archive’s, not a choice: none of the “magic frequency” arguments that have guided targeted searches — the 1420 MHz line, its multiples, the water hole, or π and e times them — falls in an ALMA band, so a transmitter deliberately placed on one is outside this experiment by construction and its absence here is not evidence against it.

6. DISCUSSION

Everything below is conditional on what the archive chose to observe. 90 per cent of the searched stars were observed for debris-disc or planet-formation science, which skews the sample young, dusty and dynamically active and puts circumstellar CO — the source of 9 of the 13 stage-1 events — in the field of a large fraction of the windows (§6.6). The quantities that follow describe *these* stars and not the 40 pc population.

6.1. What signals could ALMA archival data actually detect?

Three morphologies set the reach of the archive, each illustrated with a real window and with the completeness entries of Table 13 and Fig. 10 behind it. Denominators count the 82 systems whose deepest threshold in the class lies at or below the case power.

(i) A *continuous, frequency-stationary carrier* in a coarse channel is the best case. The UV Ceti pair at 2.7 pc has a deepest window whose trigger is 1.6×10^{13} W and whose effective threshold is 3.7×10^{13} W, and a channel-confined carrier there is recovered at near-ideal $\sqrt{N}$ gain (§G.1). Read on the effective threshold, the comparison a transmitter power calls for, 0 systems of 82 reach the numerical EIRP of the Arecibo planetary radar and 1 twice it; on the trigger the figures are 1 and 4. The survey’s best case is a few times that numerical EIRP.

(ii) A *continuously drifting carrier* in a fine 488-kHz channel is the only morphology with a measured end-to-end recovery curve (Appendix G): above $\sim 8 \times 10^{13}$ W nominal it is recovered in 83 per cent of trials with its drift inside the searched ceiling, a depth 10 systems reach in Class A.

(iii) An *intermittent carrier, on for a quarter of the track*, above $\sim 2 \times 10^{14}$ W is detected toward 28 mid-distance systems, and recovery falls rapidly at smaller dwell fractions (Table 13): below $f_{\text{dwell}} \sim 0.1$ only the nearest windows carry information.

6.2. What does the null result mean?

No persistent, independently confirmed unresolved spectral-carrier technosignature has been found within the domain searched (§5.2). *The null result primarily constrains persistent spectral carriers.* For 40 of the 82 systems the archive provides no independent epoch capable of testing recurrence at all — 27 were observed in

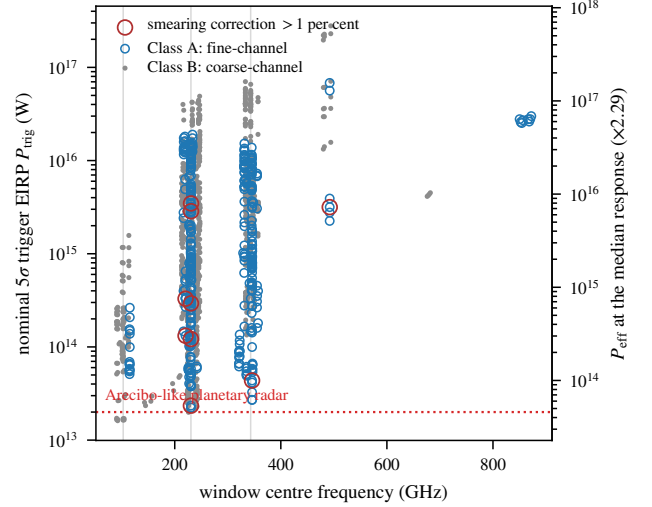


FIG. 10.— Two-dimensional sensitivity against window centre frequency for all 1655 searched windows, symbol by channelisation class, the axis along which completeness differs (Table 13). *Left axis*: the nominal trigger threshold P_{trig} , an internal detection statistic. *Right axis*: the effective threshold at the median response, $P_{\text{eff}} = 2.29 P_{\text{trig}}$, which is the power to compare with a transmitter (§4); the per-window ratio $P_{\text{eff, total}}/P_{\text{trig}}$ runs 1.33–3.98, so a single window’s threshold should be read from the catalogue, not from this axis. Red rings mark the 9 windows whose smearing correction exceeds 1 per cent; dotted, the Arecibo 2.38-GHz benchmark. Note the two decades of scatter at fixed frequency: there is no single survey sensitivity.

a single block and 13 have all their blocks within one day — so for close to half the sample the experiment places little constraint on a low-duty-cycle emitter, whatever its power. Only 23 systems can be tested for recurrence over years. The $\text{EIRP}_{\text{trig}}$ values are nominal trigger powers (§4), and completeness-calibrated exclusion holds only in Class A, the 403 fine-channel windows where the measured recovery curve applies; the 1252 coarse Class B windows provide spectral-excess coverage only and cannot discriminate drift.

The drift ceiling cuts the other way. It is derived from line-of-sight Doppler acceleration — of the 42 planets known around the 19 hosts only TRAPPIST-1 b reaches it in its edge-on worst case — and does not model transmitter-side oscillator drift, so a transmitter whose own reference drifts faster than the kinematic budget falls outside the searched grid (Fig. 6), as does one parked inside the line mask (§5.2.4).

Set against Wright et al. (2018)’s haystack, only Band 3 falls inside that axis’s 10 MHz–115 GHz span (20.6 of 115 GHz); the remaining 97.5 GHz in Bands 4–8 is new territory, though the median threshold detects a 10^{13} W transmitter only within ~ 0.9 pc. Against the sole prior ALMA-archival search (Mason et al. 2024, 28 bycatch targets at ≥ 1.01 kpc), the 82 systems here reach median per-system selection sensitivities $P_{90}^{\text{sel}} = 1.5 \times 10^{15}$ W, $\sim 459\times$ lower in EIRP, but that is almost entirely the distance term — the ratio of d^2 alone is ~ 6563 , so in *received flux* this survey is the shallower of the two. Comparison is in general better made axis by axis than on a single figure of merit: this survey differs from centimetre-wave searches in frequency, spectral resolution, number of systems, EIRP completeness, drift range and temporal coverage at once, and those axes are

coupled, so a product of marginals would misstate the overlap by orders of magnitude. Figure 1 makes the central point, that the novelty is the frequency regime and not depth; on the conventional Continuous Waveform Transmitter Figure of Merit (Appendix B) this survey is about four orders of magnitude less competitive than major centimetre-wave programmes.

How far the null reaches is set by recurrence rather than by the rank statistic. No single event in this design can clear the survey-wide threshold (Appendix J), so a genuine but weak transmitter seen once is not separable from the 4.7 chance events the survey expects — exactly the position CP–72 2713 leaves us in — and what separates them is a second epoch or an independent astrophysical identification. A repeat block is not necessarily a second epoch. Of the 82 systems, 27 were searched in a single execution block, and of the 55 searched in more than one, 13 have all their blocks within 1 day of each other, which is one observing session rather than a test of persistence. **Only 42 systems, 51 per cent, have epochs separated by more than 1 day:** 12 over days, 7 over months and 23 over years, so recurrence over years can be tested for fewer than a third of the sample. A further 10 systems hold archived blocks that were not both processed, so the test could be extended to 79 per cent of the sample with no new observing time.

ALMA has observed 90 nearby stars as a by-product of other science, across parts of the millimetre band no technosignature search had examined, and we find no channel-confined carrier the surrounding sky could not also produce. On the few gigahertz and few hours each star happened to be observed with, that says little about how common transmitters are.

6.3. What rests on the ACA windows

The control-geometry defect of §4.1 affects the 1054 ACA windows, 64 per cent of the survey, and it is worth being exact about what therefore carries a caveat and what does not.

Nothing in the headline sensitivity depends on it. The control geometry enters the *screen*, not the threshold: P_{trig} , P_{eff} and P_{90} are set by the noise in each window and by the injection campaign, and are untouched. The deepest trigger and the deepest P_{eff} in the survey are both 12 m windows.

What does depend on it is the composition of the stage-1 list. 9 of the 13 flagged windows are 12 m, where the geometry is correct and the rank is exchangeable; the remaining 4 are ACA. Three of the 4 unattributed events — 61 Vir, HD 23484 and HD 14055 — fall in ACA windows, and CP–72 2713 does not. Because contaminated controls are inflated, the bias runs *against* a window being flagged, so these three were promoted despite an adverse screen rather than because of a favourable one. That makes their disposition no stronger and no weaker: it rests, as for every event here, on the visibility test and on recurrence, neither of which uses the control annulus.

A corrected re-extraction of the ACA stratum is the single most useful piece of further work on this dataset. It would need the raw visibilities and a full re-search, and until it is done the ACA stellar ranks should be read as a prioritisation ordering with a known bias and not as a calibrated statistic.

6.4. Why no single window in this design can be significant

This structural limitation constrains what any survey built this way can claim. The rank floor $1/(512 + 1) = 1.95 \times 10^{-3}$ is 65 times coarser than the survey-wide Bonferroni scale $\alpha_{\text{survey}} = 0.05/1655$ (Eq. 2), so no single window can reach survey-level significance by rank alone, however extreme it looks — and enlarging the control ensemble does not rescue this, because the primary beam bounds how many independent control positions a field contains.

The survey’s own data show what a visibility-domain statistic buys instead. In the image plane the 13 stage-1 windows are indistinguishable: every one has zero of the 512 control positions at or above the star, so every one sits at rank 1 of $512 + 1$, equivalent to 2.9σ , with no dynamic range left to separate them. The visibility fit separates the same windows by 5.2σ , and its strongest reaches 9.5σ , a factor of 3.3 beyond anything the rank statistic can express whatever the data contain.

Hence the design adopted here, in which a technosignature claim rests on independent recurrence or independent astrophysical confirmation and the screen decides only which windows are examined. It also sets a target for successor surveys: raising the resolution floor requires either many more independent control positions per field than a single pointing affords, or a visibility-domain statistic, which does not inherit the floor at all.

6.5. The null result has not been checked against an independent pipeline

Every validation reported here tests this pipeline against itself: the injection campaign, the false-alarm calibration, the radius correction and the region-maximum audit share a single codebase with the search, so a fault common to the search and the injection code could suppress a real signal, or produce the appearance of a well-calibrated non-detection, without any internal check revealing it. No external check is available either, since the only prior ALMA-archival technosignature search (Mason et al. 2024) targeted stars beyond 1 kpc, disjoint from this 40 pc sample, and neither an independent re-implementation of the search statistic nor a blind third-party injection has been performed. **The null result reported here therefore rests on internal validation alone.**

The one thing that can be specified analytically is at least free of our code: the release includes known-answer test vectors generated from first principles — Gaussian noise plus a closed-form drifting sinusoid, with no import from the search or injection modules — together with the recovery each case should produce, so a reader who did not write this pipeline can check its stacking and normalisation against arithmetic. A blind third-party injection into raw visibilities remains the check we would most want, and we recommend it to anyone extending this work.

6.6. What this sample can and cannot speak for

Selection is on archival coverage, and ALMA’s archive was not built to sample stars. 81 of the 90 searched stars were observed under proposals categorised *Disks and planet formation* (Fig. 2), and 75 carry the *Debris*

disks science keyword. Among the 53 searched stars with a Gaia temperature, 21 are M dwarfs, 40 per cent, against 69 per cent of the reference census (Table 14), so the sample is F- and G-enriched and M-dwarf deficient; and window counts are concentrated, three systems — BD+05 1668, Proxima Cen and HD 33793 — carrying 17.2 per cent of the searched windows.

Two consequences follow, and they run in opposite directions. A debris-disc-dominated sample is the worst case for astrophysical false positives at millimetre wavelengths, which is where the stage-1 flagged windows of §5.2 concentrate, so the vetting burden here is heavier than a blind sample would carry. And M dwarfs host most of the habitable-zone planets in the solar neighbourhood while only 21 of the 5908 within 40 pc carry a searched window, so this result carries very little demographic weight for the local habitable-zone planet population. Both counts are restricted to the temperature-classified sample of Table 14, and illustrate the selection rather than measure it.

The point is sharper in targets than in percentages. Five of the searched M dwarfs host a planet that some published analysis places in the habitable zone — Proxima Centauri, TRAPPIST-1, LHS 1140, GJ 581 and HN Lib — and since habitable-zone membership is model-dependent and two of these are contested, five is an order of magnitude rather than a census. That is this survey’s coverage of the class of target most current technosignature programmes prioritise, against 5908 M dwarfs within 40 pc. Sun-like stars are sampled relatively heavily by comparison, but still through a highly non-random, disc-selected archive: the survey constrains transmitters around the stars ALMA happened to point at, not the local stellar population.

Useful subsamples still fall out of archival selection. 19 of the 90 processed stars (42 known planets) are confirmed exoplanet hosts, among them Proxima Centauri, τ Ceti, ϵ Eridani, β Pictoris and the seven-planet TRAPPIST-1 system, while HN Lib and LHS 1140, each with a habitable-zone planet, now carry an ALMA technosignature non-detection. Four are white dwarfs (Gentile Fusillo et al. 2021), all continuum non-detections and among the few mm/submm SETI constraints for that class (Huang et al. 2026).

7. CONCLUSIONS

1. **Search performed.** 403 fine-channel spectral windows toward 60 independent stellar systems, covering $\Delta\nu_A = 47.7$ GHz of unique sky frequency, searched for an unresolved narrowband carrier whose frequency drifts. A secondary coarse-channel spectral-excess search covered a further $\Delta\nu_{B\text{-only}} = 70.4$ GHz in 1252 windows but cannot discriminate drift. Both come from a uniform reduction of 484 public execution blocks, 4.1 TB of raw data.
2. **Sensitivity.** Measured by injection and recovery through the complete automated selection, not inferred from a nominal threshold: per system, on each system’s best window, $P_{90}^{\text{sel}} = 8.7 \times 10^{13} - 4.4 \times 10^{16}$ W of equivalent isotropic radiated power with a median of 1.5×10^{15} W. This is an upper bound on reach rather than a central estimate, the window-to-window transfer of the calibration being uncertain

by $\times 0.48 - \times 1.35$ and dominating the budget, and a one-sided +5 to +20 per cent decorrelation bias can only make the figure optimistic.

3. **Result.** 13 windows passed the automated screen; 9 are attributable to circumstellar or foreground carbon monoxide, among them the β Pictoris debris disc recovered blind in two transitions, the survey’s positive control. **None of the remaining 4 is independently confirmed: none is localised as a point source at the stellar position in the visibilities, and none recurs in the available repeat observations.** Localisation is the stronger of the two. The repeats available for these four are separated from the event by no more than 2 d, so their temporal sampling constrains intermittent emission only weakly, and non-recurrence is corroborating evidence rather than an exclusion in its own right. The screen itself assigns them no calibrated false-alarm probability, because its null distribution is empirically non-uniform under both the frozen and the radius-corrected normalisation (§5.2.2).

4. **Interpretation.** No persistent, independently confirmed unresolved narrowband carrier was found within the frequency, drift-rate, duty-cycle and sensitivity domain searched. **This cannot be converted into an occurrence constraint on technological civilisations within 40 pc.** The sample is archive-selected and strongly incomplete: 90 per cent of the searched stars come from disc and planet-formation programmes, only 21 of 5908 catalogued M dwarfs within 40 pc are covered, and the median system is searched over 1.84 GHz rather than the survey’s 47.7 GHz union. Occurrence figures, hosting fractions and figures of merit require a denominator this sample does not supply, and are confined to Appendix B, conditional on these stars, frequencies and epochs and on nothing wider.

These EIRP limits apply specifically to carriers unresolved by ALMA’s 15.3–1953 kHz Class A channels, and should not be read as equivalent in signal class or spectral resolution to Hz-resolution narrowband SETI limits.

In three sentences: archival ALMA observations provide a useful but strongly heterogeneous search for unresolved spectral carriers across a largely unexplored 89.6–873.1 GHz; within the Class A experiment no signal survives localisation and independent confirmation as a persistent stellar carrier; and this null constrains only the stars, frequencies, drift rates, sensitivities and epochs actually searched, not the occurrence of transmitters within 40 pc.

AUTHOR CONTRIBUTIONS

G.J.W. conceived the survey, defined the target sample and the selection criteria, and wrote the manuscript. R.D. built and operated the archival retrieval, calibration and search pipeline, implemented the spatial control statistic and the injection–recovery campaigns, and produced the released catalogue and its regeneration scripts. Both authors designed the candidate vetting procedure, carried out the statistical validation reported in §4.1 and Appendix J, and agreed the dispositions of Table 7.

FUNDING

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COMPETING INTERESTS

The authors declare no competing interests. VBRL Holdings is a privately held company through which R.D. conducts independent research, named here solely as his affiliation; it had no role in the design of the survey, the selection of targets, the analysis, the decision to publish or the content of this paper, and provided no funding for the work.

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This paper uses ALMA data from the public ALMA Science Archive. The 404 execution blocks searched span 2013 October 6 to 2025 June 11 across 36 proposal codes; their project codes and epochs are listed in the machine-readable release and should be cited from it in the standard JAO.ALMA#XXXX.X.NNNNN.S form. ALMA is a partnership of ESO (representing its member states), NSF (USA) and NINS (Japan), together with NRC (Canada), NSTC and ASIAA (Taiwan), and KASI (Republic of Korea), in cooperation with the Republic of Chile. The Joint ALMA Observatory is operated by ESO, AUI/NRAO and NAOJ. This work has made use of data from the European Space Agency (ESA) mission *Gaia*, processed by the *Gaia* Data Processing and Analysis Consortium (DPAC). This research has made use of the SIMBAD database (CDS, Strasbourg, France).

DATA AVAILABILITY

Every reduced data product, the full target-selection and analysis pipeline, the per-window catalogue, the injection records and the control statistics are sufficient to reproduce every number, table and figure in this paper. *The reference of record is the Zenodo deposit*, which is frozen, carries a DOI minted on acceptance, and is the version any reproduction should cite; the repository at <https://github.com/Tilanthi/SETI> is a working mirror and may move on. Appendix A describes the release and the checks that demonstrate its sufficiency.

SOFTWARE

The pipeline depends on CASA 6.7.6 (casatools/casatasks 6.7.6-14; CASA Team et al. 2022), Python 3 with astropy (Astropy Collaboration et al. 2022), numpy (Harris et al. 2020) and scipy (Virtanen et al. 2020), and the ALMA Science Archive and SIMBAD TAP interfaces through astroquery (Ginsburg et al. 2019); all are free and open-source, with exact versions recorded in the frozen release.

APPENDIX

WHAT THIS PAPER DID, AND WHERE THE DETAILS ARE

In plain terms, this paper asks whether any nearby star is transmitting a narrow radio signal that ALMA happened to record while observing something else. ALMA’s public archive holds thousands of hours of millimetre and submillimetre observations taken for quite different reasons — mostly studies of dust discs and planet formation

— and every one of those observations also contains a spectrum of the star at the centre of the field. A transmitter would show up in such a spectrum as a spike confined to one or two channels, sitting at the star’s position and not anywhere else in the field, and reappearing if the star were observed again. We searched the archive for exactly that, using a pipeline frozen before the search and applied without human intervention. Of the 82 systems covered, 60 have channels fine enough to resolve the drift a transmitter on a moving planet would show; those 403 windows are the primary experiment, and the remainder can register an unexplained spike but cannot tell a drifting signal from a stationary one.

Nothing survived. Thirteen windows were picked out by the automatic screen; nine are carbon monoxide from the β Pictoris debris disc, which the pipeline recovered without being told it was there and which serves as proof the method works; one is foreground gas toward HD 48370; and the remaining four are unexplained but are not independently confirmed — none is a compact source at the stellar position when the raw interferometric data are examined, and none reappears in a later observation, though the repeats available span days rather than years and so test intermittent emission only weakly. The result is therefore a non-detection, and the useful part of the paper is the calibration: how deep the search actually reached, for what kind of signal, and over what fraction of the frequencies and epochs a real transmitter would have had to use.

Two cautions matter more than the limit itself. The targets were chosen by other astronomers for other purposes, so the sample is not representative of the solar neighbourhood and cannot speak for M dwarfs, which dominate that population but are barely represented here. And the star-versus-control ranking used to pick candidates is a prioritisation tool, not a significance test: it cannot assign a probability to any individual window, which is why every disposition here rests on physical evidence instead.

The appendices are reference material and none is needed to follow the result. They cover, in order: how frequency, linewidth and smearing are defined (F); the injection campaign that measures the sensitivity (G); the per-window catalogue (H); the line mask (I); the continuum lane false-alarm accounting (J); provenance, extraction geometry and robustness checks (A); the spatial normalisation and its validation (C); the β Pictoris positive control and the epoch extension (D); the conventional figure of merit (B); and the notation (E).

REPRODUCIBILITY AND PROVENANCE

Every number, table and figure in this paper comes from the submitted-analysis snapshot, the repository commit tagged at submission, which freezes the pipeline code and the input data as run. Deleting every generated table and figure and rebuilding from the scripts alone reproduces every product exactly.

The machine-readable catalogue, `per_target_results_v3.97.csv`, carries one row per searched window, 1655 rows in 50 self-described columns: the observation; the sensitivity reached, including P_{90} with the $\times 0.48$ – $\times 1.35$ bracket the injection campaign measures; the stellar and control peak

statistics with the add-one rank under both the released and the radius-corrected normalisations, the crossing frequency, the nearest catalogued transition and, for the 10 rows that have one, the disposition verbatim from Table 9; and the control geometry. Everything quoted in this paper follows from it: the sample sizes, the candidate list under either statistic, the frequency union and every P_{90} . A script in the release opens the catalogue and nothing else, re-derives 41 of the headline numbers, compares each against the value typeset here and exits non-zero on any disagreement. The statistical unit is `system_id`, with 8 bound pairs sharing one identifier. The known-answer test vectors of §6.5 ship in `testvectors_v397/` with a manifest, checksums and the expected recovery for each case.

The prospective extension of §D.1.1 and the epoch-extension campaign of §C.1.1 lie outside the frozen release. The extension ships separately, with its own 463 window records in the same schema and the full 512-element control vector for each. Those blocks were searched after the statistic and the mask were fixed, so merging them into the catalogue would change the denominator on which every rate here is calibrated; pooling is left to the reader as an explicit choice.

Besides the catalogue, the Zenodo deposit holds the per-window search products for all 1655 windows with their 512 control statistics, so that every rank and p -value is recomputable; the line mask in frozen and repaired forms; the execution-block metadata and pointing ledger; the 1200 injection, 3888 dwell and 4032 stratified trials; the local-scramble and visibility products behind the CP-72 2713 disposition; and the change-log that timestamps the promote and retire criteria before the data were seen. Data products carry the Creative Commons Attribution 4.0 International licence (CC BY 4.0) and the pipeline code the MIT licence. Raw ALMA visibilities are public from the ALMA Science Archive at <https://almascience.org>.

No inference to the overall 40 pc stellar population is attempted in this paper. The question this experiment answers is what the existing ALMA archive already constrains, for the stars it happens to contain; the question of how common powerful transmitters are within 40 pc cannot be answered robustly from a sample selected this way, and we do not attempt it.

Extraction, control geometry and position-correlated systematics

The baseline filter takes the median in contiguous blocks of 65 channels and interpolates linearly between block centres, using $\max(2, n/65)$ blocks over n usable channels. Its effective width is 65 channels in the 134 windows long enough to hold more than two blocks and $n/2$ in the short coarse windows, capping the linewidth it passes at ≈ 32 MHz for the fine class and ≈ 0.9 GHz for the coarse one (Appendix F). The crossing channel is *not* excluded from the scale. The median runs across positions at a fixed channel, so what matters is how many of the 512 controls carry the feature: a compact feature at the star appears in none and the 50-per-cent breakdown point holds, while emission filling the primary beam appears in all of them, as for HD 48370 (§5.2).

$T_\star \equiv T(\mathbf{x}_\star)$ is evaluated at the proper-motion-propagated stellar position, and the same operation, drift

grid, channelisation and noise map give $\{T_i\}$ at all 512 control positions. `probe_positions()` in the released extraction code adds its offsets to the star’s own direction cosines, so the annulus is centred on the *star* wherever the star sits in the field. It holds 512 positions drawn uniform in area, $r = \sqrt{U(r_{\text{in}}^2, r_{\text{out}}^2)}$, and uniform in azimuth, from the fixed seed 20260825 re-applied per window, so all windows share one pattern in units of θ_{PB} . Radii run 0.14–0.78 θ_{PB} with θ_{PB} evaluated at each window’s own median frequency, and an inner check-ring of 135 positions serves the point-source test. The geometry ships as catalogue columns but is absent from the frozen products, so it is reconstructed from the code; the reconstruction recovers the check-ring’s 135 positions, and the control statistics fall with reconstructed radius where extended CO fills the beam (Spearman $\rho = -0.49$, HD 48370 Band 6).

Placing controls off-centre costs sensitivity. A star near the pointing centre sits at primary-beam response ≈ 1 while the annulus samples 0.95 at 0.14 θ_{PB} falling to 0.19 at 0.78 θ_{PB} , area-weighted mean 0.47. Star and controls are therefore not exchangeable in flux density. They remain exchangeable in signal-to-noise ratio, which is what the statistic ranks, because $T(\mathbf{x})$ is formed on uncorrected amplitudes against one shared noise scale. Extended sky brightness does not cancel: smooth emission lives on short baselines whose phases barely change under a rotation of a few arcsec, so star and annulus rise *together*, as for HD 48370, while emission offset from the star raises particular controls above it, as in the same star’s ^{13}CO window, whose ring maximum sits 9.2 arcsec from the star (§5.2). The two residual asymmetries run in opposite directions: a suppressed control makes a star-exceeds-ring outcome more likely under the null, while a multiplicative calibration error scales with the continuum flux at the extracted position and so favours the star.

Neighbouring controls are correlated below the synthesised-beam scale, so a window’s annulus carries $N_{\text{eff}} \simeq 30\text{--}43$ independent spatial trials. That is an analytic beam-geometry estimate and a worst case, the annulus containing far more resolution elements than the 512 positions drawn. The rank test needs exchangeability rather than independence, and correlation among controls costs resolution without touching the 1/513 floor. Radii are quoted in arcsec and in θ_{PB} because the frozen products carry neither the synthesised beam nor the array configuration; the 64 per cent of windows taken with the ACA 7 m array are treated in Appendix J.

Residual continuum, self-calibration residuals and position-correlated calibration error all concentrate at or near the phase centre, where the pointed science target sits, and σ is built from the annulus and carries none of it. These stars are not all blank there — an ancillary continuum lane detects 17 of 57 target/bands — and a residual bandpass ripple of fractional amplitude ϵ leaves a channel-confined artefact of order ϵS_{cont} at the extracted position and essentially nothing on the annulus. Three measurements bound the effect. The star’s own residual variance equals the controls’ to better than a per cent in the flagged windows (§K.1.2). Recomputing the noise scale locally, in a band around each feature and excluding the feature itself, gives local-to-global ratios of 0.97 (CP-72 2713), 1.29 (HD 48370), 0.93 (β Pic B3) and

1.02 (β Pic B6); the global scale is therefore good to a few per cent except towards HD 48370, where the field is filled with CO, and rescaling both the statistic and its ensemble there leaves the window flagged ($T_\star = 21.0$ against 20.8). A badly misstated σ would also displace the control distribution, which the uniform pseudo-star ranks exclude. A per-crossing local scale inside the search remains an open item (§L).

The statistic evaluates one position at the star and one at each control, because a rank means something only when the identical operation runs everywhere; §K.2 scores every window under the asymmetric region-maximum form as well and audits the difference.

Statistical units and independence

Six units appear in this paper and they are not interchangeable. A *catalogue entry* is one star as SIMBAD designates it, 90 searched. A *physical system* is the set of entries gravitationally bound together, 82 of them, the difference being seven designation-linked pairs. A *target/band* dataset is one entry in one ALMA band, 113 in all. An *execution block* is one archival observation, 404 searched, and is the unit that shares weather, calibration and correlator setup. A *spectral window* is one window of one execution block: 1655 catalogue rows, reducing to 1616 distinct (execution block, window) datasets once repeated archive products are collapsed. An *independent frequency trial* is none of these, the channel $\times$ drift grid being oversampled on both axes, so a window's 177–6 869 740 cells carry about a factor 5.3 fewer independent trials than cells (Appendix J). An *independent epoch* is a separate execution block at the same tuning; blocks of one scheduling block executed on one night are not independent epochs in any useful sense.

The survey-level flag expectation uses 1655 windows, which is conservative because 1616 is the smaller number and because windows within a block are correlated. Statements about how common transmitters are use 82 systems, never 90 entries and never 1655 windows; statements about persistence use epochs, of which most stars here have one.

Data-quality exclusions and open items

113 target/bands are searched to completion and reported here. Four classes of entry are absent from the released catalogue and are itemised in Appendix K.

A further 68 candidate target/bands were selected but never delivered a searchable product, and the cause lies in the selection step. 60 of them were never observed at all, sitting a median 1.6 degrees from the nearest field centre: those observations were pointed at solar-system targets whose quoted field of view is enormous, and a crossmatch that trusts that quantity admits stars far outside any real primary beam. The remaining 8 entries were pointed at but yielded no search product. The per-star pointing ledger is released so that the selection can be audited. 53 of the 168 candidate stars fail on pointing in every band, so the sample is 90 of at most about 115 genuinely covered stars rather than 90 of 168, and the coverage completeness of Table 2 should be read against the smaller denominator.

One automated quality check precedes any sensitivity number. The radiometer relation makes $q \equiv$

$\sigma\sqrt{t_{\text{on}}\Delta\nu_{\text{ch}}} \propto \text{SEFD}/\sqrt{N_{\text{bl}}}$ a quantity physics predicts, so a window whose q falls far below the sample median cannot be a noise fluctuation. 13 coarse time-division windows fail by about three decades, all of them channel-averaged auxiliary windows riding on a science baseband and carrying a constant WEIGHT column, so they never received that baseband's calibration. The deflation is common mode in data and weights alike, so it misstates a threshold but cannot manufacture a flag: those windows are excluded from the sensitivity accounting only, and their statistics and ranks were carried through candidate generation unchanged (Appendix K).

45 of the 1725 extracted rows are duplicate entries for one physical window, the same (star, execution block, frequency range, channel width) key appearing twice in the archive query, so one row per key is the right statistical unit. The retention rule — the key's first-listed row in the query's own identifier order — consults no measured value and cannot prefer a crossing or a deeper threshold. Repeats need not agree: two reductions of the same τ Ceti Band 6 window differ by 44 per cent in threshold, which exceeds the flux-scale systematic of §F.1 and is the most direct empirical bound here on threshold reproducibility, and the two HR 1010 Band 6 rows straddle the trigger at $T_\star = 5.10$ and 4.95, so the crossing count is rule-dependent at the level of one window. The four flagged windows are not, no discarded row being both a crossing and a stage-1 flagged window. The line-excluded continuum lane is exploratory, bears on no technosignature limit, and is reported separately.

CP–72 2713: an unconfirmed event, absent in the repeat block

CP–72 2713 is a worked example of the full vetting chain rather than a result: at the measured false-alarm rate the survey expects 4.7 such events and has 4. In Band 7 it reached $T_\star = 5.81$ against a ring maximum of 5.68, with no line within $\pm 50 \text{ km s}^{-1}$ in the stellar frame and no localisation in the visibilities ($\text{Re}/\sigma = -0.86$; §5.2.1). It acquired no astrophysical attribution and was recorded as an unattributed stage-1 flagged window under promote and retire criteria committed 9 hours before the second block was searched (2026 September 11, 119a68a6ab50).

That second block matches the first in tuning, channel width, channel count, integration count and on-source time, registers to 0.18 of a channel, and is the deeper of the two, so a persistent emitter at the first block's flux would have returned $T_\star = 6.21$. Pinned to the first block's own frequency and drift it returns 3.60, and read as two measurements of one constant amplitude the pair is inconsistent at 4.6σ . The blocks are 2.05 h apart, so this is a within-night test: it excludes a persistent emitter of that strength on that baseline and excludes nothing about an intermittent one. The disposition rests on that recurrence test and not on the rank; the frequency, drift, control and continuum diagnostics are in Appendix K.

Sample bookkeeping: entry counts, coverage criteria and pointed versus serendipitous

The 168 stars are complete against, and conditional only on, the 40-parsec reference catalogue intersected with the public-archive snapshot of 2026 September 8. Against that Gaia DR3 catalogue of 17 566 stars they are

$\lesssim 1$ per cent, and the 82 systems searched are 0.47 per cent, fewer still once §A.3 removes those never observed (Fig. 3). The two parallax cuts are not interchangeable, the sample’s binding condition being $\sigma_{\varpi}/\varpi < 20$ per cent against the census’s 10 per cent; the realised sample reaches 2.48 per cent at worst.

Stellar counts throughout are catalogue-entry counts. The 90 searched entries comprise 82 systems, differing by seven designation-linked component pairs, so the 113 star/band datasets, 1655 windows and 404 execution blocks all derive from 82 systems. The 1655 rows comprise 1616 distinct (execution block, spectral window) datasets; the larger number is the trials denominator throughout, which is conservative and makes no practical difference, predicting 3.2 chance outliers against 3.1 for the smaller.

Admission to the sample required the following.

1. At least one public execution block whose visibilities are the archive’s delivered QA2 products, taken as delivered and not re-calibrated, covering the star’s epoch-propagated position. Calibration vintage therefore varies across the 2013–2025 span; the flux-scale term this contributes is small beside the $\times 2.29$ spectral-response correction. There is no minimum integration, sensitivity, antenna count, mosaic-tile count or array configuration beyond QA2.
2. That position inside the field’s primary beam. Extraction aborts beyond $2\theta_{\text{PB}}$ and the largest retained offset is $0.46\theta_{\text{PB}}$.
3. Each spectral window admitted individually, on QA2 vetting alone.
4. No programme-category restriction: science, calibration and observatory projects enter on identical terms.
5. A fixed archive snapshot date, 2026 September 8.

Completeness with respect to the work list holds in targets and tunings, not in epochs or integration time: the survey takes at most 25 blocks per target and deduplicates spectral windows by identifier, so 404 of the 656 public execution blocks those targets hold are in this release, 61.6 per cent, with 326 further blocks reserved as an external calibration sample. ALMA project metadata quantify pointed against serendipitous coverage: almost every spectral-window-covered star is the pointed science target in at least one execution block, leaving Wolf 219 the sole serendipitous star, so the residual selection function of a work-list-complete sample is the union of ALMA target lists.

THE CONVENTIONAL FIGURE OF MERIT

These are the standard occurrence metrics, computed so the survey can be placed on the axes the literature uses. Following Enriquez et al. (2017), the Transmitter Rate is $(N_*\nu_{\text{rel}})^{-1}$ and the Continuous Waveform Transmitter Figure of Merit is

$$\text{CWTFM} = \zeta_{\text{AO}} \frac{\text{EIRP}}{N_*\nu_{\text{rel}}}, \quad \nu_{\text{rel}} = \frac{\Delta\nu_{\text{tot}}}{\nu_{\text{mid}}}, \quad (\text{B1})$$

with $\zeta_{\text{AO}} = 5.0 \times 10^{-11}$ fixed by their normalisation (CWTFM = 1 at EIRP = 10^{13} W, $\nu_{\text{rel}} = 1/2$, $N_* = 1000$) and lower values better. On the Class A experiment, $N_* = 60$ systems and $\Delta\nu_{\text{tot}} = 47.7$ GHz about $\nu_{\text{mid}} = 494$ GHz give $\nu_{\text{rel}} = 0.097$, a Transmitter Rate of

0.17 and, at the median per-system $P_{90}^{\text{sel}} = 1.5 \times 10^{15}$ W, $\text{CWTFM} = 1.3 \times 10^4$ (3.8×10^5 at the worst-case system). That figure is reported for comparison only. It uses the survey-wide frequency union, which no single star was observed over; the quantity that answers whether a transmitter’s frequency was actually observed is the per-system version below. For scale on the conventional axis, Enriquez et al. (2017) reach 0.85 and Price et al. (2020) 0.11 at the GBT. This survey is therefore some four orders of magnitude worse than cm-wave SETI on the standard continuous-carrier metric, and that is the correct conclusion: 60 systems at a 10^{15} W floor cannot compete with hundreds of stars at 10^{13} W. What the metric does not score is the frequency axis, the one on which this survey is first, since Eq. B1 credits fractional bandwidth and not location, and 47.7 GHz at 494 GHz is a small fraction of a large number.

Two caveats are not presentational. First, the transmitter-hosting fraction. Zero confirmed detections in 60 Class A systems bounds the fraction hosting a continuously transmitting carrier above each system’s own P_{90}^{sel} at $f < 5.0$ per cent, the 95 per cent Poisson limit; the $1/N$ convention common in the literature would say 1.7 per cent, and is optimistic. That bound holds only for carriers inside the searched frequency islands, drift grid, duty-cycle domain and archival epochs, on a sample ALMA selected for disc science, and it is not a statement about the 40 pc population.

Second, and more seriously, ν_{rel} in Eq. B1 is a survey-level quantity, and *a civilisation around one star does not have access to the union of frequencies observed toward all 82 systems*. System by system the median Class A coverage is 1.84 GHz, from 0.11 to 19.08 GHz over the 60 systems that have any — 3.9 per cent of the survey’s own union. Recomputing on that median gives $\nu_{\text{rel}} = 0.0037$ and $\text{CWTFM} = 3.3 \times 10^5$, worse by $\times 26$. We therefore take the per-system value as the scientifically meaningful one and the survey-union figure as a literature convention: it measures aggregate bandwidth, not the probability that a given transmitter frequency was observed. Under any frequency prior broader than the searched islands the occurrence bound vanishes, which is a property of the archive and not of the method.

THE SPATIAL NORMALISATION: EVIDENCE AND VALIDATION

This appendix collects the three measurements behind §5.2.2 and the prospective test of the correction.

Prospective validation of the radius correction

The radius-corrected normalisation was derived from the calibration data, so on its own it establishes nothing about data it has not seen. We therefore froze it — the probe radii, the bin edges and the standardisation are those of Appendix K, with no free parameter to tune — and applied it unmodified to 778 windows that played no part in its construction: the pre-specified hold-out and the prospectively searched extension of §D.1.1. The level for consistency with uniformity was fixed at $p \geq 0.05$ before the test was run, and nothing was retuned afterwards.

Two rank distributions can be tested and they answer different questions. The *pseudo-star* distribution ranks an inner-annulus probe against the other probes;

it passes both before and after correction (uncorrected $D = 0.005$ at $p = 0.16$, corrected $D = 0.006$ at $p = 0.09$, over 49792 probes). That is expected and is not evidence for the correction: every probe is drawn from the same radial distribution, so probe-against-probe ranks are near-uniform however they are normalised. The discriminating test is the *stellar* distribution, because the star lies at the phase centre, inside the probe annulus, and that asymmetry is what the correction exists to repair.

On that test the correction fails. Out of sample the uncorrected stellar rank has median 0.409 against the expected 0.5, with $D = 0.105$ at $p = < 10^{-4}$; after correction the median is 0.407 with $D = 0.097$ at $p = < 10^{-4}$. The departure from uniformity is essentially undiminished, and the median moves further from the exchangeable value. The result holds separately in both samples (hold-out 0.405→0.392, extension 0.409→0.417).

The interpretation we draw is narrow and, we think, unavoidable. The radial gradient is real and is one mechanism behind the displacement, but removing it does not leave an exchangeable statistic, so the residual is not radial. Neither normalisation yields a calibrated tail probability, and the spatial screen is therefore a candidate-generation device under both. **We consequently retain the frozen screen as primary — it at least has the virtue of having been fixed in advance — and rest every disposition on the physical evidence rather than on either rank.**

The external calibration sample

A sample of 326 execution blocks toward 56 stars was reserved by a rule fixed and committed before the archival search finished and before any reserved block was searched, so that the statistic could be checked on data it had never seen. Appendix K.4.1 gives the reservation rule, its commitment dates and the full result. Three findings carry into the main line of the paper.

First, the pipeline behaves on unseen data: over 1322 windows it returns 7 stage-1 flagged windows, of which 4 are β Pictoris CO and 3 are unattributed, against 3.7 expected at the measured tail rate. Second, and adversely, the stellar rank is *not* uniform out of sample: its median is 0.405 where 0.5 is expected, with a block-clustered 95 per cent interval of 0.341–0.458 and $D = 0.109$ at $p0.0011$ over 315 windows. A displacement that survives a hold-out chosen before the data is a property of the statistic and not of this sample, which is why the spatial rank is used as a screen and never as a significance (§4.1). It reproduces a third time, at 0.409 over 463 prospectively searched windows (§D.1.1). Third, the false-alarm tail factor — the ratio of the measured rate at which a no-signal position ranks first to the rate exchangeability would give — is 1.2 on these blocks, with a bootstrap over the 77 reserved blocks giving 0.6–1.9. That factor and its width set the accuracy of every chance expectation in this paper.

Where the displacement comes from: radius, not stars

The displacement has a mechanism, and it is instrumental. Images are corrected for the primary-beam response, so the true noise rises with distance from the pointing centre, while the search divides every position

by one scale estimated over the whole window. The control probes therefore sit at systematically different effective noise from the star, which is at the phase centre. Appendix K.4.2 sets out the evidence: the control level varies with radius across the annulus, pseudo-stars carrying no star at all are displaced as far as real ones, and rescaling the survey with the gradient removed changes which windows are flagged but not what the paper concludes.

Two numbers matter here. Standardising each control probe within its own radius bin — the probes come from one fixed seed, so their radii are known, and the correction has no free parameter — takes the stage-1 list from 13 windows to 10, losing 4 and gaining 1, all of them marginal and all already dispositioned on other grounds. And the survey median add-one rank does *not* move toward 0.5 under the correction: it goes from 0.445 to 0.432, slightly further away. The radial term is one demonstrated mechanism for the displacement, not the whole of it, so we do not adopt the corrected statistic as primary and we do not present it as a validated null.

EXTERNAL CHECKS AND THE POSITIVE CONTROL

No independent pipeline check

The consequence of validating this pipeline against itself is stated in §6.5; the measurement behind it is the absence of shared data with the only prior ALMA-archival technosignature search. The 28 “stellar bycatch” targets of Mason et al. (2024) all lie at or beyond 1.01 kpc while every star here is within 40 pc, so the two distance ranges are disjoint by a factor of about 25 and no target, hence no execution block, is common to both. The 112 Band 3 windows searched here are in their band but in different fields. Neither of the two external checks that would carry weight — an independent re-implementation of the search statistic, or a blind injection by a third party before delivery — has been performed; the second would be the more informative, and is the check we would recommend to anyone extending this work.

A prospective epoch extension

The work list was built against an earlier archive map, which had not exposed 177 public in-scope execution blocks. 115 of them have since been searched with the frozen pipeline, *after* the detection statistic and the molecular-line mask were fixed (snapshot 2026-09-22 08:18 UTC). They are reported here and not merged into the science sample: every statistic above rests on a denominator and a candidate list fixed before the data were examined, and merging would destroy that property. The frozen sample is unchanged.

The extension is 463 windows toward 16 stars in 16 systems, 67 fine-channel and 396 coarse, 57.6 GHz of frequency of which 13.2 GHz is Class A. Every block is repeat coverage of a star and a tuning the survey had already searched, so it adds no star, no system and no frequency; what it adds is epochs. 32 further windows in 8 blocks were withheld under the survey’s own rule, being ϵ Eridani Band 6, where the star sits near 0.7 primary-beam FWHM and the Gaussian beam form used for the correction does not hold.

The epoch gain is small. The number of systems with

TABLE 11

THE PROSPECTIVE EPOCH EXTENSION AGAINST THE FROZEN SCIENCE SAMPLE. EVERY FROZEN-COLUMN NUMBER IS COMPUTED FROM THE RELEASED CATALOGUE BY THE SAME DEFINITION APPLIED TO THE EXTENSION, AND THE GENERATOR CROSS-ASSERTS THEM AGAINST THE MACROS THE REST OF THE PAPER USES, SO THE TWO COLUMNS ARE LIKE FOR LIKE. THE EXTENSION IS *not* POOLED INTO THE SAMPLE: IT WAS SEARCHED AFTER THE STATISTIC AND THE MASK WERE FIXED. THE LAST THREE LINES CARRY THE RESULT — THE ATTRIBUTED EVENTS ARE ALL THE β PICTORIS DISC, THE UNATTRIBUTED COUNT IS ZERO, AND THE ADD-ONE RANK IS DISPLACED FROM 0.5 BY ABOUT AS MUCH AS IN THE SAMPLE ITSELF, ON WINDOWS THE STATISTIC HAD NEVER SEEN.

	frozen sample	extension
execution blocks	404	115
windows	1655	463
Class A	403	67
stars	90	16
systems	82	16
threshold crossings	75	13
rank-first windows	16	4
stage-1 events	13	3
attributed to CO	9	3
unattributed	4	0
median add-one rank	0.446	0.409

more than one searched epoch rises from 55 to 56 of 82, the number with a single epoch falls from 27 to 26, and 1 system acquires a second epoch: the extension deepens the time coverage of systems that already had several rather than repairing the confirmation gap of §6.6.

Three results carry. The departure from exchangeability reproduces on data the statistic had never seen: the median add-one rank is 0.409 over 463 windows where 0.5 is expected, with $D = 0.116$ at $p = < 0.01$. That is a third independent measurement of the effect §C.1.1 and §C.1.2 report, and it is why the spatial rank is used as a screen and never as a significance. The positive control recovers: 3 windows reach stage 1 and all 3 of them are the β Pictoris disc in CO(1–0), at $T_\star = 23.17$, 19.39 and intermediate against ring maxima below 6.21, within 8.8–9.0 MHz of the transition, in three *different* previously unsearched execution blocks. And no new unattributed event appears: 13 threshold crossings (2.81 per cent of windows), 4 rank-first against 1.3 expected, and 0 unattributed stage-1 events against 0.21 expected, that expectation computed on the 78 extension windows whose control maximum reaches the trigger (§5.2). The count of unattributed events in this paper therefore remains 4 (Table 11).

An astrophysical positive control: β Pictoris CO

A search that finds nothing must show that it could have found something, and two demonstrations are needed. The injection campaign of Appendix G establishes that the pipeline recovers a drifting narrowband carrier, which is the searched morphology. β Pictoris CO, resolved, several km s^{-1} wide and effectively undrifting, establishes the complementary property: a real, spatially localised, narrow spectral feature at a stellar position, recovered blind by the frozen pipeline, in two transitions, in 8 execution blocks, over 8.4 yr from 2013-10-06 to 2022-03-03, and at a drift consistent with none — the Band 3 crossings agree across blocks to 1.00 channel once each block’s own tuning is removed, which bounds any apparent drift far below the grid the window searches. It

is therefore a positive control for localisation and recurrence, not for the carrier morphology. HD 48370 is the complementary case, an extended foreground line that fills the annulus and so is rejected by the same spatial statistic that promotes β Pictoris; CP–72 2713 and HD 23484 (Table 26) are the third regime, marginal single cells with no recurrence in the repeats available.

β Pictoris is also the site of a radio detection of a very different kind. Ortiz Ceballos et al. (2026) report recurring, highly circularly polarised bursts together with persistent emission from the planet β Pic b over 0.85–3.5 GHz with MeerKAT, which they attribute to the electron-cyclotron maser instability and use to infer a field of $\gtrsim 1.25$ kG at the planet. That emission is inaccessible to this survey through the physics of the mechanism rather than through our depth. Cyclotron maser emission cannot exceed the electron cyclotron frequency at its source, $2.8 \text{ MHz G}^{-1} \times B$; the quoted field reproduces their own 3.5 GHz upper edge, and reaching the lowest frequency searched here, 89.6 GHz, would require 32 kG, about 26 times the β Pic b field, with Band 7 requiring 99 times it. Their detection sits 1.4 decades below our coverage, so nothing in this paper bears on it and our null is not evidence against it. It does settle one argument of §4 in both directions. Strong circular polarisation is measurable from a planet-mass companion in one of the very systems searched here, so the polarisation discrimination we forgo is real rather than hypothetical; and it is not by itself a mark of artificiality, since a natural magnetosphere produces it. Polarisation would separate compact coherent emission from the thermal dust and line emission that fills this archive, not engineering from magnetospheres.

The working behind the attribution is as follows. The Band 6 window is a line tuning, 53.9 MHz wide at 15.26 kHz, so it holds 3533 channels like a wide window and searches 1944 trial drift rates against a survey median of 4, the largest trials load in the survey at $\eta_{\text{drift}} = 422$: this rank is harder to earn than a typical one, not easier. Carried through the frame chain of §F.1, the Band 3 crossing lies -0.39 and the Band 6 crossing -0.29 km s^{-1} from the stellar systemic velocity, in the two lowest CO rotational transitions of a star with mapped circumstellar CO (Matrà et al. 2017; Dent et al. 2014), and a third, unflagged window six years earlier shows the same component at -0.46 km s^{-1} ; all three lie within 0.46 km s^{-1} of systemic (Table 9). The full frame audit, the flux ratio and the excitation argument are in Appendix K, with the per-window numbers in Table 20.

Attributing the pair to CO does not diminish it. Against the survey background the Band 3 peak $T_\star = 30.76$ is matched or exceeded by 0 of the 1655 per-window control maxima, and the Band 6 peak $T_\star = 11.68$ by 16 of them. Those are ranks in a reference set this paper shows is not exchangeable with the stellar position, so they are quoted as ranks and not as p -values; the attribution rests on the velocity agreement and the recurrence, neither of which needs them, and the alternative reading needs a transmitter placed by coincidence at matched offsets from two CO transitions in one star.

A non-recurrence elsewhere means nothing unless the same machinery recovers a repeat when one exists, so the recurrence test was run on emission already believed in. Every β Pictoris CO feature this survey flagged is recov-

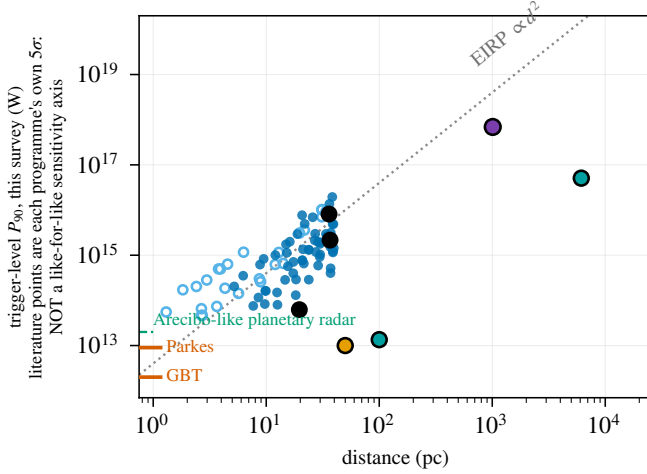


FIG. 11.— EIRP against distance for this survey and for published centimetre-wave searches. Blue: this survey, plotted as the trigger-level P_{90} for the deepest window per star; orange: Enriquez et al. (2017); aqua: Margot et al. (2023); violet: Mason et al. (2024); left-edge ticks: Price et al. (2020). Grey dotted: $\text{EIRP} \propto d^2$ at constant received flux. Green dashed: the Arecibo 2.38-GHz benchmark, a power scale only. The vertical coordinates are commensurable as total powers for a sub-channel carrier; what differs between programmes is the detection threshold and the completeness level each adopts. This survey’s value is a 90 per cent recovery through the complete selection, a stricter convention than the nominal thresholds most of the comparison programmes quote.

ered in an independent execution block: the Band 3 unit is exhaustive at 4 of 4 blocks with the line present in every one, the Band 6 recurrence unit likewise, and the stage-1 Band 6 window recurs in its unit’s second block at $T_* = 8.95$. Three things make this a recurrence and not a coincidence: the peaks agree to about one channel once each block’s own tuning is removed, the barycentric velocities agree to 0.7 km s^{-1} of systemic across the full baseline, and the lines are $10.8\text{--}15.2 \text{ km s}^{-1}$ wide, an order of magnitude broader than the single channel a carrier occupies. Appendix K gives the block-by-block numbers, including the warning that the spatial screen loses power when the CO belt fills the control annulus.

NOTATION

FREQUENCY CONVENTIONS, LINEWIDTH READING, AND SMEARING

Frequency and velocity reference frames

Every frequency axis here — searched spectra, tabulated ranges and crossing frequencies alike — is *topocentric* as delivered by the ALMA correlator, with no velocity-frame shift applied at any stage. Velocities enter only in the vetting stage’s line mask. The stacking is drift-aligned and inverse-variance weighted: coherent in frequency trajectory, not in electric-field phase.

The native channel width is also the linewidth convention for every EIRP threshold. Channel dilution spreads a sub-channel carrier’s total flux across the channel, so $\text{EIRP}_{\text{trig}}$ is the minimum *total* power of a channel-confined carrier, independent of the transmitter’s own width below the channel, and never a spectral power density. The Hz-resolution surveys’ limits are total powers in the same sense, each the best case of its own channelisation, which makes the axes of Fig. 1 commensurable. One number fixes the scale, illustratively and not

TABLE 12

NOTATION. EVERY POWER SCALE AND EVERY TERM THIS PAPER USES IN A PARTICULAR SENSE IS DEFINED HERE; LATER SECTIONS USE THEM WITHOUT RE-EXPLANATION. THE ABSTRACT AND CONCLUSIONS QUOTE P_{90}^{sel} ; THE RELEASED CATALOGUE CARRIES ALL OF THE POWER QUANTITIES PER WINDOW.

Quantity	Meaning and use
T_*	the search statistic at the stellar position (Eq. 3): a matched-filter amplitude divided by a noise estimate. Dimensionless, and <i>not</i> quoted in units of σ , its null distribution being neither assumed nor shown to be Gaussian
T_{trig}	the trigger, $T_{\text{trig}} = 5$: a fixed level on T_* , not a tail probability. Throughout, σ is reserved for genuine noise scales — injected amplitudes, image rms, fit uncertainties (§4.1)
<i>The power hierarchy, in order.</i>	<i>Only the last is a sensitivity.</i>
P_{trig}	the power at which the search statistic reaches T_{trig} in one native channel: an internal trigger, not a detection significance and not a completeness level (§4)
C_{resp}	spectral-response correction, $\times 2.29$ median, window-specific (Appendix F)
C_{smear}	intra-integration drift smearing, $1/\eta_{\text{smear}}$ (Appendix F)
P_{eff}	$P_{\text{trig}} C_{\text{resp}} C_{\text{smear}}$: the power a transmitter must radiate to reach the trigger (Fig. 5). Every EIRP quoted here is the <i>total</i> radiated power of an unresolved monochromatic carrier, not the power falling in the peak ALMA channel
P_{50}, P_{90}	the injected amplitudes the <i>trigger</i> recovers half and nine-tenths of the time, $1.2 P_{\text{eff}}$ at 90 per cent: completeness of one stage, not of the selection (§G.1, Appendix G)
P_{90}^{sel}	the amplitude the complete automated selection recovers nine-tenths of the time, trigger <i>and</i> control-ring gate together (§5.3): the headline sensitivity, in which every limit in the abstract, discussion and conclusions is expressed
Class A/B	fine-channel / coarse-channel windows; different experiments, never pooled in a completeness statement (§4, Table 1)
dwel	recovery against the fraction of the track a signal is present (§G.1)
candidate	a stage-1 flagged window that also recurs or is independently confirmed; this survey reports none (§4.1)
pseudo-star	an inner-annulus control probe ranked against the others; it contains no star, so a first rank is a false alarm by construction (§C.1.2)
debit $m(0)$	the radial correction subtracted from the stellar statistic before ranking (§C.1.2)
stratum	a subset of windows sharing an instrumental property (band, array, channelisation) analysed separately (Appendix G)
ring max	the largest control statistic in a window’s annulus (§4.1)
R_σ	the star’s robust residual scale over the controls’ median; unity means no noise peculiar to the stellar position (§K.1.2)
hold-out	blocks withheld from the analysis by a rule fixed in advance, used only to check a threshold and never to set one (§C.1.1)
drift	the rate at which a carrier’s observed frequency changes, $\dot{\nu} = \nu a/c$ for line-of-sight acceleration a . Quoted in Hz s^{-1} or kHz s^{-1} for one window, and in $\text{Hz s}^{-1} \text{ GHz}^{-1}$ ($= \dot{\nu}/\nu$, so a/c)

achievably: the median Class A channel is 488 kHz, so rechannelising the same data to the 2.79 Hz of a dedicated narrowband search would, on $P_{\min} \propto \sqrt{\Delta\nu}$, divide the threshold by 418 and put the median Class A selection power at $\sim 6.8 \times 10^{12}$ W. ALMA cannot do this: the correlator delivered the channelisation the archive holds, and no reprocessing recovers resolution that was never recorded.

Every spectral window of an observation enters the spectral-carrier search, capped at 8 per target to bound compute cost; the cap bound one target/band, 61 Vir Band 7, and three further targets are represented by their deepest single window and marked in the released catalogue. Each searched window yields its own EIRP_{trig} row, and Figure 1 plots only the deepest per star.

Primary-beam response. Carrier-lane thresholds and continuum fluxes are corrected for ALMA’s primary-beam response at the star’s offset from the phase centre, negligibly for most targets (median factor 1.00003, 90th percentile 1.002), though 87 of the 1655 retained windows exceed 2 per cent and 73 exceed 10 per cent. A window is kept only where that response is at least 0.5, a correction of at most $\times 2$, since at low response a small error in the beam model becomes a large error in inferred flux. The offsets are bimodal and the floor sits in the gap: the last retained window has response 0.55 at $0.46 \theta_{\text{PB}}$ (LP 736-15 Band 7, $1.81\times$), the first window beyond it 0.35 at $0.62 \theta_{\text{PB}}$ ($2.88\times$), and nothing lies between, so the rule binds prospectively and excludes none of the 1655 released windows. Beyond it are the 12 ϵ Eri Band 6 windows at $1.0\text{--}3.5\times$, also past the first sidelobe transition where the Gaussian form fails; the two criteria are independent and select the same windows. They are *withheld*, enter no number here and are absent from the released catalogue, although under the nominal rule they would have covered $3.7 \times 10^{13}\text{--}1.1 \times 10^{15}$ W at 3.22 pc, among the deepest thresholds in the survey but resting on a beam model that cannot be defended. The annulus geometry keeps the pipeline’s own $\theta_{\text{PB}} = 1.22 \lambda/D$, an Airy first-null coefficient; recomputing it on ALMA’s FWHM $1.13 \lambda/D$ at each window’s own dish diameter moves 79 of 1655 windows by more than 1 per cent, extremes -7.3 to $+10.3$ per cent, and no headline threshold moves.

Transmitter power benchmarks

The benchmark is frequency-matched and purely geometric. For a filled circular aperture of diameter D radiating P_{tx} at wavelength λ ,

$$G = \frac{4\pi A_{\text{eff}}}{\lambda^2}, \quad \text{EIRP} = P_{\text{tx}} G, \quad (\text{F1})$$

so a 12-m aperture radiating 1 MW at 230 GHz with a realistic aperture efficiency of 0.7 has $G = 5.9 \times 10^8$ and an EIRP of 5.9×10^{14} W; a perfect aperture, $\eta = 1$, would give 8.4×10^8 and 8.4×10^{14} W, and the realistic case is the one quoted. The median Class A effective threshold, 1.9×10^{15} W, misses such a transmitter by a factor 2.3.

The Arecibo-like planetary radar used for comparison (Drake 1974; Sheikh et al. 2025, EIRP 2.0×10^{13} W) is an 2.38 GHz *power* scale carrying no transmitter model. Gain scales as ν^2 , the same equation at fixed A_{eff} , so the same aperture and transmitter power at 230 GHz would

radiate $2.0 \times 10^{13} \times (230/2.38)^2 \simeq 2 \times 10^{17}$ W. The comparison is between physical transmitter powers, so it uses P_{eff} and not the trigger: that difference carries the deepest window from below the Arecibo figure to 3.7×10^{13} W above it (§6.1).

Reference frames. The line mask’s chain runs topocentric $\rightarrow$ barycentric (Earth ephemeris at the execution-block mid-time) $\rightarrow$ stellar rest frame (catalogued systemic radial velocity), with JPL/CDMS rest frequencies (Pickett et al. 1998; Müller et al. 2005); in the LSR frame it stops at the standard solar motion, not the systemic velocity (§5.2.4). The residual error is the catalogue radial-velocity uncertainty alone, a few km s^{-1} within 40 pc; second-order relativistic terms at the 50 km s^{-1} scale are $\lesssim 5$ kHz at 350 GHz, below the finest native channel searched (15.3 kHz). The β Pictoris and AU Mic attributions use the same chain, so their offsets are comparable to the exclusion tolerance.

Channel widths, and the instrumental response. Channel widths come from the measurement sets’ own spectral window tables (CHAN_WIDTH). A missing “EFFECTIVE_BANDWIDTH” column is not evidence of an absence of online spectral averaging, MS v2 defining no such column, so the quoted channelisations are not claimed to be noise bandwidths.

ALMA applies online Hanning smoothing by default, so the spectral response is ~ 2 channels wide and neighbouring channels are correlated. For the 0.25/0.5/0.25 kernel the coefficients are $\rho_1 = 2/3 = 0.667$ and $\rho_2 = 0.167$, with $\sigma_{\text{smoothed}}/\sigma_{\text{raw}} = 0.612$: the correlation is not confined to adjacent channels, the lag-two term being a sixth. The peak-channel fractions 0.38–0.50 that follow, median 0.44, and the resulting $\times 2.00\text{--}\times 2.67$ optimism of the nominal thresholds are given in §4; the released catalogue carries the nominal threshold and both corrected values per window, and the sensitivity summary quotes the worst case. The injections deposit an unsmoothed delta-in-channel tone, so the recovery curves of §G.1 and Appendix G carry the same bias and the end-to-end check (Eq. G1) cannot see it.

Two measurements fix the smoothing state. The retained per-integration residuals of the second-stage local null (§5) have a measured lag-one spectral autocorrelation of 0.65 against the Hanning prediction 0.667, so that correlation is instrumental and the residual spectra are smoothed and un-averaged. And the archive’s reported effective spectral resolution per window, against each measurement set’s own channel separation, is exactly 2.00 in 1509 of the 1655 windows, in both correlator modes and including 1220 of the 1252 coarse time-division-mode (TDM) ones; the other 146 report a smaller ratio, the signature of online channel averaging, where the peak-channel loss is smaller, and the 0 Band 9 and 10 windows, absent from the metadata snapshot, take ALMA’s default. The blanket factor is right for the great majority of both classes. Thresholds are not rescaled: each is nominal and up to $\times 2.67$ optimistic. The correlation model is truncated at lag two, and the baseline step — a per-integration median in blocks of up to 65 channels — could imprint correlation on longer scales, which the released products retain no residual spectra to measure. Such a term would make the independent-trials count of Appendix J an over-estimate

and the achievable per-window resolution coarser still.

Linewidth convention. The statistic is the single-channel flux density after drift correction, so the quoted $\text{EIRP}_{\text{trig}}$ applies to any transmitter narrower than the native channel and is flat in the assumed linewidth. Two sub-channel qualifications attach: flatness is exact only for a centred tone, a boundary-straddling one splitting its power between two channels at worst by a factor of two; and the channel-to-channel transition follows the correlator’s passband shape, which the frozen products do not retain. Thresholds therefore carry a sub-channel-position systematic of tens of per cent, and EIRP values are quoted to two significant figures.

Transmitters wider than one channel. The baseline filter (§4.1) is flat for features narrower than roughly half its effective width W and falls to zero above W , so the survey’s sensitivity is to channel-confined morphologies and W is the linewidth ceiling: 65–74 channels over the fine class and 30–59 over the coarse one, that is ≈ 32 MHz at 488 kHz and ≈ 0.9 GHz at 15.62 MHz. Within that ceiling, a top-hat transmitter spanning N_{ch} native channels has per-channel flux the total divided by N_{ch} at unchanged noise, so to first order

$$\text{EIRP}_{\text{trig}}(N_{\text{ch}}) \simeq N_{\text{ch}} \times \text{EIRP}_{\text{trig}}(1), \quad N_{\text{ch}} \lesssim W/2, \quad (\text{F2})$$

the rescaling the main text applies to signals broader than one channel.

Worked example: from flux to EIRP. Take UV Ceti (G 272-61B) Band 3, the deepest of its 4 windows in that band (execution block A002_X877e41_X14c1, lower edge 91.565 GHz). Its per-channel rms is 0.241 mJy, so $S_{\text{min}} = 5 \times 0.241 = 1.21$ mJy in one 15.625-MHz native channel, or $1.21 \times 10^{-29} \text{ W m}^{-2} \text{ Hz}^{-1}$ at 2.67 pc, and $\text{EIRP}_{\text{trig}} = 4\pi d^2 S_{\text{min}} \Delta\nu = 1.6 \times 10^{13} \text{ W}$.

Intra-integration smearing. The frequency excursion at the maximum searched drift rate across one integration, $\dot{\nu}_{\text{max}} t_{\text{int}}$, is at most 1.10 channels across the 1655 windows (median 0.001, 90th percentile 0.06). A linear drift smeared by Δ channels within an integration keeps a fraction $\eta_{\text{smear}} = \text{sinc}(\pi\Delta/2)$ of the peak-channel amplitude, so the effective threshold is $\text{EIRP}_{\text{trig}}/\eta_{\text{smear}}$; a pure occupancy argument would retain $\min(1, 1/\Delta)$ instead, giving 1.0 rather than 0.64 at $\Delta = 1$, so the sinc form is the conservative reading. The 9 windows with $\eta_{\text{smear}} < 0.99$, as star, band, (η_{smear} , threshold multiplier), are HIP 10679 Band 6 (0.574, $\times 1.74$), β Pic Band 6 (0.574, $\times 1.74$), HD 172555 Band 6 (0.574, $\times 1.74$), HIP 10679 Band 6 (0.606, $\times 1.65$), HD 172555 Band 6 (0.606, $\times 1.65$), η Crv Band 8 (0.865, $\times 1.16$), η Crv Band 7 (0.932, $\times 1.07$), HD 48370 Band 6 (0.979, $\times 1.02$) and CP-72 2713 Band 6 (0.979, $\times 1.02$); the remaining 446 agree with their nominal thresholds to better than 1 per cent. Compounding the worst smearing penalty with the worst Hanning factor gives $\times 4.6$ on the nominal threshold for those five windows, and no released column carries that product.

INJECTION-RECOVERY SENSITIVITY VALIDATION

Dwell-fraction completeness

How long a carrier occupies one native channel defines the searched class, so recovery is measured against it directly. A carrier was injected into the retained per-

integration spectra of 12 real windows, six coarse and six fine, at amplitudes of 2–20 times the per-integration noise and dwell fractions $f_{\text{dwell}} = 0.10$ –1.00, six realisations each: 3888 trials, released with one row per injection (Table 13). The detection criterion, a logged per-integration S/N of five, re-derived from the released rows agrees with the logged flag 3888 times in 3888 trials, with 0 mismatches across a clean separating gap. Recovery rises monotonically with dwell: pooling amplitudes at or above 4σ , coarse windows recover 100 per cent of continuous carriers, 100 per cent at $f_{\text{dwell}} = 0.25$ and 83 per cent at 0.10, and fine windows 100, 91 and 71 per cent. Appendix G.3 gives the end-to-end check, the campaign’s scope and the representativeness of the injected windows.

The trigger acts on the stacked statistic and the de-drift step is an inverse-variance-weighted sum over integrations, so a carrier of per-integration amplitude A present for a fraction f_{dwell} of a track of N_{int} integrations accumulates to

$$a = A f_{\text{dwell}} \sqrt{N_{\text{int}}} \quad (\text{G1})$$

in units of the per-integration noise. Recovery depends only on a , so binning the released trials in a places all 12 injected windows on one axis; they span a factor of 63 in integration count over 11 targets, two resolution classes and Bands 3, 6 and 7. In the 6 windows whose amplitude grid straddles the trigger, the 50 per cent recovery point lies at $a = 5.3$ –8.3, the trigger itself, independent of channelisation, integration count and band. In the other 6 the weakest injection already stacks to $a = 3.4$, consistent but uninformative about the curve’s position. The pooled curves (Table 13) agree between classes within the sampling noise.

The drift ceiling carries one caveat. The searched value is 12.0 – $13.3 \text{ Hz s}^{-1} \text{ GHz}^{-1}$, line-of-sight $|a| = 3.60$ – 4.00 ms^{-2} , raised above that generic value only in the 56 windows where a known planet’s own bound requires it. It clears Earth-rotation and Earth-orbit drifts by two orders of magnitude but is marginal against a close-in planet on a TRAPPIST-1 b-like orbit, so the limits of §6 are conditional on it (Figure 6).

Interference rejection is an operative gate. The highest frequencies allocated to any satellite constellation are the E-band feeder links of the second-generation systems, 71–76 and 81–86 GHz (SpaceX 2024), below the 89.6 GHz start of the searched union, and ITU Radio Regulations No. 5.340 prohibits all emissions in bands covering 13.3 GHz (11.2 per cent) of it. Allocation tables exclude neither harmonics nor intermodulation products nor near-field emitters, so no exclusion claim rests on one; the operative defences are the control ensemble, the visibility phase test and recurrence, measured in Appendix M.

What was searched, by class. Class A, the fine-channel drifting-carrier search: 26 frequency islands totalling 47.7 GHz between 114.0 and 873.1 GHz, channels 0.02–1.95 MHz, accelerations $|a| \leq 3.6$ – 4.0 ms^{-2} . Class B, persistent channel-confined excess with no drift discrimination: 35 islands totalling $\Delta\nu_B = 90.5$ GHz between 89.6 and 682.8 GHz, overlapping Class A by 20.1 GHz and so adding $\Delta\nu_{B-\text{only}} = 70.4$ GHz, channels 7.81–31.25 MHz. Both cover only the archival epochs and only

TABLE 13

COMPLETENESS MATRIX. (a) WHAT IS MEASURED FOR EACH SEARCHED MORPHOLOGY AND WHERE: ‘e2e’ IS INJECTION INTO VISIBILITIES THROUGH THE UNMODIFIED PIPELINE, ‘POST’ INJECTION INTO RETAINED PER-INTEGRATION SPECTRA AFTER THE BASELINE STEP, AND ‘CFG’ THE NUMBER OF WINDOW CONFIGURATIONS THE CELL SPANS. THE TWO CAMPAIGNS MEASURE DIFFERENT QUANTITIES: 1200 INJECTION TRIALS (APPENDIX G) AND 3888 DWELL TRIALS OVER 12 REAL WINDOWS. THE 0 OF 500 COARSE-WINDOW ENTRY IS A PROPERTY OF THE RECOVERY CRITERION, WHICH REQUIRES THE RECOVERED DRIFT TO MATCH THE INJECTED DRIFT AND SO CANNOT BE SATISFIED ON A CHANNEL WIDER THAN THE WHOLE DRIFT TRACK (§G.1).

Morphology	e2e	cfg	thresh.	dwell	drift
Class A drifting	yes	1	yes ^a	no	yes
Class A near-static	post	6	no ^b	yes ^c	n/a ^c
Class B stationary	yes	6	yes ^d	yes ^e	none
Class B drifting	yes	6	yes ^d	yes ^e	none ^c
Intermittent, either class	post	12	no	yes ^e	by class

^a42 per cent at the trigger, 83 at twice it, one configuration only (AU Mic, Band 6, 488 kHz), transfer to the other 403 Class A windows measured at 0.64–1.13× on 28 configurations (Fig. 12; the breakdown by drift stratum is Table 24). ^bAfter the baseline step only. ^cSub-channel over the track by definition. ^dHalf recovery at stacked $a = 5.3$ –8.3 in the 6 windows straddling the trigger. ^e83–100 per cent (coarse), 71–100 per cent (fine), over $f_{\text{dwell}} = 0.10$ –1.00.

(b) completeness at the trigger, against stacked amplitude a (Eq. G1), pooled over the 12 injected windows; trial counts in parentheses.

stacked a	Class B (coarse)	Class A (fine)
< 3	0% (30)	0% (72)
3–4	0% (12)	7% (42)
4–5	3% (30)	12% (42)
5–6	46% (24)	33% (48)
6–8	58% (60)	70% (96)
8–12	100% (108)	99% (174)
> 12	100% (1680)	100% (1470)

carriers one native channel or narrower, over 113.3 GHz of the 118.1 GHz union after molecular-line masking, at nominal trigger thresholds whose measured recovery is 42^{+16}_{-14} per cent. Inside that box the non-detection is a constraint. Outside it, the exclusions are those of §5.3: the trigger is not a significance; the spatial screen is empirically calibrated, and holds for isolated point sources rather than for beams filled by resolved circumstellar or foreground emission; the 1252 coarse windows carry no drift discrimination and no survey-threshold completeness curve; coverage is the 46 islands of Fig. 4 less the 4.1 per cent of the union removed by the $\pm 50 \text{ km s}^{-1}$ line mask; drift above the ceiling and emission broader than one native channel lie outside the searched morphology; duty cycles below ~ 0.1 are essentially unconstrained; and the search is Stokes- I only, so a per-hand search of the 198 blocks that deliver both parallel hands would gain up to $\times 1.41$ on a polarised carrier.

Amplitude completeness from visibility-domain injections

Synthetic tones of known amplitude were injected into real visibilities, using the extraction code’s own mechanism as far upstream of the statistic as the retained products allow, and the unmodified pipeline’s return measured. Six configurations span three bands and both classes: three coarse Band 7 windows of AT Mic, one coarse window each in Bands 3 and 6 of AU Mic, and

AU Mic’s 488-kHz flagband window with its 49-trial drift grid. Each receives a complex tone at the star position at 4–10× the statistic’s own noise, over drift fractions $f_{\text{drift}} = 0$ –1.0 of the ceiling, on- and half-grid rates and channel-centred and boundary placements: 200 trials per configuration, 1200 in all. The recovery criterion was frozen in advance: $T \geq 5$, peak channel within 3 of the injection, peak drift within one grid step.

Coarse windows recover 0 of 500 injections from $T_{\star} = 4$ to 10 under that criterion, which they cannot satisfy, every in-grid drift being sub-channel over the track so that the reported peak drift is arbitrary; the fine window fails it the same way. Scored on detection alone the same windows recover essentially everything (Appendix G.3). Class B limits therefore bound carriers of any dwell fraction, with the completeness of §G.1; what those windows lack is drift discrimination.

On the fine window, drifting carriers ($f_{\text{drift}} \geq 0.5$, pooled over grid phase and sub-channel placement) are recovered 17, 42, 58, 75 and 83 per cent at $T_{\star} = 4, 5, 6, 8$ and 10, saturating by twice threshold power (Fig. 12). With the tone injected after the baseline step the machinery alone is amplitude-faithful to within $-9/+14$ per cent for channel-centred tones. Boundary-straddling placement costs 0.6–0.8 in the statistic and recovers 0 per cent at the trigger against 83 per cent for centred tones, crossing 50 per cent near $1.4\times$ threshold amplitude. Survey completeness uses the placement-pooled curve, conservative for the boundary-straddling case: $\text{EIRP}_{50} = 1.10 \text{ EIRP}_{\text{trig}}$ and $\text{EIRP}_{90} > 2.0 \text{ EIRP}_{\text{trig}}$, a bound only.

That curve is measured on one window and is not a survey completeness function. Of the 403 Class A windows, 197 (49 per cent) differ from the calibration configuration on at most one axis, the mismatch dominated by band (200 windows) and drift-trial count (167) rather than channel width (30), integration time (87) or beam offset (42). Those are the axes the stratified campaign spans; other target environments and edge-phased drift grids are not spanned at all.

The stratified campaign. A drifting carrier was injected into the retained per-integration spectra of 28 further windows, one per star, spanning Bands 3–10, 15.3–977 kHz channels, 45–485 integrations, 7–1186 drift trials and both the 12 m array and the ACA, and pushed through the pipeline’s own baseline removal and de-drift stack on the search’s drift grid: nine amplitudes, 4032 trials. Recovery is a threshold crossing at the injected frequency and not the global maximum of the grid, since requiring an injection to beat a window’s brightest cell measures that window’s worst artefact. Pooled recovery is 34, 61, 89 and 99 per cent at $T_{\star} = 4, 5, 6$ and 8, with $P_{50} = 4.6\sigma$ and $P_{90} = 6.1\sigma$ (Fig. 12). The spread across configurations sets the transfer bracket: P_{50} runs 0.64–1.13× its median and P_{90} 0.48–1.35× its own, so the assumed 0.5–2× bracket is conservative. Band and array matter least, P_{50} varying only 4.4–4.8σ across bands (Table 24).

Drift rate, resolved. All 4032 trials carry a non-zero drift, drawn over 9–10344 Hz s^{-1} , so the pooled curve is a drifting-carrier curve. A pooled curve could hide a deficit at the edge of the grid, where a carrier crosses the most channels and the de-drift stack has the least signal in any one of them, so Table 24 splits the campaign by drift rate

as a fraction of each window’s own ceiling. P_{90} varies by 10 per cent across the three thirds with no trend, and the outer third is not the worst; at a fixed 6σ injection they recover 89, 84 and 93 per cent. The rate is recovered as well as the frequency: 92 per cent of detections above 6σ return the injected drift trial itself, median error 0.00 grid steps and 0.5 at the ninetieth percentile. Carrier sub-channel phase, stratified separately over 5 positions and 400 trials, is the one axis that does matter, and is why the threshold carries the $\times 2.69$ – $\times 2.02$ response correction rather than a single factor.

How far the grid reaches. 28 configurations are a sparse grid for 403 windows. It spans Bands 3, 6, 7, 8 and 10, channel widths 15.3–977 kHz, 45–485 integrations and both arrays, and every Class A window lies in a band it samples. The configuration needed for the comparison — band, channel width and integration count together — is recorded for 132 of the 403; the peak-frequency freeze that carries the integration count does not cover the 271 blocks searched since, which are never assumed to be inside the grid. Of the 132 that can be judged, 113 lie inside the grid on all three axes at once, 86 per cent of them and 28 per cent of all 403. The remaining 19 are extrapolations, 18 in integration count alone and 1 in channel width. The transfer bracket is measured for the 113 windows and assumed for the rest. The five deepest Class A thresholds are all interpolated, so the headline sensitivity does not rest on an extrapolation.

The spectral response, measured. The $\times 2.29$ correction of §4 is analytic, the injected tones being unsmoothed. A tone injected at fractional position ϕ , split between neighbouring channels as the delivered channelisation requires and convolved with the 0.25/0.5/0.25 kernel, has a peak $\times 2.02$ that of a channel-centred unsmoothed tone of equal total flux at $\phi = 0$ and $\times 2.69$ at the channel edge, median $\times 2.31$ over 400 trials, every point within 1.1 per cent of the analytic value. Both campaigns ship trial-level records.

Dwell campaign: end-to-end check, scope and representativeness

A zero-drift, always-on tone injected into the calibrated visibilities of a coarse Band 7 window at 1, 3 and 10 times the per-integration rms returns $T_\star = 20.5, 65.3$ and 217.7, each within one channel of the injection. The ideal stacking gain over its 639 integrations is $\sqrt{639} = 25$, and the measured 20.5 is that value less weighting and flagging losses. This is both the end-to-end check through the unmodified pipeline and what a coarse window returns when detection alone is scored: Class B detects excess power, and cannot tell whether the excess drifted.

Two distinctions apply. The dwell campaign injects into retained per-integration spectra, after the baseline step, while the end-to-end check injects into visibilities before it; and one cell, the through-pipeline recovery of near-static carriers on Class A windows, is measured only in part.

How representative the injected windows are of the survey is set out in §G.2: the end-to-end demonstration is made on one configuration, about half the Class A windows differ from it on at most one axis, and the stratified campaign measures the transfer to the rest and states which windows that bracket interpolates and which it extrapolates.

TABLE 14

SELECTION FUNCTION OF THE SEARCHED SAMPLE AGAINST A VOLUME-LIMITED REFERENCE CENSUS: GAIA DR3 SOURCES WITH $\varpi > 0$, $\sigma_\varpi/\varpi \leq 0.1$ AND $1/\varpi \leq 40$ PC (17566 OBJECTS, OF WHICH 8594 CARRY A **TEFF_GSPPHOT** ESTIMATE, 49 PER CENT), AGAINST THE 90 STARS WITH AT LEAST ONE SEARCHED WINDOW. TEMPERATURES ARE AVAILABLE FOR 53 OF THE 90 (ALL 90 MATCHED TO THE 168-ENTRY ALMA-COVERED CENSUS, 55 BY EXACT NAME; 53 OF THOSE MATCHES CARRY A TEMPERATURE), SO THE TEMPERATURE PANEL IS NORMALISED OVER CLASSIFIED OBJECTS ONLY. “RATIO” IS THE SAMPLE COLUMN FRACTION OVER THE CENSUS COLUMN FRACTION; > 1 IS OVER-REPRESENTATION.

Property	Census	Searched	Ratio
Distance (all 90 searched stars; 17566 census objects)			
$d \leq 5$ pc	48 (0.3%)	12 (13.3%)	48.8
5–10 pc	240 (1.4%)	13 (14.4%)	10.6
10–20 pc	2069 (11.8%)	18 (20.0%)	1.7
20–30 pc	5391 (30.7%)	19 (21.1%)	0.7
30–40 pc	9818 (55.9%)	28 (31.1%)	0.6
T_{eff} class ^a (53 searched; 8594 census)			
A (≥ 7500 K)	24 (0.3%)	1 (1.9%)	6.8
F (6000–7500 K)	402 (4.7%)	9 (17.0%)	3.6
G (5300–6000 K)	860 (10.0%)	16 (30.2%)	3.0
K (3900–5300 K)	1400 (16.3%)	6 (11.3%)	0.7
M (< 3900 K)	5908 (68.7%)	21 (39.6%)	0.6
Other axes			
Known planet hosts ^b	20 (11.9%)	19 (21.1%)	1.8
Searched stars / systems	–	90 / 82	–

^aBins are temperature bins, not MK types: A ≥ 7500 K, F 6000–7500, G 5300–6000, K 3900–5300, M < 3900 K. Gaia **teff_gspphot** is unavailable for about half the reference census, preferentially for the coolest and faintest objects, so the census M fraction is a lower limit and the earlier-type ratios are correspondingly upper limits. White dwarfs are not separable in this parameterisation and fall wherever their (unreliable) photometric temperature places them.

^bThe Gaia reference census carries no planet information; the census column here is instead the 168-entry ALMA-covered census, of which 20 entries are known planet hosts. The 90 searched stars include 19 hosts of 42 known planets.

The injected carriers do not drift, so what is measured across 12 configurations is the persistent and partial-dwell non-drifting class. On Class B windows that is the searched class; on Class A windows the searched class is the drifting carrier, whose curve rests on the single calibration configuration of §G.2. The stacked amplitude a is a derived combination, justified by the form of the statistic and by the collapse of 12 configurations onto one curve, not measured independently at every (A, f_{dwell}) pair.

PER-TARGET RESULTS

The per-window results ship as the comma-separated catalogue of the Data Availability statement. Its **EIRP_{trig}** column uses the definition of §4, on the same threshold and primary-beam correction as the continuum column, so a row’s two limits are comparable; its per-channel flux-density threshold uses the carrier lane’s per-channel noise rather than the tabulated **rms_mJy**, the two differing by $5 \times G(r)$. Bands 6 and 7 supply 91 per cent of the windows.

THE VELOCITY-SPACE LINE MASK

A one-channel line check is too tight for circumstellar gas: a native channel spans a fraction of a km s^{-1} , while real emission is shifted by many channels (§5.2). The mask tolerance is set by the kinematic budget of circumstellar gas: Keplerian motion where

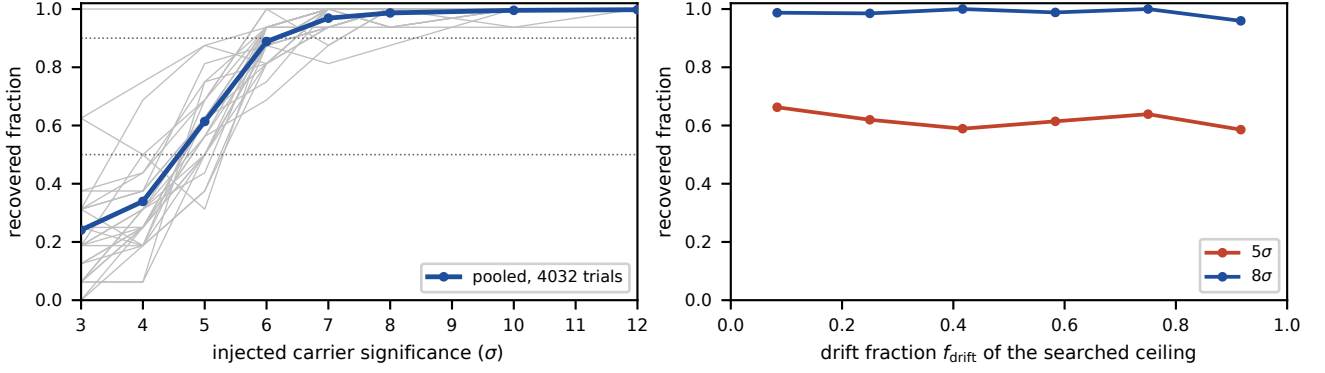


FIG. 12.— Stratified injection recovery. Grey, each of the 28 configurations separately; blue, the pooled curve over 4032 trials. Dotted lines mark the 50 and 90 per cent levels. The spread between grey curves is the completeness transfer, measured rather than assumed. Right, recovery against drift fraction of the searched ceiling at $T_\star = 5$ and 8.

TABLE 15

PER-BAND SURVEY CHARACTERISATION OF THE 1655 SEARCHED WINDOWS. “Sys.” COUNTS UNIQUE STELLAR SYSTEMS SEARCHED IN THE BAND, SO A SYSTEM OBSERVED IN TWO BANDS IS COUNTED IN BOTH ROWS (82 SYSTEMS, 90 STARS IN TOTAL); “EBs” THE CONTRIBUTING EXECUTION BLOCKS, ATTRIBUTED BY THE PROVENANCE OF THE DE-DUPLICATED WINDOWS; $\Delta\nu_{\text{ch}}$ THE NATIVE CHANNEL WIDTH IN MHz (MEDIAN, RANGE IN BRACKETS); “UNION” THE UNIQUE SKY FREQUENCY SEARCHED IN THE BAND AFTER MERGING OVERLAPPING WINDOWS; $\text{EIRP}_{5\sigma}$ THE MEDIAN NOMINAL THRESHOLD ACROSS THE BAND’S WINDOWS; $\dot{\nu}_{\text{ceil}}$ THE RANGE OF DRIFT-RATE CEILINGS IN Hz s^{-1} PER GHz OF CARRIER FREQUENCY. THE “ALL” ROW IS COMPUTED OVER THE WHOLE SAMPLE, NOT SUMMED DOWN THE COLUMNS: EBs AND SYSTEMS RECUR ACROSS BANDS AND THE FREQUENCY UNIONS ARE DISJOINT BY CONSTRUCTION.

Band	Sys.	EBs	Win.	$\Delta\nu_{\text{ch}}$ (MHz)	Union (GHz)	Med. $\text{EIRP}_{5\sigma}$ (W)	$\dot{\nu}_{\text{ceil}}$
3	8	27	112	15.63 (0.244–15.63)	20.6	1.3×10^{14}	12.0–13.3
4	1	1	4	15.63	6.9	2.6×10^{13}	12.0
5	1	1	4	15.63	6.9	4.4×10^{13}	12.0
6	55	265	1072	15.63 (0.015–31.25)	29.4	5.2×10^{14}	12.0–13.3
7	35	102	427	15.63 (0.061–15.63)	26.7	3.6×10^{15}	12.0–13.3
8	3	6	24	15.63 (0.061–15.63)	6.7	4.2×10^{16}	12.0
9	1	1	4	15.63	6.9	4.3×10^{15}	12.0
10	1	1	8	0.977	13.8	2.7×10^{16}	12.0
All	82	404	1655	15.63 (0.015–31.25)	118.1	7.1×10^{14}	12.0–13.3

TABLE 16

BAND-BY-BAND COST OF THE MOLECULAR-LINE EXCLUSION. UNION: UNIQUE SKY FREQUENCY SEARCHED IN THE BAND. MASKED: THE PART OF IT INSIDE A $\pm 50 \text{ km s}^{-1}$ STELLAR-FRAME TUBE AROUND A CATALOGUED TRANSITION OF ONE OF THE MASKED SPECIES. LEFT: WHAT A TECHNOSIGNATURE SEARCH CAN STILL OCCUPY. THE MASK IS EXCLUDED SEARCH SPACE AND NOT A VETO (§5.2.4); THE LOSS IS CONCENTRATED IN BAND 6, WHERE $\text{CO } 2 \rightarrow 1$ SITS.

Band	Union (GHz)	Masked (GHz)	Left (GHz)	Lost (%)
3	20.64	0.272	20.37	1.3
4	6.94	0.049	6.89	0.7
5	6.94	0.065	6.87	0.9
6	29.45	1.374	28.07	4.7
7	26.74	1.589	25.15	5.9
8	6.68	0.164	6.52	2.5
9	6.94	0.454	6.48	6.5
10	13.80	0.844	12.96	6.1
All	118.13	4.811	113.31	4.1

these molecules survive ($\lesssim 13 \text{ km s}^{-1}$ beyond 10 au), linewidths $\lesssim 1 \text{ km s}^{-1}$, and an uncorrected barycentric term bounded by $\pm 30 \text{ km s}^{-1}$. The result is empirically conservative: the largest attributed circumstellar component (β Pic, within 0.46 km s^{-1} of systemic) sits at 0.9 per cent of it, and *zero* crossing frequencies lie $13\text{--}50 \text{ km s}^{-1}$ from their nearest *masked* transition (the wider catalogues are another matter: §5.2), so a $\pm 13 \text{ km s}^{-1}$ Keplerian-only mask would have suppressed and released the same windows: the wider tolerance costs no yield and buys robustness against radial-velocity catalogue error. The alternative selection quoted in §5.2.4 is every Splatalogue transition over the 46 islands with

$E_u \leq 150 \text{ K}$ and $\log A_{ij} \geq -5$, 5,114 of 235 carriers; HCN, CS, SiO, CN and H_2CO are in the adopted list already, and no coverage figure and no disposition changes.

The repaired mask. The four defects and their repair are in §5.2.4. The rounding defect is why the released $\text{CO}(1\rightarrow 0)$ offset column, referred to the frozen mask’s rounded 115.271000 GHz entry, reads 0.202 MHz (0.53 km s^{-1}) smaller than the laboratory-referred value of Table 20. A cheaper rule exists: the archive records the rest frequency each execution block was tuned to, so masking each window’s own tuned line, inside an LSR-frame tube of a few $\times 10 \text{ km s}^{-1}$, catches the β Pictoris, HD 48370 and [C I] windows automatically.

The crossing ledger, once: of the 75 windows with an on-star crossing, 13 also beat their control ring and three of those are accounted for by the mask. Two further crossing windows fall inside the mask without beating their ring (the unflagged β Pic Class A window at -0.46 km s^{-1} ; HD 285968 at $+8.7 \text{ km s}^{-1}$, already rejected by its ring, maximum 10.9 against a star peak of 6.5), and the remainder lie far outside any circumstellar tolerance (AU Mic $+379$, LHS 1140 $+115 \text{ km s}^{-1}$). The other 16 are ordinary noise crossings: in each, the window’s own control ring reaches or exceeds the stellar value (ring-maximum median 5.84 against a crossing median 5.18), the expected Gaussian-tailed population of moderate $T_\star = 5\text{--}5.5$ crossings with comparably strong peaks among 512 controls. But the transitions that make astrophysical false positives cheap are also natural ref-

erence frequencies for a deliberate communicator (Cocconi & Morrison 1959), so the released products keep a known-line-coincident list: every on-source threshold crossing inside the masked intervals, with its offset from each nearby transition.

FALSE-ALARM ACCOUNTING

Table 17 collects every false-alarm expectation quoted in this paper. Only the last row conditions on the trigger, and it alone is compared with an observed count (§5.2); the other six predict rank-first windows and are diagnostics of the pipeline.

The per-window rank-first rate is $1/513$ and the survey expectation $1655/513 = 3.23$. The floor falls only if the control ensemble grows, which the primary beam bounds. Windows are not independent trials either, grouping into 404 execution blocks that share weather, calibration and correlator setup. Drawing one control index per block and applying it to all of that block's windows gives the clustered distribution in closed form: mean 3.23 rank-first windows and, under the trigger, 0.92 stage-1 flagged windows, with $P(\geq 1) = 0.60$ against the 4 unattributed events observed. Clustering changes little because the *position* of the control maximum is not shared across a block's windows.

Table 18 gives the trials budget in cells and decomposes the on-star excess. A window holds 177×6869740 channel $\times$ drift cells (median 468), the survey 1.32×10^8 in total. Correlation cannot move the expected *number* above the trigger, which is Np whatever the correlation, only its clumping, so the excess over an iid-Gaussian budget decomposes into two multiplicative parts: cell multiplicity within a crossing window, a factor 3.0 which is what an over-count of 2.67 on the frequency axis predicts, one real threshold excursion being counted about three times; and the count of crossing windows, a factor 1.9 whose one-sided Poisson probability is < 0.01 . Once the four dispositioned windows are set aside, the residual excess is consistent with chance at that level.

The control ensembles calibrate the budget without a Gaussian assumption. The same arithmetic predicts 477 of the 1655 windows carrying a control above T_{trig} and 471 do, agreeing to 1 per cent, with per-window control maxima tracking the iid-Gaussian prediction at a median ratio 0.996 (5th–95th percentile 0.90–1.12). Grid cells are not independent trials: Hanning smoothing correlates neighbouring channels at $\rho_1 = 2/3$ and $\rho_2 = 1/6$, and the drift grid is oversampled two to one, giving the combined over-count 5.3 of Table 18. Correcting for it leaves the observed control maxima +6 and +8 per cent above prediction in the fine and coarse classes independently. Non-Gaussianity of that size is why the operative statement is the empirical rank and not the Gaussian budget. The residual on-star excess is not zero, runs one way and carries the same sign at the controls, so it is reported as an open systematic: the crossing list is dominated by CO(2 $\rightarrow$ 1) at 230.5 GHz toward disc hosts, and a residual excess of chance crossings at the tens-of-per-cent level cannot be excluded.

All of this fixes what a single window can ever mean. The rank floor is $1/(N_{\text{ctrl}} + 1) = 1.95 \times 10^{-3}$ against a survey-wide Bonferroni scale of 3.0×10^{-5} , a factor 65, and correlation widens it: with $N_{\text{eff}} \simeq 30\text{--}43$ indepen-

dent spatial trials the ensemble resolves only 0.023–0.032. Drawing more controls would not help, the primary beam bounding the annulus. This is the design's central limitation, and is why authentication rests on recurrence and astrophysical attribution rather than on rank.

The control geometry uses a 12-m θ_{PB} for every window, so the 1054 ACA 7-m windows carry a known geometric error. It runs in the conservative direction: on the true 7-m beam those annuli span 0.08–0.46 of θ_{PB} at a median area-weighted gain of 0.75 against 0.47 for the 12-m windows, so exchangeability in flux density improves there. Table 19 gives the two strata side by side. The 12m stratum holds 9 of the 13 stage-1 flagged windows, the one lost being HD 48370's foreground CO, and the deepest trigger and the deepest P_{eff} are both 12 m windows, so no survey-level statement changes. Re-extracting the ACA windows on the correct 7 m beam would remove the error rather than bound it, and needs the raw visibilities and a full re-search. All 4 unattributed events lie outside the β Pictoris windows, against 3.13 expected there.

Rank calibration and exchangeability

The drift grid and the thousands of channels apply identically at the star and at every control, so the per-window trials factor is absorbed empirically: under exchangeability one designated position of 513 ranks first with probability exactly $1/513$ whatever the noise distribution, and a heavy tail cannot by itself produce an excess of rank-first windows. The survey expectation is 3.23 and 16 are observed; the execution-block clustered permutation returns $P(\geq 16) = 2.2 \times 10^{-1}$, and the excess is astrophysical, celestial line emission sitting at the stellar position in 3 of the 16 (Table 9) and leaving 2 against 3.23 expected. A prespecified Monte Carlo that promotes one control per window to pseudo-star and runs it through the operative gates expects 0.20 flagged pseudo-target windows, in agreement.

Ranking against 512 controls cannot resolve a probability below $1/513$, so a local scramble test carries the four stage-1 windows past that floor. It scrambles each integration independently in channel and re-runs the identical search with the same noise map, weights and drift grid, destroying anything coherent along a drift track while preserving each integration's own spectral autocorrelation and trials factor. Three of the four pass validation. The two β Pictoris windows behave as positive controls, no draw of 10^5 reaching T_* in either ($p < 1 \times 10^{-5}$ and $p < 5 \times 10^{-5}$); for CP–72 2713 the null resolves the value itself, 157 draws of 10^5 equalling or exceeding T_* , with an extreme-value fit to that window's own control maxima agreeing to a factor 1.3. HD 48370 Band 6 fails outright, 73 per cent of its real controls exceeding the maximum of 10^5 null draws, because the field carries extended emission and imaging residuals that are coherent across integrations and that the scramble destroys by design, so no scrambled-noise probability is quoted there. That failure follows from retaining full dynamic spectra for only 8 of the 512 control positions. No number quoted here rests on a parametric extrapolation of a null tail, and the diagnostic has not been re-run on the windows the completed sweep added (§L).

Exchangeability is the statistic's central assumption.

TABLE 17

EVERY FALSE-ALARM EXPECTATION QUOTED IN THIS PAPER, AND WHAT EACH ONE PREDICTS. THE FIRST SIX PREDICT THE NUMBER OF WINDOWS IN WHICH A NO-SIGNAL POSITION WOULD *rank first* AMONG THE CONTROL ENSEMBLE; THEY DIFFER IN SAMPLE, DENOMINATOR AND WHETHER THE MEASURED TAIL EXCESS AND THE WITHIN-BLOCK CLUSTERING ARE INCLUDED, AND THEY ARE PIPELINE DIAGNOSTICS. A STAGE-1 EVENT ALSO REQUIRES THE POSITION TO REACH THE TRIGGER, SO ONLY THE LAST ROW, WHICH IMPOSES THAT CONDITION, MAY BE COMPARED WITH AN OBSERVED COUNT; IT IS THE COMPARISON USED IN THE ABSTRACT, IN §5.2 AND IN THE CONCLUSIONS. SQUARE BRACKETS GIVE A BOOTSTRAP INTERVAL ON THE *expectation*, ROUND BRACKETS THE 95 PER CENT INTERVAL OF THE PREDICTED *count* UNDER THAT EXPECTATION.

Sample	Windows	Rate per window	Predicts	Expected	Obs.
All searched windows, exchangeable	1655	$1/(512 + 1)$	rank-first windows	3.2	–
Distinct (block, window) datasets	1616	$1/(512 + 1)$	rank-first windows	3.1	–
All windows, measured tail rate	1655	$1.4/(512 + 1)$	rank-first windows	3.9	–
All windows, block-resampled	1614	$1.4/(512 + 1)$	rank-first windows	4.4	–
Pseudo-star first-rank rate	1655	measured on inner probes	rank-first windows	4.7	–
Class A windows, block-resampled	394	$1.4/(512 + 1)$	rank-first Class A windows	1.1 [0.5–1.4]	–
Class A windows, block-resampled, trigger required	394	$1.4/(512 + 1)$ if $\max c \geq 5$	unattributed stage-1 events	1.07 (0–4)	4

TABLE 18

THE ON-STAR CELL EXCESS, DECOMPOSED. TOP: WHAT IS OBSERVED AGAINST AN IID-GAUSSIAN BUDGET. MIDDLE: THE TWO MULTIPLICATIVE PARTS OF THAT FACTOR. BOTTOM: THE SAME CORRELATION CORRECTION APPLIED TO THE CONTROL MAXIMA, WHICH ARE MEASURED IN THE SAME UNITS AND CALIBRATE THE BUDGET EMPIRICALLY.

Quantity	Value	Factor
On-star cells $\geq T_{\text{trig}}$	229	
in the stage-1 windows	174	
remaining, unattributed	55	
iid-Gaussian expectation (cells)	37.8	1.5
Cells per crossing window (obs./exp.)	3.4/1.1	3.0
predicted by Hanning correlation	2.67	
Crossing windows (obs./exp.)	62/33.2	1.9
one-sided Poisson p	<0.01	
Frequency-axis over-count $1 + 2p_1 + 2p_2$	2.67	
Drift-grid oversampling	2	
Combined cell over-count		5.3
Control max., fine: pred. $\rightarrow$ corrected	5.74 $\rightarrow$ 5.45	
observed	5.79	+6 %
Control max., coarse: pred. $\rightarrow$ corrected	4.47 $\rightarrow$ 4.10	
observed	4.44	+8 %

TABLE 19

ARRAY-SPLIT ROBUSTNESS. THE CONTROL GEOMETRY USES A 12-M θ_{PB} FOR EVERY WINDOW, SO THE ACA 7-M STRATUM CARRIES A KNOWN GEOMETRIC ERROR. EXPECTED RANK-FIRST COUNTS ARE $N_{\text{win}}/(512 + 1)$ UNDER EXCHANGEABILITY, AND ARE THEREFORE LOWER BOUNDS: THE MEASURED TAIL RATE IS A FACTOR 1.4 HIGHER (§K.1.3), WHICH BRINGS THE 12 M STRATUM’S 9 OBSERVED WITHIN ONE OF EXPECTATION. NO DISPOSITION AND NO HEADLINE THRESHOLD DEPENDS ON THE ACA WINDOWS.

	12 m	ACA 7 m
Windows	601	1054
Execution blocks	139	265
Stars; systems	70; 62	38; 38
Threshold crossings	22	53
Rank-first windows	9	7
expected	1.17	2.05
Stage-1 outliers	9	4
Stellar-rank $D(p)$	0.027 (0.77)	0.10 (<0.001)

For a debris-disc archive its dominant failure is resolved astrophysical emission filling the annulus, seen in HD 48370’s CO ring and in β Pictoris’ recurring Band 6 window, where the ring outshines the star by $1.5\times$. That suppresses a star-exceeds-ring outcome instead of manufacturing one, so it costs sensitivity and nothing else, and the repeat block of the *flagged* Band 6 window measures the cost (§D.1.2). The primary beam is settled by construction, the statistic being formed on *uncorrected*

amplitudes whose thermal noise does not depend on position with $\sigma(t, \nu)$ one scale shared by the star and every control (§4.1). The 513 positions are therefore approximately exchangeable under a noise-only null, a statement that covers the thermal noise and the beam but not residual calibration or continuum systematics, which can occur preferentially where the pointed science target sits.

Four measurements bound what is left, and all four are null. Ranking each control against the other 512-1 by the rule used for the star gives 220 672 pseudo-star ranks, uniform to the resolution the ensemble can express ($p = 0.37$) in every band and at both channelisations. That geometry cannot by itself detect a centre-versus-annulus asymmetry, so the test that can is the two-sample comparison of the 1655 real stellar ranks against the pooled pseudo-ranks, which returns $p = 0.38$: on these data the stellar position behaves like one more position in its own annulus, though out of sample it does not (§C.1.1). The Spearman correlation of control statistic against reconstructed radius has median 0.006 over the 1165 crossing-free windows, bounding any systematic radial suppression of the controls below 13 per cent in the mean. And the star’s own residual variance is measured directly against the controls’ in §K.1.2. Per-control noise records and gain are absent from the frozen products, so those two stratifications are bounded only through these rank distributions.

One stratum departs. The fine-channel windows fail the $U(0, 1)$ rank test at $p0.015$ ($n = 403$), because 13 of them contain an on-star peak above their own control maximum against 0.79 expected, and those are precisely the flagged windows of Table 9. The coarse stratum, three times larger and predicting 2.44 such windows, contains 0: every stage-1 flagged window in the survey is drift-resolving. A 15.6–31.25 MHz channel dilutes a narrow circumstellar line, so the class that can resolve one is the class that finds it, and the same reasoning says a genuine narrowband emitter would appear in Class A first. The departure is nonetheless genuine, and it survives removal of every fine window carrying a stellar crossing (Fig. 13), so fine-window rank probabilities are optimistic for the class as a whole and not only where a crossing sits. This is the quantitative reason the spatial comparison is used as a prioritisation and never

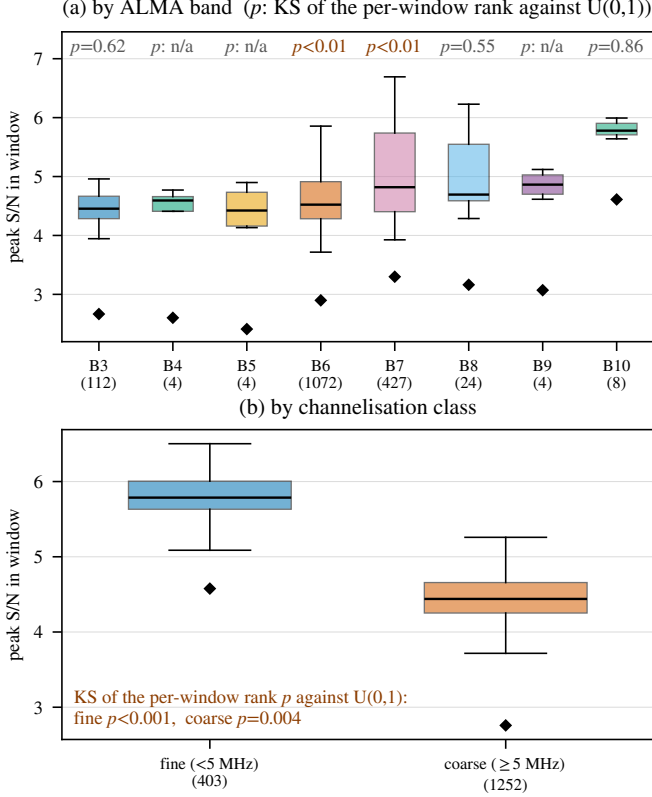


FIG. 13. — Exchangeability of the 512-position control ensemble by stratum, over 1655 windows. Boxes give the per-window control maximum, black diamonds the median on-star peak, and p the Kolmogorov-Smirnov test of the per-window rank against $U(0,1)$. (a) by ALMA band, with the window count under each label; (b) by channelisation class, fine (< 5 MHz) against coarse. The ensemble tracks the on-star statistic in every band, and the fine class sits higher than the coarse, 5.79 against 4.44, as its larger trial count predicts. Both depart from uniformity: fine $D = 0.150$, $p < 10^{-4}$; coarse $D = 0.050$, $p = 0.004$. The departure is not the flagged windows’ doing — removing all 71 fine crossings lowers the statistic only to 0.103 — and §C.1.2 identifies the radial noise gradient as the cause.

as a test of the null (§4.1).

A fully symmetric region-max variant, which maximises identically at the star and at every control, can be run on only seven locally reprocessed windows, the frozen release not carrying the products it needs; in all seven the star is exchangeable with its ring (Appendix K). The single-position calibration above remains the release-wide criterion.

ANALYSIS PROVENANCE AND ROBUSTNESS CHECKS

The β Pictoris frame audit

Table 20 carries the frame chain for the line-attributed windows: frequencies match the catalogued rest values to the chain’s stated tolerance, velocities are concordant across epochs and transitions, the emission localises against the control ensemble, and the drift class is zero to within the grid. Two further properties fit CO and not a carrier. The stage-1 Band 6 window carries 114 threshold crossings where a channel-confined carrier admits one, a floor of 2.26 km s^{-1} on the feature’s extent; and the peak flux densities, 76 and 78 mJy per channel, are the right order for this belt’s spatially integrated CO, their ratio depressed below optically thin LTE by beam

dilution on emission resolved in both beams and by the subthermal excitation of Matrà et al. (2017).

Stellar emission, the competing explanation towards an active young star, is suppressed twice: chromospheric and coronal gyrosynchrotron and free-free emission are broadband — the AU Mic flares of MacGregor et al. (2020) are coherent across the full 8 GHz — whereas the block-median baseline of §4.1 removes everything broader than its 65-channel passband ($\sim 32 \text{ MHz}$ at the fine class), and the inverse-variance stack over all integrations dilutes time-confined flares. What survives is the multiplicative route of §4.1, a flare continuum times a spectral calibration residual, bounded there and measured for CP–72 2713 in §5.2.

The carbon tuning agrees by omission. All 24 Band 8 windows belong to 3 stars and are [C I] $^3P_1-^3P_0$ (492.160651 GHz) tunings. HD 48370’s on-star crossing reaches $T_\star = 5.05$ against a ring maximum of 6.23 and lies -129 km s^{-1} from [C I], so it is an ordinary noise crossing and the window shows no line at the star in this disc’s carbon tuning, as Cataldi et al. (2023) report. The crossing peak in the flagged window is 0.82 Jy in one 122-kHz channel, with the ring at the same level.

The survey rescored under a repaired screen

Two corrections were built from the stored control vectors and rescored over the 447 windows of the 1655 searched that retain a control vector, and the 4 stage-1 windows inside them. *Radius matching* ranks the star against only the 54 probes with $u \leq 0.30$, at a cost in resolution; *detrending* keeps all 512 probes and subtracts the pooled radial profile $m(u)$ measured on the calibration windows, at unchanged resolution, its one free choice being the stellar debit $m(0)$ between 0.15 and 0.32 σ . Table 22 therefore reports the debit each flagged window survives rather than adopting a value.

Neither correction is adopted, for two reasons. Neither promotes a window: detrending returns the released set unchanged at 4 windows (0 added, 0 lost) and radius matching returns 5 (Table 22), and the unattributed count is one for every repair and every debit between zero and 0.61 σ . And neither makes the rank distribution uniform: detrending moves the calibration-sample median from 0.429 to 0.437 ($D = 0.091$ to 0.081) and the survey median from 0.466 to 0.474 ($p = 0.26$ to 0.51), while a leave-one-star-out fit over the 27 calibration stars returns median 0.437 with $D = 0.081$ against 0.437 and 0.081 in sample, so the profile is neither over-fitted nor sufficient to restore exchangeability. The survey’s false-alarm budget is therefore taken from the measured tail rate below.

Noise at the stellar position

The noise scale of Eq. 3 is built across the annulus, so it carries no variance peculiar to the stellar position, where residual continuum, deconvolution or self-calibration residuals and position-correlated calibration error would concentrate. That is the method’s largest untested assumption, and the retained products bound it. R_σ , the star’s robust residual scale over the 4 stage-1 windows that retain full control vectors, measured against the median of the 8 retained control positions, runs 0.995 to 1.001: the star agrees with its ring to better than 0.7 per cent at 95 per cent confidence. That matters

TABLE 20

LINE-ATTRIBUTION AUDIT THROUGH THE FRAME CHAIN OF §F.1: THE THREE LINE-ATTRIBUTED WINDOWS, AND A CLASS A β PICTORIS BAND 6 WINDOW THAT CARRIES NO STAGE-1 FLAG YET SHOWS THE SAME CO COMPONENT. TOPOCENTRIC OFFSETS ARE OBSERVED MINUS CATALOGUED LABORATORY REST FREQUENCY (PICKETT ET AL. 1998), SO A NEGATIVE OFFSET MEANS THE OBSERVED FREQUENCY LIES BELOW REST. THE BARYCENTRIC CORRECTION IS THE EARTH-EPOCHERIS TERM AT THE EXECUTION-BLOCK MID-TIME. THE AU MIC SYSTEMIC VELOCITY IS CATALOGUED (FOUQUÉ ET AL. 2018). FOR β PICTORIS WE ADOPT $v_{\text{sys}} = +20.0 \pm 0.7 \text{ km s}^{-1}$ (MATRÀ ET AL. 2017) RATHER THAN THE *Gaia* DR3 VALUE OF $+16.84 \text{ km s}^{-1}$ (GAIA COLLABORATION ET AL. 2023), BECAUSE THE STAR IS AN A6V ROTATOR INSIDE THE HOT-STAR TEMPLATE REGIME IN WHICH DR3 RADIAL VELOCITIES CARRY A KNOWN SYSTEMATIC (BLOMME ET AL. 2023).

Flag	UTC start	f_{obs} (GHz)	Transition	Topo. offset (MHz; km s^{-1})	Bary. corr. (km s^{-1})	v_{sys} (km s^{-1})	Stellar-frame offset (km s^{-1})
β Pic B3	2022-03-03 21:49	115.260326	CO(1 $\rightarrow$ 0)	-10.88; -27.46	-7.90	+20.0	-0.39
β Pic B6	2019-03-12 00:14	230.516128	CO(2 $\rightarrow$ 1)	-21.87; -28.44	-8.15	+20.0	-0.29
AU Mic B6	2014-08-18 04:01	230.825230	CO(2 $\rightarrow$ 1)	+287.23; +373.51	-10.02	-4.71	+378.82
β Pic B6 [†]	2013-10-06 06:23	230.528134	CO(2 $\rightarrow$ 1)	-9.87; -12.83	+7.63	+20.0	-0.46

[†]Class A Band 6 window of a different execution block, without a stage-1 flag (Appendix H), shown because it independently shows the same CO component.

TABLE 21

THE STAGE-1 STAR-BAND PAIRS, SCORED UNDER BOTH STATISTICS. THE REGION-MAXIMUM STATISTIC FLAGS 66 WINDOWS IN 41 STAR-BAND PAIRS ACROSS THE SURVEY; LISTED HERE ARE THE PAIRS EITHER STATISTIC PROMOTES TO STAGE 1. S_{reg} : REGION MAXIMUM OF EQ. 3 OVER $n_{\text{src}} = 135$ POSITIONS; $T_{\star}$: THE SYMMETRIC SINGLE-POSITION STATISTIC; ‘CTRL MAX’, ‘RING MAX’: THE LARGEST CONTROL STATISTIC UNDER EACH. THE ν COLUMN IS EACH WINDOW’S *centre*; THE CROSSING FREQUENCY DIFFERS BY UP TO 0.85 GHz, AND TABLE 9 CARRIES ν_{cross} . FOR CP-72 2713 THE REGION MAXIMUM ALREADY LAY ON THE STAR, SO THE TWO COINCIDE.

Star	Band	ν (GHz)	region-max statistic		symmetric statistic			Symmetric outcome
			S_{reg}	ctrl max	$T_{\star}$	ring max		
HD 48370	6	230.7305	27.95	26.83	27.10	26.83	flagged	
β Pic	3	115.2605	20.31	7.27	14.65	7.27	flagged	
β Pic	6	230.5164	12.78	9.12	11.68	9.12	flagged	
AU Mic	6	230.5384	5.97	5.85	5.22	5.85	not flagged	
ALMA J1537-3319	6	217.0183	5.83	5.63	5.09	5.63	not flagged	
CP-72 2713	7	345.1152	5.81	5.68	5.81	5.68	flagged	
HD 14055	7	345.1377	5.80	5.74	5.22	5.74	not flagged	

TABLE 22

THE SURVEY RESCORED UNDER BOTH REPAIRS. “SURVIVES” IS THE LARGEST STELLAR DEBIT $m(0)$, IN UNITS OF EACH WINDOW’S OWN CONTROL SCALE, AT WHICH THE STAR STILL EXCEEDS EVERY DETRENDED CONTROL; IT IS A PROPERTY OF THE WINDOW AND ASSUMES NEITHER DEBIT. FOR REFERENCE THE CONSERVATIVE DEBIT IS 0.15σ AND THE DEBIT THAT CENTRES THE CALIBRATION-SAMPLE RANK DISTRIBUTION IS 0.32σ , AND A WINDOW SURVIVES WHICHEVER DEBIT IS SMALLER THAN ITS ENTRY. NO WINDOW IS ADDED BY EITHER REPAIR.

Window	$T_{\star}$	Survives (σ)	Rad.-m.	Detr.
β Pic B3	14.65	25.77	yes	yes
β Pic B6	11.68	10.80	yes	yes
HD 48370 B6	27.10	0.21	yes	yes
CP-72 2713 B7	5.81	0.61	yes	yes
Flagged, whole survey			5	4
Added			1	0
Expected by chance			8.1	0.9

most for CP-72 2713, where an unmodelled broadband excess of 2.2 per cent at the star would push $T_{\star}$ below the ring maximum; the bound is a factor of three smaller, so that explanation is disfavoured without being excluded. On a seeded random sample of 25 *unflagged* windows the same test returns 0.997 (0.968–1.004), every one within 3 per cent of unity. It tests variances and not means, and is blind to the one- or two-channel artefact the search is built to find, which the local null and recurrence address instead.

The measured false-alarm rate

Because the body of the rank distribution is displaced, $1/(512 + 1)$ is the rank *resolution* of the ensemble and not a per-window null probability, and the survey’s false-alarm budget is quoted from measurement. Two independent measurements agree: of the 599 calibration win-

dows carrying no attributed line, 1 exceed all 512 of their controls, one in 599 or 0.9 times exchangeable, and the pseudo-stars, which cannot contain a signal at all, rank first in 0.28 per cent of out-of-sample trials against 0.19 per cent exchangeable, a factor 1.4.

The uncertainty on that factor is not binomial, because the 16 inner probes of one window share its noise realisation and its execution block. Over all 1326 calibration windows the stored control vectors give 63 first ranks in 2126 trials, a factor 1.5, and resampling whole windows, or whole execution blocks (327 of them), widens the 95 per cent interval barely at all, 1.2–1.9 and 1.2–1.9 against 1.2–1.9 binomial. The factor lies between 1.2 and 1.9, with the released-product value 1.4 inside that interval, and no disposition changes anywhere in it.

Propagated over the survey’s 1655 windows, the pseudo-star rate predicts 4.7 unattributed stage-1 flagged windows, $P(\geq 1) = 99$ per cent, against 3.23 under exchangeability and 4 observed. Every per-window probability in this paper should be read as multiplied by 1.4, and none was claimed significant after the survey-wide correction of Appendix J. The tail is mildly anti-conservative where the body is strongly displaced, and it is the tail the candidate screen uses. The displacement is not one configuration’s doing: 23 of 27 stars have a median rank below 0.5 (Fig. 14). The calibration sample covers 56 stars and windows within a block are not independent epochs, so the sizes above carry that caveat; at 578 null windows the test resolves a shift of this size ($D \geq 0.06$ against 0.08 observed).

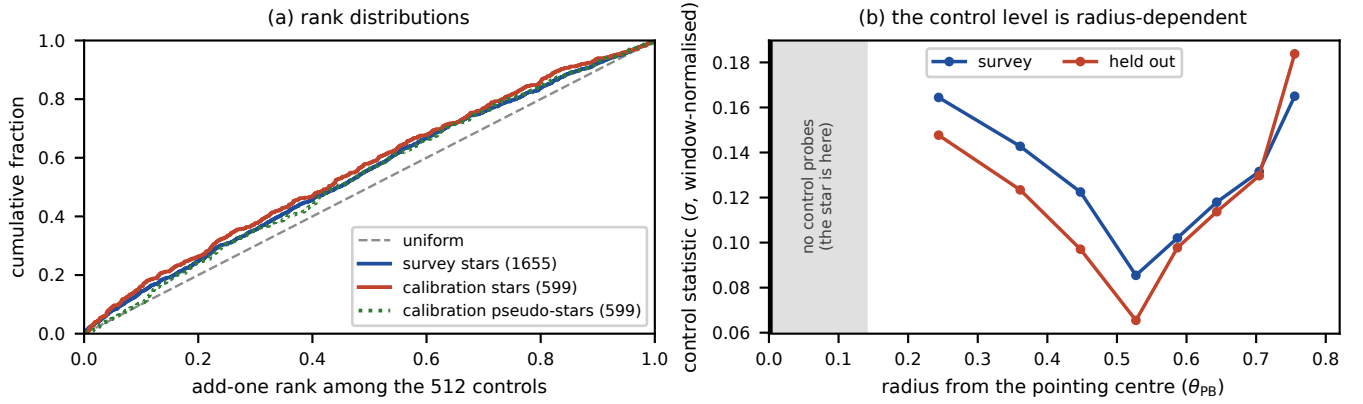


FIG. 14.— Why the spatial control ensemble is a calibrated screen and not an exchangeable rank test. (a) add-one rank of the star among its 512 controls, for the survey and for the calibration windows carrying no attributed line: both lie above the uniform diagonal. The dotted curve is the *pseudo-stars*, control positions at the inner edge of the annulus ranked against the other controls, which contain no star and are displaced just as far (§C.1.2), so the displacement is instrumental. (b) the control statistic itself against radius from the pointing centre, in units of each window’s own noise scale. It is lowest at mid-annulus and rises towards both edges; the shaded strip inside $0.14\theta_{PB}$ holds no control probe at all, and that is where the star sits (median $0.0032\theta_{PB}$).

Audit against a region-maximum statistic

The statistic of Eq. 3 evaluates star and controls identically. A region maximum over $n_{\text{src}} = 135$ positions at the star, ranked against single-position controls, is not exchangeable with them and ranks first with probability $n_{\text{src}}/(n_{\text{src}} + 512) = 20.9$ per cent rather than $1/(512 + 1)$, a property of the estimator class. All 1655 windows are scored under both forms and both columns are released per window. The symmetric criterion flags a strict *subset*, 13 windows against 66 in 41 star-band pairs, with no window flagged by the symmetric form alone (Table 21); those that drop out do so because the stellar position never exceeded its own control ensemble. AU Mic is the clearest case: the excess lies in the field around the star rather than at it.

50 stars hold further public blocks this survey did not take (§3), which §6 turns into an epoch-activity constraint.

Entries excluded from the released catalogue

Four classes of entry are excluded. (i) Four target/bands (Barnard’s Star [B7], Wolf 359 [B7], SCR J1845–6357, Wolf 358) yielded no carrier result, each keeping a valid continuum non-detection. (ii) Giclas 9–38A and B [B3], the pair’s only matched pointing lying ~ 52 arcmin from either star, $35\times$ its primary-beam FWHM. (iii) 13 windows failing the noise test of §A.3, whose noise sits a factor 388–743 above that of the shallowest sibling of the same block and channel width. (iv) mislabelled entries such as “ γ Lupi”, which resolves to HD 139664 at 17.40 pc; these are relabelled and no measured value changes. The 12 Band 9/10 windows toward AU Mic and HD 61005 are included and none is a detection or spatial outlier ($T_{\star} \leq 4.91$).

The archival sweep also searched 198 windows in 47 execution blocks toward 22 stars between 40.8 and 49.8 pc, queued by a beam-matching step that did not check distance. They lie outside the sample this paper reports and enter no number in it, but they went through the identical frozen pipeline and influenced no threshold, mask or statistic decision, which makes them an external null: their stellar add-one ranks have median 0.437 against the 0.5 expected under exchangeability ($D = 0.11$, $p = 0.02$), and they contain 0 stage-1 flagged windows against 0.4 expected.

Validation checks in detail

Each of the three checks summarised in §C.1.1, §C.1.2 and §5.2.4 is given here with the numbers behind it.

The hold-out reservation

The star-versus-control rank prioritises candidates and provides no detection p -value, because exact star/control exchangeability fails in the prospective hold-out sample. Every disposition in §5.2 therefore rests on astrophysical evidence or on recurrence, never on a rank alone.

The statistic and the mask were settled on the 1655 windows above and cannot also calibrate themselves. A hold-out was therefore reserved by a rule committed on 2026-09-14 (c75069040eab), before the archival search finished and before any reserved block was searched: a block is held out if and only if $\text{sha256}(\text{UID})_{[.8]} \bmod 5 = 0$, subject to two guards that use identifiers alone — blocks

TABLE 23

THE CP–72 2713 REPEAT TEST, BOTH BLOCKS SIDE BY SIDE, SEPARATING “NOT DETECTED AGAIN” FROM “A PERSISTENT SIGNAL OF THE ORIGINAL STRENGTH WOULD HAVE BEEN RECOVERED”. THE REPEAT BLOCK IS THE DEEPER OF THE TWO, MATCHES THE FIRST IN TUNING, CHANNELISATION AND INTEGRATION, AND ITS FREQUENCY AXIS REGISTERS TO A FIFTH OF A CHANNEL. A SOURCE AT THE FIRST BLOCK’S FLUX WOULD HAVE RETURNED $T_{\star} = 6.21$ THERE; THE MEASURED VALUE IS 4.70. INTERMITTENT EMISSION IS NOT EXCLUDED BY THIS TEST, AND NO STATEMENT HERE ASSUMES OTHERWISE.

	first block	repeat block
Execution block	Xff0235_X4a6d	Xff0235_X502d
Start-to-start separation (h)	2.05	
On-source time (s)	2994	2994
Channel width (kHz)	488.28	488.28
Channel rms (mJy)	2.097	1.964
Primary-beam FWHM (arcsec)	18.2	18.2
Frequency axes registered to	0.18 channel	
Test frequency (GHz)	344.2697	
Test drift (kHz s ^{−1})	−2.06	
Drift track sweeps (channels)	15.8	
Flux at that cell (mJy)	11.3 $\pm$ 2.0	−1.0 $\pm$ 1.8
$T_{\star}$ in this window	5.81	4.70
Largest control statistic	5.68	5.93
$T_{\star}$ expected if persistent	—	6.21
Pair inconsistent at	4.6 σ	
First flux taken as exact	6.8 σ	

TABLE 24

THE INJECTION CAMPAIGN RESOLVED BY CONFIGURATION. P_{50} AND P_{90} ARE THE INJECTED AMPLITUDES, IN UNITS OF THE LOCAL NOISE, AT WHICH HALF AND NINE-TENTHS OF TRIALS ARE RECOVERED. NO AXIS SHOWS A DEFICIT LARGE ENOUGH TO MOVE THE QUOTED THRESHOLD OUTSIDE THE WINDOW-TO-WINDOW BRACKET OF $\times 0.48$ – $\times 1.35$; THE SUB-CHANNEL-PHASE ROWS ARE MEAN RESPONSE FACTORS, NOT RECOVERY AMPLITUDES, AND ARE THE REASON THE THRESHOLD CARRIES A RESPONSE CORRECTION.

Axis	Level	Trials	P_{50} (σ)	P_{90} (σ)
ALMA band	B3	432	4.8	6.7
	B6	1872	4.7	6.1
	B7	1152	4.4	6.0
	B8	432	4.4	6.0
	B10	144	4.5	5.9
Channel width (kHz)	15.3	144	4.2	5.8
	61.0	288	3.0	4.6
	122.1	432	4.2	5.8
	244.1	864	4.7	6.5
	488.3	1152	4.7	6.2
Array	976.6	1152	5.0	6.0
	12 m	3600	4.6	6.0
	ACA	432	4.8	6.7
Drift rate / ceiling	<1/3	1341	4.6	6.1
	>2/3	1410	4.5	5.9
	1/3–2/3	1281	4.6	6.5
Integrations	<150	720	4.6	6.4
	>350	1872	4.5	5.9
	150–350	1440	4.6	6.4
Sub-channel phase	0.000	80	2.0	—
	0.125	80	2.2	—
	0.250	80	2.3	—
	0.375	80	2.5	—
	0.500	80	2.7	—

in the previous release stay in the survey, and no system loses every block — so that the hold-out assignment can be recomputed from the UIDs alone. The three choices that can manufacture a candidate were fixed first: the detection statistic of Eq. 3 on 2026-09-09 (0c465f2661bd), the promote/retire criteria on 2026 September 11 (119a68a6ab50), and the hold-out rule 5 days after the statistic, with the first reserved block searched 3 days later. Because local commit times-

TABLE 25

DATA PARTITIONS USED IN THIS STUDY. “SEEN BEFORE THE STATISTIC FROZE” RECORDS WHETHER THE DATA WERE AVAILABLE TO THE CHOICES THAT DEFINE THE DETECTION STATISTIC AND THE MASKS; THE HOLD-OUT WAS RESERVED BY THE RULE OF §C.1.1, COMMITTED BEFORE ANY OF ITS BLOCKS WAS SEARCHED. ONLY THE SCIENCE SAMPLE CONTRIBUTES TO THE HEADLINE SEARCH RESULT.

Partition	Blocks	Windows	Stars	Purpose	Seen before the statistic froze?	In the headline result?
Science sample	404	1655	90	the search re-	yes	yes
Pre-specified hold-out	77	315	36	ported here calibrates rank, tail and radial behaviour	no	no
Out-of-sample, beyond 40 pc	47	198	22	external null	no	no
No surviving window	3	–	–	failed the quality cut	n/a	no

tamps can be rewritten, the dates are corroborated by GitHub server-side push times and by the Zenodo deposit’s minted date. The diagnostics built afterwards — the radius-corrected normalisation, the visibility-domain test, the drift-resolved injection analysis and the P_{90} budget — cannot promote a window.

Of 484 blocks processed, 77 are reserved (15.9 per cent), giving 315 windows toward 36 stars with no star and no system lost from the headline sample. Four quantities are calibrated on those blocks and on nothing else: the rank displacement, the false-alarm tail factor, the radial behaviour of the control ensemble, and the claim that stage-1 status alone carries no candidate significance.

The ranks are displaced out of sample by very nearly the amount measured in sample: median add-one rank 0.405 against 0.5 (block-clustered 95 per cent 0.341–0.458), $D = 0.109$ at $p = 0.0011$ over 315 windows. The false-alarm tail factor, the measured pseudo-star first-rank rate divided by the exchangeable rate 1.95×10^{-3} , is 1.2 on 5040 trials, with a bootstrap clustered over the 77 reserved blocks giving 0.6–1.9, and the reserved blocks contain 0 stage-1 outliers against 0.61 expected. The standardised control level runs +0.12, +0.10, +0.12, +0.16 from the inner to the outer edge of the annulus, a mean per-window slope of $+0.067 \pm 0.033$: the displacement reproduces out of sample, but the steep inner-edge excess seen in sample does not, so the radial correction of §C.1.2 is one demonstrated mechanism for the displacement rather than the whole of it.

The older, processing-order calibration set is retained because it carries repeat observations the hold-out does not. Table 26 gives what the frozen pipeline returned on it, with the 4 unattributed outliers and their repeat blocks, and Table 23 the CP–72 2713 repeat test in full. The positive control recovers (β Pictoris CO at the stellar position in 4 of the 7 stage-1 windows), the measured tail rate predicts 3.7 unattributed events over 1322 windows, and none of the 4 recurs: each is a single cell a few per cent above the largest of 512 controls that falls back into the body of its own control distribution when re-searched at its own tuning. Recurrence rather than rank settles a marginal event.

The rank distribution of the remaining 578 windows is adverse to the method. The star’s median add-one rank is 0.405 where 0.5 is expected, with 14 per cent of ranks below 0.1 and 6 per cent above 0.9 against 10 per cent each, so the distribution is displaced rather than growing a tail. The window-level figure ($D = 0.08$, $p < 0.01$) treats windows sharing a calibration as independent, but clustering removes neither the shift nor its significance:

TABLE 26

THE EXTERNAL CALIBRATION SAMPLE AS SEARCHED AT 2026 SEPTEMBER 17. (a) WHAT THE FROZEN PIPELINE RETURNED ON 1322 WINDOWS IT HAD NEVER SEEN. (b) THE 4 UNATTRIBUTED OUTLIERS, ALL OF THEM FROM THE SURVEY SAMPLE, EACH A SINGLE CELL A FEW PER CENT ABOVE THE LARGEST OF 512 CONTROLS, AND NONE RECURRING AT ITS OWN TUNING IN A FURTHER BLOCK. THE REPEAT ROW GIVES THE LARGEST STATISTIC THE FURTHER BLOCK RETURNS ANYWHERE ON ITS OWN GRID; THE *pinned* VALUE AT THE FIRST BLOCK’S OWN FREQUENCY AND DRIFT, WHICH IS THE STRONGER TEST, IS 3.60 FOR CP–72 2713 (§A.3.1).

(a) Calibration sample, as searched

Execution blocks / stars	326 / 56
Spectral windows	1322
Median add-one rank of the star	0.419
against uniform	$D = 0.095$
Trigger crossings at the star	60
Stage-1 outliers	7
β Pictoris CO at the star	4
no astrophysical attribution	3
Expected unattributed at the measured tail rate	3.7

(b) The four unattributed single-epoch outliers

	CP–72 2713	61 Vir	HD 14055	HD 23484
Sample	survey	survey	survey	survey
T_*	5.81	6.16	6.03	5.30
Ring maximum	5.68	5.88	5.96	5.25
Largest T_* in a further block at the same tuning	4.70	4.83	5.42	4.99
controls above the star there	many	119	20	21
Recurs?	no	no	no	no

99 of 149 blocks have a mean rank below 0.5 (sign test $p < 0.001$), and a star-clustered bootstrap over 578 windows places the median at 0.440 (95 per cent 0.42–0.46). The survey’s own 1655 windows lean the same way without resolving it (median 0.45, star-clustered 95 per cent 0.42–0.47) and the two distributions are indistinguishable ($D = 0.03$, $p = 0.78$), so the in-sample checks were underpowered rather than contradicted. The control ensemble is an empirically calibrated screen for extreme values, not an exactly exchangeable rank test.

The radial gradient

The cause is the control geometry, not the primary beam. Earlier versions of this paper argued that primary-beam-corrected noise rising with radius left the probes at a different effective noise from the star. That explanation cannot be right: the statistic is formed on uncorrected amplitudes, whose thermal noise does not depend on position, and a primary-beam effect would act on both arrays, whereas the departure is confined to the ACA stratum. The extraction code takes the primary beam from a fixed 12 m dish for every window, so an ACA 7 m window has its control annulus placed at 58 per cent of

the correct radius; at 230 GHz the inner edge lands at 3.8 arcsec instead of 6.6 arcsec, inside a typical ACA synthesised main lobe of 7.3 arcsec. Those controls carry the star’s own flux, stand high, and push the star low — the direction observed.

Three tests establish it, all from the released products. The control probes are drawn once from a fixed seed, so fractional radii are identical in every window and each stored control statistic pairs with its radius. *First*, the excess is confined to the affected stratum and sits at the inner edge of the annulus: pooling by fractional radius, the innermost bin stands +0.11 above the outer half in the 529 ACA windows with stored control vectors, against +0.01 in the 299 12 m windows. Pooled over both arrays the profile runs +0.16 at $u = 0.24$, falls to +0.09 near mid-annulus and returns to +0.17 at the outer edge — the U shape that a monotonic radial noise gradient does not predict and that contamination of the innermost probes does. *Second*, matching on radius removes most of the departure: ranking the star against only the 54 probes with $u \leq 0.30$ moves the survey median from 0.444 to 0.455 and $D = 0.081$ ($p = 8 \times 10^{-8}$) to 0.066 ($p = 3 \times 10^{-5}$). *Third*, a position with no star in it behaves the same way: an innermost probe treated as a *pseudo-star* and ranked against the other 512–1 probes is displaced as far as the real ones out of sample, median 0.443 against 0.429 ($p = 0.003$). The displacement is a property of *position in the field*, not of stars.

The repair belongs at the measurement stage: standardising each probe within its own radius bin, and the star on the innermost bin, removes the gradient by construction, with no free parameter. Run over all 1614 windows it changes the list — 4 windows lose the flag and 1 gains it, so the count falls from 13 to 10. We do not adopt the radius-corrected statistic as primary, and the reason is empirical: applied prospectively to 778 windows it played no part in building, it leaves the stellar rank inconsistent with uniformity and moves the median further from the exchangeable value (§5.2.2). Since the correction removes the radial component by construction, what remains is larger than what it removed and is not radial. The frozen statistic therefore defines every count reported here; under the corrected one the survey would have 10 stage-1 windows, 8 attributed to CO and 2 unattributed — different windows, the same conclusion — and the released catalogue carries both.

The same mechanism touches completeness, which the injections alone would not reveal because they are placed at the stellar position only. Extrapolating the out-of-sample radial profile to the stellar radius gives an offset of $+0.08\sigma$, so an injected signal is divided by too much and the quoted recovery fractions are pessimistic by of order 2 per cent of the threshold: small, and in the safe direction, but not zero.

The general point is that spatial control positions prioritise candidates soundly in targeted interferometric data but cannot establish significance in them, because the field around a chosen target is not blank sky. The remedy is physical rather than statistical — a test for a point source at the stellar position, which the visibilities answer directly (§L).

The line mask

The searches as executed matched crossings to catalogued rest frequencies within one spectral channel at zero drift. The β Pictoris windows showed that inadequate: both fell outside that tolerance, by 10.6 and 21.9 MHz, while lying within 3.6 km s^{-1} of the systemic velocity of a resolved, mapped disc. Every crossing was therefore reclassified with a velocity-space mask, $\pm 50 \text{ km s}^{-1}$ around each catalogued transition — a post-hoc classification criterion, never a detection threshold, whose half-width was adopted after the β Pictoris crossings had been seen. Reclassifying all 75 crossings at ± 20 to $\pm 100 \text{ km s}^{-1}$ costs 2.1 to 10 per cent of gross bandwidth and admits no new unattributed outlier.

The mask holds 17 transitions of 11 species at full JPL/CDMS precision (Pickett et al. 1998; Müller et al. 2005). The species criterion is the one objective rule statable in advance: the transitions ALMA’s band definitions are built around — the CO isotopologues, HCN, HCO^+ , CS, SiO, SO, CN, H_2CO and [C I] — which dominate the emission of disc-selected fields. The choice is consequential: a full Splatalogue selection would mask 75.1 per cent of the union against the 4.1 per cent the adopted list masks, so “coincident with a known molecular line” is a statement about a catalogue rather than a binary property. Since no masked crossing exceeds its own control ensemble, no plausible widening or narrowing of the species list changes a disposition.

The cost is 4.1 per cent of the unique-frequency union and 4.7 per cent of window-summed gross bandwidth, CN and CO dominating. It is not uniform, so Table 16 gives it band by band: it reaches 6.5 per cent in Band 9 and stays below 1.5 per cent in Bands 3–5. Nor is it shared equally between the two experiments — 7.4 per cent of the Class A union against 3.1 per cent in Class B — so the fine-channel experiment pays 2.4 times the fractional price, its coverage being concentrated in the line-tuned windows a molecular mask is designed to remove. The two do not sum to the total because the classes overlap in frequency.

Masking is a design selection effect separating *discovery* from *classification*. A carrier at a line frequency is as detectable as anywhere else, but it cannot be classified, because the astrophysical prior at a catalogued transition is overwhelming. Removing those frequencies from the searched domain is the conservative choice, at the price that *we place no constraint at all on artificial emission lying inside the masked regions, at any signal strength*. A better design searches every frequency and attaches a *lower prior* to a line-coincident event instead, since these are exactly the frequencies a communicator wanting to be found near a natural reference might choose. The released channels support that second search, for which the β Pictoris crossings are the test case.

Additional robustness checks

The checks in Table 27 support the conclusions without changing them; the numbers behind each are in the released products, and the regeneration script reproduces them.

TABLE 27
FURTHER ROBUSTNESS CHECKS. EACH WAS RUN TO COMPLETION; NONE ALTERS A DISPOSITION.

Check	Outcome
R_σ residual scale at the four unattributed positions	Consistent with the control median to better than 1 per cent; the stellar positions carry no noise peculiar to themselves
Fully symmetric region-maximum statistic	Flags a strict subset of the frozen screen's windows; no window is flagged by the symmetric form alone
Region-maximum audit, all windows	Both statistics released per window; the dependence of the candidate list on the choice is reproducible from the catalogue
Polarisation states	198 of 200 resolvable blocks deliver both parallel hands; a per-hand search remains available at no observing cost
CP-72 2713 diagnostics and second epoch	The event does not recur (§5.2); no diagnostic changes that
β Pictoris recurrence block by block	The disc CO recurs in every block whose tuning covers it, which is what makes it the survey's positive control

RECOMMENDATIONS FOR A NEXT-GENERATION
ALMA TECHNOSIGNATURE SEARCH

ALMA’s advantages here are its mm-wave sensitivity, an essentially unsearched tuning range, the spatial discrimination an interferometer supplies, and a public archive that can be reprocessed without new allocations. Six changes would enlarge what this experiment can say, given in priority order. The first two would change a conclusion; the rest widen a domain.

(1) *Use the visibilities.* A celestial point source at the stellar direction $\mathbf{x}_\star$ contributes

$$V_{ij}(\nu, t) = A(\nu, t) \exp(-2\pi i \mathbf{u}_{ij} \cdot \mathbf{x}_\star), \quad (\text{L1})$$

so rotating the phase centre to the star leaves a coherent signal entirely in the real part, with an imaginary component consistent with zero on every baseline. That asks whether the data contain a point source *where the star is*. The control ranking asks only whether one extraction is unusually large compared with its neighbours, which is a weaker question and one whose null depends on where in the field those neighbours sit. Applied to every window it would demote the control ensemble to a calibration layer and give each candidate a physically specific test rather than a rank. It requires retaining the visibilities, which this survey did not do.

(2) *Search the remaining public blocks these stars already hold.* §D.1.1 reports 115 of the 177 searched with the frozen pipeline, at no telescope cost. The gain is narrower than it sounds: every block is repeat coverage of a star and tuning already searched, so they add no star, no system and no frequency, and they extend only the time axis. They are also concentrated, attaching to 29 of the 90 stars. Of the 27 systems with a single searched epoch only 5 would gain a second block, and the extension moved 1 system out of the single-epoch class. A second block often supplies a baseline of hours rather than years, which one archive query settles before any download.

(3) *Retain more control spectra.* Full dynamic spectra survive for only 8 of the 512 control positions, which is what breaks the second-stage null in a structured field. Retaining all of them costs storage and nothing else.

(4) *Drift discrimination on coarse windows.* Every in-grid drift on a 15.625-MHz window is sub-channel over a track, so 1252 of 1655 windows measure amplitude but cannot distinguish a drifting carrier from a stationary one. Finer channelisation, or a drift model spanning several blocks, would recover that axis.

(5) *Extend the injection campaign and add polarisation.* The stratified campaign of Appendix G spans band, channel width, integration time and array; what remains unspanned is other target environments and edge-phased drift grids. A per-hand search buys partial discrimination against weakly polarised backgrounds at no sensitivity cost, but ALMA’s linear feeds mean it is blind to circular polarisation; a cross-hand extraction is what a circularly polarised carrier would require.

(6) *A boundary on drift linearity.* The search assumes linear drift over a track, and curvature matters once $\frac{1}{2}\ddot{\nu}T^2$ exceeds half a native channel, which only the shortest-period orbits reach: $P \lesssim 1.6$ d for a typical Class A window. Among the 42 known planets of the 19 hosts in the sample, 1 lies inside that boundary,

TRAPPIST-1 b at 1.51 d. The grid is built at constant acceleration, so the boundary is one number across the band, 3.60 ms^{-2} (Fig. 6); it clears the accelerations a transmitter on a rotating, orbiting planet cannot avoid by a wide margin, exceeding Earth’s rotation by a factor of approximately 106 (a factor of about one hundred, not 10^6).

Finally, disc-CO false positives should be budgeted for, but the targets should not be pre-judged. 9 of the 13 stage-1 flagged windows are molecular emission toward disc hosts, arising in 2 of the 69 debris-disc systems searched: one flagged system in 34. Any future search on ALMA, NOEMA or the JVLA inherits that rate. The remedy is procedural — measure the continuum at each target *before* promoting a window, which would have caught HD 48370 and both β Pictoris windows at the first gate. We do *not* recommend lowering the threshold toward disc hosts: a detection statistic should not know whether its target is astrophysically interesting, or a real transmitter would be penalised for the company it keeps.

RADIO-FREQUENCY INTERFERENCE

Millimetre interferometry is an unusually poor environment for terrestrial interference, though not an empty one, and an earlier version of this statement overreached. The searched range is not above all allocated services: ITU Radio Regulations Article 5 carries active allocations throughout 90–275 GHz, and WRC-19 identified parts of 275–450 GHz for land-mobile and fixed use. The specific hazard in this band is the 94.0–94.1 GHz Earth exploration-satellite (active) allocation used by spaceborne cloud radar, CloudSat and, since 2024, Earth-CARE, which is a documented contaminant of ALMA Band 3. 1 of the 1655 searched windows overlaps that band, and it carries 0 threshold crossings and 0 stage-1 events, so no result here rests on it and no overpass timing needs to be checked. What does favour this band is that terrestrial transmitters are sparse compared with centimetre wavelengths, that the array sits at 5000 m on a site chosen partly for radio quietness, and, decisively, that an interferometer is not a total-power detector. A signal that is not in the far field at the phase centre does not fringe-track: it acquires a rapidly varying geometric phase across the array and is suppressed in the correlation rather than imaged. Three tests are applied to every stage-1 event: the spatial control screen, which asks whether the excess is at the star or in the field; the visibility fit of §5.2.1, which asks whether it has the phase signature of a source at the phase centre; and recurrence at the same tuning in an independent epoch. Each is, for this reason, also an interference test.

Three measurements support that reasoning. First, interference is a property of the observatory and not of the star, so a terrestrial carrier should appear at the same *sky* frequency in every window that covers it. Of the 4 unattributed events, 1 has a released crossing frequency; 86 windows toward 6–39 other stars in 276 other blocks cover the frequencies of the four, and 0 of them carries a crossing in the same channel. For the other 3 the pipeline stored no peak frequency, so this test cannot be run on them and we do not claim it was. Second, over all 451 released crossing frequencies in the survey there is no clustering in sky frequency: binned at 1 MHz the largest

multiplicity is 2, against 1.9 for a null that keeps each crossing in its own window and randomises its position within it ($p = 0.90$). Third, an artefact of the signal chain would sit at a fixed intermediate frequency rather than a fixed sky frequency, and the crossings do not: their fractional positions within their own windows have median 0.50 and are uniform to $D = 0.023$ at $p = 0.97$ over 451 crossings. CP-72 2713's crossing, the one with a released frequency, lies 35 channels from its nearest window edge.

None of this is a substitute for the ON/OFF cadence a dedicated search uses, and we do not present it as one. Accordingly we claim only that *no identified interference mechanism accounts for these events*; allocation tables cannot exclude harmonics, intermodulation products, unconventional emitters or near-field sources, and an archival dataset without a simultaneous off-source strategy cannot do better than that. What it establishes is that the crossing population carries none of the three signatures interference leaves: fixed sky frequency, fixed

intermediate frequency, or presence in unrelated targets. The phase test each event failed is also the test a near-field interferer would fail.

HD 48370, in one line. $T_\star = 27.10$ against a ring maximum of 26.83, a margin of 0.27, the smallest in the survey; CO(2→1) at $v_{\text{bary}} = +41.2 \text{ km s}^{-1}$, matching the foreground cloud Cataldi et al. (2023) report; and the visibility fit puts the emission *off* the star, with an imaginary part 7.9σ from zero. *Disposition: foreground carbon monoxide*, on three independent lines of evidence, none of them the rank.

Three further checks are reported in Appendix D: a prospectively searched extension of 115 public blocks, which recovers the β Pictoris disc CO in three previously unsearched epochs and adds no unattributed event; the β Pictoris recovery itself, which is the survey's astrophysical positive control for both the spatial and the spectral machinery; and the absence of any shared data with the only prior ALMA-archival search, which leaves the null result resting on internal validation alone.

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